

Success



PREMIUM
IITJEE MATERIALS



JEE ADVANCED
WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr.Super60-I

2011-P1

Dt: 29-07-17

Time: 07:30 AM to 10:30 AM

WAT-12

Max.Marks: 240

Name of the Student: _____

H.T. NO:

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29-07-17_ Jr.Super60-I_WAT-12

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as :
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

SYLLABUS

MATHS: (As per New Schedule)

SYLLABUS COVERED FROM_ 24-07-17 TO 28-07-17 (WAT-12)

PHYSICS: (As per New Schedule)

SYLLABUS COVERED FROM_ 24-07-17 TO 28-07-17 (WAT-12)

CHEMISTRY: (As per New Schedule)

JEE-ADVANCE-2011-P1-Model**Time: 3:00****IMPORTANT INSTRUCTIONS****Max Marks: 240****CHEMISTRY**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 8 – 11)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 12 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 17 – 23)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 24 – 30)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 31 – 34)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 35 – 39)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 40 – 46)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

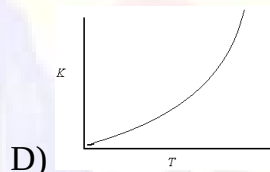
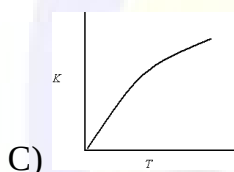
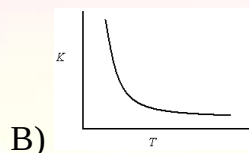
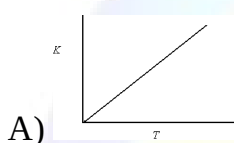
MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 47 – 53)	Questions with Single Correct Choice	3	-1	7	21
Sec – II(Q.N : 54 – 57)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III(Q.N : 58 – 62)	Questions with Comprehension Type (2 Comprehensions – 2 + 3 = 5Q)	3	-1	5	15
Sec – IV(Q.N : 63 – 69)	Questions with Integer Answer Type	4	0	7	28
Total				23	80

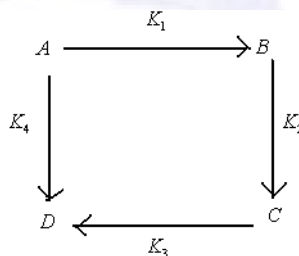
SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct

1. The rate constant of most of the reactions increases with increase in temperature. According to Arrhenius equation, $k = Ae^{-E_a/RT}$. The curve of rate constant k against temperature T will be



2. Consider an elementary reaction sequence shown in the figure which of the following equations is/are correct



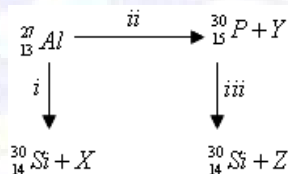
A) $\frac{d[A]}{dt} = -K_1[A] + K_4[D]$

B) $\frac{d[C]}{dt} = K_2[B] - K_3[C]$

C) $\frac{d[D]}{dt} = -K_4[D] + K_3[C]$

D) $\frac{d[A]}{dt} = K_2[C] - K_4[D]$

3. For the formation of B from A, heat liberated is 20 kJ/mol and required activation energy of 100 kJ/mol . Then activation energy (in kJ/mol) for the reaction $A \rightarrow B$ is :
 A) 120 B) 100 C) 80 D) 60
4. The nuclide sulphur-35, is neutron rich therefore, it is likely to undergo radioactive decay by
 A) β^- - emission B) electron capture C) β^+ - emission D) neutron capture
5. ${}_{24}\text{Cr}^{53} + x \rightarrow y + {}_{26}\text{Fe}^{56}$ The correct combination of x, y is
 A) ${}_0^1\text{n}, {}_2^4\text{He}$ B) ${}_2^3\text{He}, {}_1^2\text{H}$ C) ${}_2^4\text{He}, {}_0^1\text{n}$ D) ${}_2^3\text{He}, {}_0^1\text{n}$
6. Bombardment of aluminium by α -particle leads to list artificial disintegration in two ways, i and ii as shown, and ${}_{15}^{30}\text{P}$ is disintegrating to give ${}_{14}^{30}\text{Si}$. Products x, y and z respectively are



- A) Proton, neutron, positron B) neutron, positron, proton
 C) proton, positron, neutron D) positron, proton, neutron
7. The rate constant of the reaction varies with temperature as :



T(K)	200	400
$k(\text{s}^{-1})$	2.0×10^{-11}	1.0×10^{-11}

Assuming that the frequency factor does not change in this temperature range, the arhenius activation energy for the reaction is: ($\ln 2$ is 0.693)

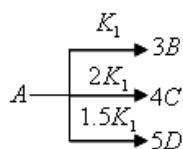
- A) 1.650 kJ/mol B) 2.3046 kJ/mol C) 23.046 kJ/mol D) 16.5 kJ/mol

SECTION – II**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

8. According to Arrhenius equation, which of the following is /are correct Statement(s)
- A) a high activation energy usually implies a fast reaction
- B) rate constant increases with increase in temperature. This is due to greater number of collisions whose energy exceeds the activation energy
- C) the pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.
- D) higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant.
9. Choose the correct statement(s) :
- A) If rate of formation of NO is $3.6 \times 10^{-3} M / \text{Sec}$ in $4NH_{3(g)} + 5O_{2(g)} \rightarrow 4NO_{(g)} + 6H_2O_{(g)}$ then the rate of formation of H_2O is $5.4 \times 10^{-3} M / \text{sec}$
- B) The plot of $\ln \frac{a}{a-x}$ against 't' is a straight line then the reaction may be decomposition of H_2O_2 to H_2O and O_2 (a= initial conc, a-x is conc at time 't',)
- C) The value of rate constant, predicated by Arrhenius equation is 'A', if $T \rightarrow \infty$
- D) The rate constant k_1 and k_2 of two reactions are in the ratio 2 : 1. The corresponding energies of activation of the reactions will be related as $E_2 > E_1$. If A value is same for two reactions
10. A plot of number of neutrons (N) against the number of protons (P) of stable nuclei exhibits upward deviation from linearity for atomic number, $Z > 20$, For an unstable nucleus having $\frac{N}{P}$ ratio less than 1, the possible mode (s) of decay is (are) :
- A) β^- -decay (β -emission) B) neutron emission
- C) orbital or K-electron capture D) β^+ -decay (positron emission)

11.



All reactions are of first order. At time $t=t_1$ ($t_1 > 0$) $\frac{[B]}{[C]} = \alpha$; at time $t=t_2$ ($t_2 \geq t_1$), $\frac{[C]}{[D]} = \beta$

Which of the following is/are correct ?

A) $\alpha = 0.5$

B) $\beta = \frac{4}{3}$

C) $\alpha = 0.375$

D) $\beta = \frac{16}{15}$

SECTION – III

(COMPREHENSION TYPE)

This section contains 2 groups of questions. One group has 3 question and another group has 2 multiple choice questions based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Passage - I

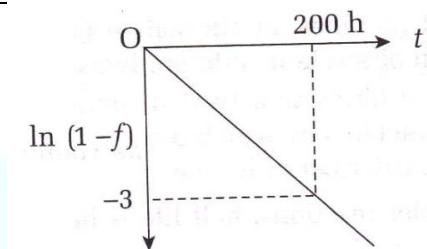
Oxidation of metals is generally a slow electrochemical reaction involving many steps. These steps involve electron transfer reactions. A particular type of oxidation involve overall first order kinetics with respect to fraction of unoxidised metal surface thickness $(1-f)$ relative to the maximum thickness (T) of oxidized surface, when metal surface is exposed to air for a considerable period of time.

Rate law: $\frac{df}{dt} = k(1-f),$

Where $f = x/T$, x = thickness of the oxide film at time 't'

And T = thickness of the oxide film at time $t = \infty$

For the oxidation of a particular metal, a graph is shown in the figure.



12. The time taken for thickness to grow 50% of T is
 A) 23.1 h B) 46.2 h C) 100 h D) 92.4 h
13. The exponential variation of f with $t(h)$ is
 A) $f = (1 - e^{-3t/200})$ B) $f = e^{-3t/200} - 1$ C) $f = e^{-3t/200}$ D) $f = e^{3t/200}$

Passage - II

In any nuclear reaction, the total mass of product is found to be less than the total mass of reactants. The difference in mass is due to the conversion of some mass into energy and hence, energy is produced in the nuclear reactions.

14. The nuclide Sc^{50} , mass 49.9516 u, is neutron rich. It decays to form Ti^{50} , mass 49.94479 u. If the emitted β - particle has a kinetic energy of 0.80 MeV, what is the kinetic energy of the neutrino emitted simultaneously?
 A) 2.87 MeV B) 5.54 MeV C) 9.36 MeV D) 6.58 MeV
15. Calculate the binding energy per nucleon for the nuclide $_{17}Cl^{37}$, if its mass is 36.9659 u (Given m (neutron) = 1.0087 u, m (proton) = 1.0078 u)
 A) 317.53 MeV B) 8.58 MeV C) 7.689 MeV D) 7.88 MeV
16. A nuclear explosion has taken place leading to increase in concentration of C^{14} in nearby areas. C^{14} concentration is C_1 in nearby areas and C_2 in areas far away. If the age of the fossil is determined to be T_1 and T_2 at the places respectively, then

A) The age of the fossil will increase at the place where explosion has taken and

$$T_1 - T_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

B) The age of the fossil will decrease at the place where explosion has taken and

$$T_1 - T_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

C) The age of fossil will be determined to be same

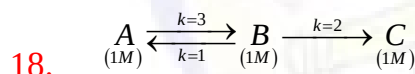
D) $\frac{T_1}{T_2} = \frac{C_1}{C_2}$

SECTION -IV

(INTEGER ANSWER TYPE)

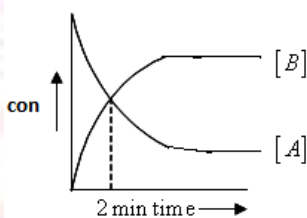
This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

17. The human body contains about 18% carbon by weight. Of total carbon, $1.50 \times 10^{-10}\%$ is C^{14} (by weight). The number of disintegrations occurring per minute in a person of 70 kg is $x \times 10^y$. The value of (x+y) is nearly (half-life of C^{14} is 5570 years)



If all the reactions in the above system give first order kinetics what is the rate of disappearance of 'B' in the given instant

19. For a first order reaction $A_{(g)} \rightarrow 3B_{(g)}$ the concentration verses time graph is given below



What is the half life in minutes (answer to the nearest integer) ?

(log 2 = 0.3, log 3 = 0.48)

20. ${}_{92}\text{U}^{238}$ disintegrates to give 4α and 6β particles. The group number of the daughter element in modern periodic table is

21. For a reaction $A \rightarrow B$, the rate law expression is $\frac{-d[A]}{dt} = K[A]^{1/2}$. If initial concentration of A is A_0 . The number of correct statements are

i) The plot of $\sqrt{A}$ against 't' will be linear

ii) The half life period is $t_{1/2} = \frac{\sqrt{2}(\sqrt{2}-1)}{K} \sqrt{A_0}$

iii) The integrated form of rate expression is $\sqrt{A} = \frac{-K}{2}t + \sqrt{A_0}$

iv) Time taken for reaction is decreased for second half life than first half life period

v) When concentration of reactant is doubled rate increases to 4 times at that instant

22. The rate constant of a reaction at 27°C is $3 \times 10^{-3} \text{ min}^{-1}$. The fraction of molecules crossing the energy barrier of the reaction at that temperature is $10^{-3}\%$. If temperature is increased, the maximum value of rate constant k can be attained is $x \times 100 \text{ min}^{-1}$. The value of x is

23. Half life of a radioactive nuclide is 20 years. If a sample of this nuclide has activity of 6400 dpm today, its activity (in dpm) after 100 years would be $x \times 100 \text{ dpm}$; the value of 'x' is :

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct

24. A smooth wire frame in the form of a parabola $x^2 = cy$ is rotated about its vertical axis (y-axis) at angular speed ω . A bead of mass m is at rest relative to wire frame at $(-a, b)$. Then the value of ' ω ' is :
- A) $\sqrt{\frac{2g}{a}}$ B) $\sqrt{\frac{2g}{c}}$ C) $\sqrt{\frac{2g}{b}}$ D) $\sqrt{\frac{2gc}{ab}}$
25. A small block of mass 1 kg is connected by a light string passing through a hole in a smooth table with a block of mass 5kg which hangs vertically. Find the angular velocity of the block on the table moving in a circle of radius 0.1 m in order to keep the other block at rest?
- A) 10rad/s B) $10\sqrt{5} \text{ rad/s}$ C) $10\sqrt{2} \text{ rad/s}$ D) $\frac{10}{\sqrt{2}} \text{ rad/s}$
26. A thin rod of mass m and length ' ℓ ' is rotated in a horizontal plane about one of its ends with a constant angular speed ω in gravity free space. Then the tension at the midpoint of the rod is :
- A) $\frac{m\ell\omega^2}{2}$ B) $\frac{m\ell\omega^2}{4}$ C) $\frac{m\ell\omega^2}{8}$ D) $\frac{3}{8}m\ell\omega^2$
27. A small sphere of mass 0.2 kg is attached to a light string of length 1.3 m whose upper end is fixed to a ceiling. The sphere is made to describe a horizontal circle of radius 0.5 m. Then the time taken to complete one revolution is :
- A) $\frac{2\pi}{5}\sqrt{3} \text{ sec}$ B) $\frac{\pi}{5}\sqrt{3} \text{ sec}$ C) $\frac{\pi}{5} \text{ sec}$ D) $\frac{2\pi}{5} \text{ sec}$

28. A smooth horizontal circular table is rotating at an angular speed ω about its vertical axis. A groove is made on the surface along a radius. A particle is gently placed inside the groove at a distance ' a ' from the center. The speed of the particle w.r.t table when its distance from the center is ' x ' is :

A) $\omega\sqrt{x^2 - a^2}$ B) $\frac{\omega}{2}\sqrt{x^2 - a^2}$ C) $\omega\sqrt{x^2 + a^2}$ D) $\frac{\omega}{2}\sqrt{x^2 + a^2}$

29. A car is moving in a circular horizontal track of radius 10 m with a constant speed of 10 m/s. A plumb bob is suspended from the roof of the car by a light string. The angle made by the string with the vertical (in steady state) is ($g = 10 \text{ m/s}^2$)

A) zero B) 30° C) 45° D) 60°

30. A point moves along an arc of a circle of radius R . Its velocity depends from upon the distance covered s as $V = a\sqrt{s}$, where a is constant. The angle θ between the vector of total acceleration and tangential acceleration is

A) $\tan \theta = \sqrt{\frac{s}{R}}$ B) $\tan \theta = \sqrt{\frac{s}{2R}}$ C) $\tan \theta = \frac{2s}{R}$ D) $\tan \theta = \frac{s}{2R}$

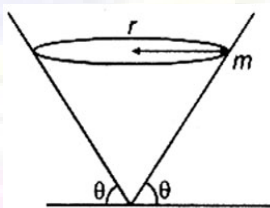
SECTION – II
(MULTIPLE CORRECT CHOICE TYPE)

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

31. The position of a particle varies with time as $\vec{r} = a \sin 2t\hat{i} + a \cos 2t\hat{j} + bt\hat{k}$. Then the

- A) trajectory of particle as seen by an observer moving with a velocity $b\hat{k}$ is a circle.
- B) trajectory of the particle w.r.t ground observer is a circle.
- C) Angle between velocity and acceleration is $\pi/2$
- D) Angle between velocity and acceleration is constant.

32. A block of mass 1kg is placed on a horizontal turn table rotating at a constant angular velocity ω rad/s. at a distance 1m from the axis by connecting it to a string of length 1m whose other end is fixed at the centre of table. The value (/s) of ω for which tension in string is zero is/are : [the coefficient friction between block and surface is 0.4]
- A) 2rad/s B) 1rad/s C) 0.5rad/s D) 4 rad/s
33. A particle is moving along y-axis with a speed of $10m/s$. At some instant of time it is at $(0, 4m)$. The magnitude of its angular velocity about a point,
- A) $(0, 0)$ is zero B) $(0, 0)$ is infinity
C) $(3m, 0)$ is $1.2rad/s$ D) $(-3m, 0)$ is $1.2rad/s$
34. A ball of mass m is rotating in a circle of radius r with uniform speed v inside a smooth cone as shown in figure. Let N be the normal reaction on the ball by the cone then choose the correct option



- A) $N = mg \cos \theta$ B) $g \sin \theta = \frac{v^2}{r} \cos \theta$ C) $N \sin \theta = \frac{mv^2}{r}$ D) None of these

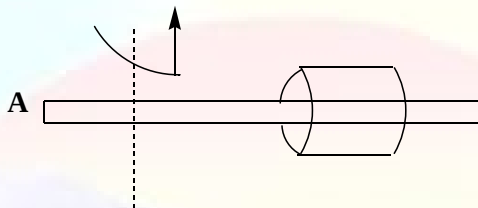
SECTION – III (PARAGRAPH TYPE)

This section contains 2 paragraphs. Based upon one of the paragraphs 3 multiple choice questions based on the 3 multiple choice questions have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Passage-1

The figure shows a rod which starts rotating with angular acceleration α about vertical axis passing through one of its end (A) in horizontal plane. A bead of mass m just fits

the rod and is situated at a distance r from end A. Friction exist between rod and the bead with coefficient μ . As the angular velocity of rod increases the bead starts sliding over the rod (say after time t_0) Based on above information answer the following questions)



35. The normal force acting on bead at time $t(< t_0)$ is

- A) mg B) $mr(\alpha t)^2$ C) $m\sqrt{g^2 + r^2(\alpha t)^4}$ D) $m\sqrt{g^2 + (r\alpha)^2}$

36. Friction force acting on bead at time $t(< t_0)$ is given by

- A) μmg B) $mr(\alpha t)^2$ C) $m\mu\sqrt{g^2 + (r\alpha)^2}$ D) $\mu m\sqrt{g^2 + r^2(\alpha t)^4}$

37. If the bead start sliding at $t = t_0$ then value of t_0 is given by

- A) $\sqrt{\frac{\mu\sqrt{g^2 + (r\alpha)^2}}{r\alpha^2}}$ B) $\sqrt{\frac{mg}{r\alpha^2}}$ C) $\sqrt{\frac{\mu\sqrt{g^2 + (r\alpha)^2}}{r\alpha^2}} - \sqrt{\frac{mg}{r\alpha^2}}$ D) None of these

Passage-2

The role of friction is very important in determining the range of speeds of a vehicle on a banked road. It can change both in magnitude and direction to prevent the vehicle from slipping or skidding. A turn of radius 20m is banked for vehicles going at a speed of 36 kmph. The coefficient of static friction between road and tyre is 0.4. Then mark the correct options.

38. The maximum speed of vehicle without skidding ?

- A) 36 kmph B) 54 m/s C) 54 kmph D) 72 kmph

39. The minimum speed of vehicle without slipping is (approximate value)

- A) 4 m/s B) 5 kmph C) 36 kmph D) 6 m/s

SECTION -IV
(INTEGER ANSWER TYPE)

This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. **The appropriate bubbles below the respective question numbers in the ORS have to be darkened.**

40. A particle moves on a circular path of radius 0.2m with an angular acceleration 0.25 rad./s^2 . The particle starts from rest at $t = 0$. The time at which magnitude of radial acceleration and tangential acceleration are same is _____ sec.

41. A circular ring of mass 3.14 kg and radius 0.05 m is rotating with a constant angular velocity 10 rad/s about its vertical axis on a smooth horizontal table. Then the tension in the ring is $10/n$ N, then n is _____.

42. A long horizontal rod has a bead which can slide along its length and is initially placed at a distance L from one end A. the rod is rotated in a horizontal plane about end A, with a constant angular acceleration $\alpha = 0.1 \text{ rad/s}^2$. The coefficient of friction between rod and bead is $\mu = 0.4$. Neglecting gravity, the time after which the bead starts slipping is _____ sec.

43. A small block slides on a frictionless horizontal table. It is constrained to move inside a ring of radius 1m fixed to the table. At $t = 0$ the block is moving along the circumference inside the ring with speed 10 m/s. the coefficient of friction between block and ring is 0.1. the speed of the block at $t = 1$ sec is : _____ m/s
44. A particle moves in the x-y plane with velocity $\vec{v} = a\hat{i} + bt\hat{j}$. At the instant $t = \frac{a\sqrt{3}}{b}$ the magnitude of radial acceleration is $\frac{b}{n}$. find n =?
45. A particle of mass 1kg is moving in x-y plane with a velocity of 6 m/s at angle of 53° with x-axis. A force of 15N is applied along x-axis at $t = 0$ on it. The radius of curvature of the path of the particle just after application of force is _____ m.
46. A particle of mass 1 kg is at origin. At $t = 0$. It is given a velocity of 1m/s at angle of 30° with y-axis in x-y plane. It is moved under the influence of a force $\vec{F} = \vec{V} \times \vec{A}$, where $\vec{A}$ is constant vector of magnitude π units directed along positive x-axis. The x-coordinate of particle at $t = 2$ s is: _____ m ($\vec{V}$ - instantaneous velocity)

SECTION – I**(SINGLE CORRECT CHOICE TYPE)**

This section contains 7 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct

47. If $p(1 + \frac{t}{\sqrt{2}}, 2 + \frac{t}{\sqrt{2}})$ is any point on a line, then the range of the value of t for which the point P lies between the parallel lines $x + 2y = 1$ and $2x + 4y = 15$ is
- A) $-4\sqrt{2}/3 < t < 5\sqrt{2}/6$ B) $0 < t < 5\sqrt{2}/6$
 C) $4\sqrt{2} < t < 0$ D) none of these
48. If the lines $ax + y + 1 = 0$, $x + by + 1 = 0$ and $x + y + c = 0$ (a, b, c being distinct and different from 1) are concurrent, then $\left(\frac{1}{1-a}\right) + \left(\frac{1}{1-b}\right) + \left(\frac{1}{1-c}\right) =$
- A) 0 B) 1 C) $1/(a+b+c)$ D) none of these
49. The straight line $x + 2y - 9 = 0$, $3x + 5y - 5 = 0$ and $ax + by - 1 = 0$ are concurrent, if the Straight line $35x - 22y + 1 = 0$ passes through the point
- A) (a, b) B) (b, a) C) (-a, -b) D) none of these
50. The locus of the image of the point (2, 3) in the line $(x - 2y + 3) + 9\lambda(2x - 3y + 4) = 0$ is ($\lambda \in R$)
- A) $x^2 + y^2 - 3x - 4y - 4 = 0$
 B) $2x^2 + 3y^2 + 2x + 4y - 7 = 0$
 C) $x^2 + y^2 - 2x - 4y + 4 = 0$
 D) $x^2 + y^2 - 2x - 4y + 3 = 0$
51. The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation $x - 7y + 5 = 0$. The equation of the other line is
- A) $3x + 3y - 1 = 0$ B) $x - 3y + 2 = 0$ C) $5x - 5y - 3 = 0$ D) $5x + 5y - 3 = 0$

52. If the pairs of lines $3x^2 - 2pxy - 3y^2 = 0$ and $5x^2 - 2qxy - 5y^2 = 0$ are such that each pair bisects the angle between the other pair, the pq equals to
- A) -1 B) -7 C) -9 D) -15
53. The equation $a(x^4 + y^4) - 4bxy(x^2 - y^2) + 6cx^2y^2 = 0$ represents two pairs of perpendicular lines. The two pairs will coincide if
- A) $b^2 = a + 3ac$ B) $a^2 = b^2 - 3ac$ C) $a^2 + b^2 = 3ac$ D) $2b^2 = a^2 + 3ac$

SECTION - II
(MULTIPLE CORRECT CHOICE TYPE)

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which ONE OR MORE is/ are correct

54. If the chord $y = mx + 1$ of the circle $x^2 + y^2 = 1$ subtends an angle measure 45° at the major segment of the circle, then the value of 'm' can be
- A) 2 B) 1 C) -1 D) none of these
55. If (α, α^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0, x + 2y - 3 = 0, 5x - 6y - 1 = 0$ then
- A) $2\alpha + 3\alpha^2 - 1 > 0$ B) $\alpha + 2\alpha^2 - 3 > 0$ C) $\alpha + 2\alpha^2 - 3 < 0$ D) $6\alpha^2 - 5\alpha + 1 > 0$
56. The sides of a triangle are the straight lines $x + y = 1, 7y = x$ and $\sqrt{3}y + x = 0$. Then which of the following is an interior point of the triangle?
- A) Circumcenter B) Centroid C) Incenter D) Orthocenter
57. The distance of any point (x, y) from the origin is defined as $d = \max\{|x|, |y|\}$ then the distance of the common point for the family of lines $x(1 + \lambda) + \lambda y + 2 + \lambda = 0$ (λ being parameter) from the origin is
- A) 1 B) 2 C) $\sqrt{5}$ D) 0

SECTION – III
(COMPREHENSION TYPE)

This section contains 2 paragraphs. based upon one of the paragraphs 3 multiple choice questions based on the 3 multiple choice questions have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) for its answer, out of which ONLY ONE is correct.

Passage - 1

Let L be the line belonging to the family of straight lines

$$(a + 2b)x + (a - 3b)y + a - 8b = 0, a, b \in R \text{ which is the farthest from the point } (2, 2)$$

58. The equation of line L is

A) $x + 4y + 7 = 0$ B) $2x + 3y + 4 = 0$ C) $4x - y - 6 = 0$ D) none of these

59. Area enclosed by the line L and the coordinate axes is

A) $4/3$ sq. units B) $9/2$ sq. units C) $49/8$ sq. units D) none of these

Passage - 2

A(1, 3) and C $(-2/5, -2/5)$ are the vertices of a triangle ABC and the equation of the internal angle bisector of $\angle ABC$ is $x + y = 2$

60. The equation of side BC is

A) $7x + 3y - 4 = 0$ B) $7x + 3y + 4 = 0$ C) $7x - 3y + 4 = 0$ D) $7x - 3y - 4 = 0$

61. The coordinates of vertex B are

A) $(3/10, 17/10)$ B) $(17/10, 3/10)$ C) $(-5/2, 9/2)$ D) $(1, 1)$

62. The equation of side AB is

A) $3x + 7y = 24$ B) $3x + 7y + 24 = 0$ C) $13x + 7y + 8 = 0$ D) $13x - 7y + 8 = 0$

SECTION -IV
(INTEGER ANSWER TYPE)

This section contains 7 questions. The answer to each of the questions is a single digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened.

63. In a plane there are two families of lines: $y = x + r, y = -x + r$, where $r \in \{0, 1, 2, 3, 4\}$. The Number of the squares of the diagonal of length 2 formed by these lines is
64. The straight lines $L \equiv x + y + 1 = 0$ and $L_1 \equiv x + 2y + 3 = 0$ are intersecting, 'm' is the Slope of the straight line L_2 such that L is the bisector of the angle between L_1 and L_2 . The value of m^2 is
65. Let 'P' any point on $x - y + 3 = 0$ and 'A' be fixed point (3,4). If the family of lines given by $(3\sec\theta + 5\operatorname{cosec}\theta)x + (7\sec\theta - 3\operatorname{cosec}\theta)y + 11(\sec\theta - \operatorname{cosec}\theta) = 0$ are concurrent at B for all permissible values of ' θ ' and maximum of $1|PA - PB| = 2\sqrt{2n} (n \in \mathbb{N})$ then $n =$
66. If the point $P(\tan^2\theta, \tan\theta)$ lies in the region corresponding to the acute angle between the lines $2y = x$ and $4y = x$, then $\left[\frac{\tan\theta}{2}\right]$ is equal to (where $[.]$ denotes greatest integer function)
67. The difference of the slopes of the lines represented by $x^2(\tan^2\theta + \cos^2\theta) + 2xy\tan\theta + y^2\sin^2\theta = 0$ is
68. If the area of the triangle formed by the lines $3x^2 - 2xy - 8y^2 = 0$ and the line $2x + y - k = 0$ is 5 sq. units. then ' $|k|$ ' is
69. For all real values of a and b, lines $(2a + b)x + (a + 3b)y + (b - 3a) = 0$ and $mx + 2y + 6 = 0$ are concurrent. Then $|m|$ is equal to



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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr.Super60-I

2011-P1

Date: 29-07-17

Time: 07:30 AM to 10:30 AM

WTA-12

Max.Marks:240

KEY SHEET

CHEMISTRY

1	D	2	B	3	A	4	A	5	C	6	A
7	B	8	BCD	9	ABCD	10	CD	11	CD	12	B
13	A	14	B	15	B	16	A	17	7	18	0
19	5	20	3	21	4	22	3	23	2		

PHYSICS

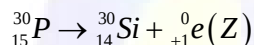
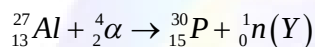
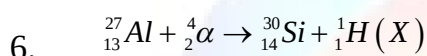
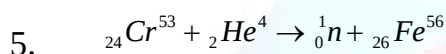
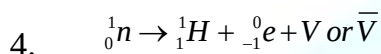
24	B	25	B	26	D	27	A	28	A	29	C
30	C	31	ACD	32	ABC	33	ACD	34	BC	35	D
36	B	37	A	38	C	39	A	40	2	41	4
42	2	43	5	44	2	45	3	46	1		

MATHS

47	A	48	B	49	A	50	D	51	D	52	D
53	D	54	BC	55	ACD	56	BC	57	B	58	A
59	C	60	B	61	C	62	A	63	9	64	4
65	5	66	1	67	2	68	5	69	2		

SOLUTIONS**CHEMISTRY**

1. rate constant increases exponentially with temperature.
2. For elementary reactions order is equal to molecularity.
3. Conceptual



$$9. \quad +\frac{1}{4} \frac{d(NO_2)}{dt} = +\frac{1}{6} \frac{d(H_2O)}{dt}$$

$$11. \quad \frac{[B]}{[C]} = \frac{3K_1}{8K_1} = \frac{3}{8} = \alpha; \frac{[C]}{[D]} = \frac{8K_1}{7.5K_1} = \frac{8}{7.5}$$

$$12 \text{ \& } 13. \quad \frac{df}{dt} = k(1-f)$$

$$\frac{df}{(1-f)} = kdt$$

On integration

$$-\ln(1-f) = kt + c$$

But C = 0

$$k = -\frac{2.303}{t} \log(1-f)$$

$$k = \frac{-2.303}{200} \times -3$$

$$= \frac{2.303 \times 3}{200}$$

At half life

$$\frac{-2.303 \times 3}{200} = -\frac{2.303}{t_{1/2}} \log \left(1 - \frac{T}{2T} \right)$$

$$-\frac{3}{200} = \frac{1}{t_{1/2}} \log \frac{1}{2}$$

$$t_{1/2} = 46.2 \text{ hrs}$$

14.

$$\Delta M \times 931.5 = kE_1 + kE_2$$

$$15. \quad \Delta m = (20 \times 1.0087 + 17 \times 1.0078)u; \overline{B.E} = \frac{37.3066 - 36.9659}{37} = 8.58 \text{ MeV}$$

$$16. \quad T_1 - T_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$

$$17. \quad \text{weight of } C^{14} = 70 \times \frac{18}{100} \times \frac{1.56 \times 10^{-10}}{100} = 1.9656 \times 10^{-11} \text{ kg}$$

$$\text{Number of } C^{14} \text{ atoms} = \frac{1.9656 \times 10^{-11} \times 10^3}{14} \times 6.023 \times 10^{23}$$

$$= 8.54 \times 10^{14}$$

$$\text{Activity} = \lambda N = \frac{0.693}{5570 \times 365 \times 24 \times 60} \times 8.54 \times 10^{14}$$

$$= 2 \times 10^5 \text{ disintegrations/min}$$

$$18. \quad \frac{d(B)}{dt} = 2(B) + 1(B) - 3(B) = 2 \times 1 + 1 \times 1 - 3 \times 1 = 0$$

19.



$$\text{At } t = 0 \quad a \quad 0$$

$$\text{At } t = t \quad a - x \quad 3x$$

$$\text{At time } t, (a - x) = 3x$$

$$a = 4x$$

$$k = \frac{2.303}{t} \log \frac{4x}{3x} \Rightarrow \frac{0.693}{t_{1/2}} = \frac{2.303}{t} \log \frac{4}{3}$$

$$t_{1/2} = 4.82 \text{ min}$$

$$\text{no. of } \alpha = 4 \quad A = 238 - 16 = 222$$

$$20. \quad \text{no. of } \beta = 6 \quad Z = 92 - 2 = 90$$

${}_{90}\text{Th}^{232}$ belongs 3^{rd} group

21. i, ii, iii, iv, v are correct statements

$$22. \quad f = e^{-E_a/RT} = \frac{10^{-3}}{100} = 10^{-5}$$

$$K = A e^{-E_a/RT}$$

$$3 \times 10^{-3} = A \cdot 10^{-5}$$

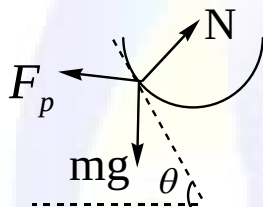
$$A = \frac{3 \times 10^{-3}}{10^{-5}} = 300 \text{ min}^{-1}$$

$$23. \quad \text{No. of half lives (h)} = \frac{100}{20} = 5$$

$$A = \lambda N = \lambda \frac{N_0}{2^n} = \frac{6400}{2^5} = 200$$

PHYSICS

24.



$$N \cos \theta = mg$$

$$N \sin \theta = m r \omega^2$$

$$\tan \theta = \frac{r \omega^2}{g}$$

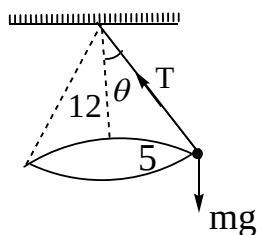
$$\frac{dy}{dx} = \frac{r \omega^2}{g}$$

$$\omega = \sqrt{\frac{2g}{c}}$$

$$25. \quad m_1 r \omega^2 = m_2 g$$

$$26. \quad T = \int_{r=\frac{2}{2}}^2 d m r \omega^2$$

27.



$$T \cos \theta = mg$$

$$T \sin \theta = mr\omega^2$$

$$\text{Time taken} = \frac{2\pi}{\omega}$$

$$28. \quad a = r\omega^2 = v\omega = v \frac{dv}{dx}$$

$$29. \quad \tan \theta = \frac{g}{a} = \frac{gr}{v^2}$$

$$30. \quad N = mr\omega^2 \text{ \& } mg \sin \theta - f = ma_t$$

$$\therefore f = mg \sin \theta = 5N$$

$$31. \quad \vec{S}_{rel} = a \sin 2t \hat{i} + a \cos 2t \hat{j}$$

$$\vec{a} \perp \vec{v}$$

$$32. \quad T + f = mr\omega^2$$

$$T = 0 \Rightarrow f = mr\omega^2 \leq \mu mg$$

$$33. \quad w = \frac{v_{\perp}}{r}$$

$$34. \quad N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{r}$$

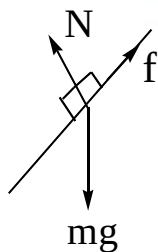
$$35-37. \quad N_1 = mg \text{ \& } N_2 = ma_t = mr\alpha$$

$$N = \sqrt{N_1^2 + N_2^2}$$

$$\text{For } t < t_0, f = mr\omega^2 = mr(\alpha t)^2$$

$$\text{At } t = t_0 = f = \mu N$$

38-39.



$$N \sin \theta \mp f \cos \theta = \frac{mv^2}{r}$$

$$N \cos \theta \pm f \sin \theta = mg$$

$$40. \quad R\alpha = R\omega^2, \omega = \alpha t$$

$$41. \quad T = m r \omega^2 / 2\pi$$

$$42. \quad N = m a_t = m r \alpha$$

$$f = m r \omega^2$$

$$t = \sqrt{\frac{\mu}{\alpha}}$$

$$43. \quad N = m r \omega^2$$

$$f = \mu N$$

$$f = \mu m r \omega^2 = -m r \alpha$$

$$\frac{d\omega}{dt} = -\mu \omega^2$$

$$\int \frac{d\omega}{\omega^2} = \int -\mu dt$$

$$44. \quad a = \sqrt{a_t^2 + a_r^2}$$

$$45. \quad R = \frac{v^2}{a_N} = \frac{mv^2}{F \sin \theta}$$

$$46. \quad \text{path followed is helix, with its axis along x-axis. X-coordinate} = \text{pitch} = v v \cos 60^\circ t = 1$$

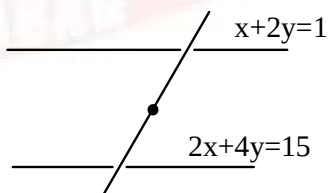
MATHS

$$47. \quad \text{Let P be on } x+2y=1, \text{ then}$$

$$1 + \frac{t}{\sqrt{2}} + 2 \left(2 + \frac{t}{\sqrt{2}} \right) - 1$$

$$\text{Let P be on } 2x+4y=15, \text{ then}$$

$$2 \left(1 + \frac{t}{\sqrt{2}} \right) + 4 \left(2 + \frac{t}{\sqrt{2}} \right) - 15$$



Since the point lies between the lines and $x=t, t \in \left(\frac{-4\sqrt{2}}{3}, \frac{5\sqrt{2}}{3}\right)$

48. If the given lines are concurrent, then
$$\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 - C_1$

$$\text{or } a(b-1)(c-1) - (c-1)(1-a) - (b-1)(1-a) = 0$$

Dividing by $(1-a)(1-b)(1-c)$

Adding 1 on both sides we get $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$

49. The three lines are concurrent if
$$\begin{vmatrix} 1 & 2 & -9 \\ 3 & 5 & -5 \\ a & b & -1 \end{vmatrix} = 0 \quad \text{or } 35a - 22b + 1 = 0$$

Which is true if the line $35x - 22y + 1 = 0$ passes through (a, b)

50. The family of lines $(x - 2y + 3) + \lambda(2x - 3y + 4) = 0$ is concurrent at point $P(1, 2)$

If the images of point $A(2, 3)$ in the above variable lines is $B(h, k)$, then

$$AP = BP$$

$$\text{Or } (h-1)^2 + (k-2)^2 = (2-1)^2 + (3-2)^2$$

51. The family of line through the given lines is $L \equiv x - 7y + 5 + \lambda(x + 3y - 2) = 0$

For line $L=0$ in the diagram, the distance of any point say $(2, 0)$ on the line $x + 3y - 2 = 0$ from the line $x - 7y + 5 = 0$ and line $L=0$ must be the same, therefore

$$\frac{|2+5|}{\sqrt{50}} = \frac{|2+2\lambda+5-2\lambda|}{\sqrt{(1+\lambda)^2 + (3\lambda-7)^2}}$$

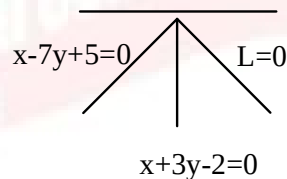
$$\text{Or } 10\lambda^2 - 40\lambda = 0$$

$$\text{i.e } \lambda = 4 \text{ or } 0$$

$$\text{Hence, } L = 0, \lambda = 4$$

Therefore, the required line is $5x + 5y - 3 = 0$

52. Use angular bisector formulae



53. Conceptual

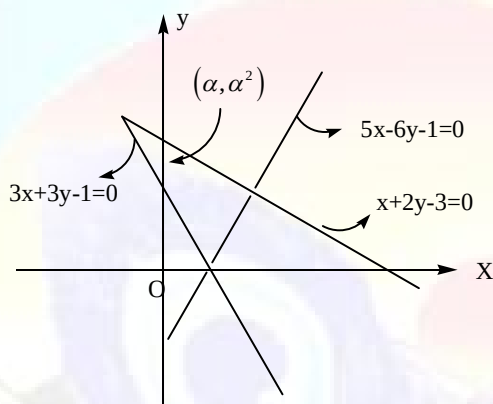
54. The chord subtends 90° at the center $(0,0)$. Making $x^2 + y^2 = 1$ homogeneous in $y = mx + 1$, we get

$$x^2 + y^2 = (y - mx)^2$$

$$\text{Or } (1 - m^2)x^2 + 2mxy = 0$$

The angle between these lines is 90° if $1 - m^2 + 0 = 0$ i.e $m = \pm 1$

55.



O and the point (α, α^2) lie on the opposite sides with respect to $2x+3y-1=0$ Hence,

$$2\alpha + 3\alpha^2 - 1 > 0 \dots\dots\dots(i)$$

O and the point (α, α^2) lie on the same side with respect to $x+2y-3=0$, Hence

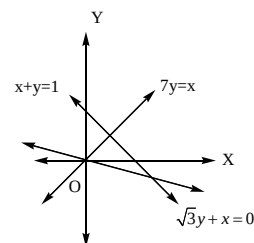
$$\alpha + 2\alpha^2 - 3 < 0 \dots\dots\dots(2)$$

Again O and the point (α, α^2) lie on the same side with respect to $5x-6y-1=0$

Hence

$$5\alpha - 6\alpha^2 - 1 < 0 \text{ or } 6\alpha^2 + 5\alpha + 1 > 0$$

56. Clearly from the figure, the triangle is obtuse-angled. Hence the centroid and incenter lie inside the triangle. The orthocenter and circumcenter lie outside the triangle. Therefore, it is an obtuse-angled triangle.



57. Conceptual

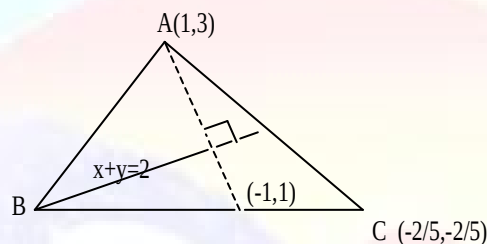
58. Given lines $(x+y+1) + b(2x-3y-8) = 0$ are concurrent at the

Point of intersection of the lines $x+y+1=0$ and $2x-3y-8=0$ which is $(1, -2)$ Now, the line through $A(1, -2)$ which is farthest from the point $B(2, 2)$ is perpendicular to AB . Now, the slope of AB is 4 . Then the required line is $y+2 = -(1/4)(x-1)$ or $x+4y+7=0$

59. Also, the line $x+4y+7=0$ meets the axes at $C(-7,0)$ and $D(0, -7/4)$. Then the area of triangle COD is $(1/2) |(0)(-7/4) - (-7/4)(0)|$ or $49/8$

60. The image of $A(1, 3)$ in line $x+y=2$ is $(1-2(2)/2, 3-2(2)/2) = (-1, 1)$

So line BC passes through $(-1, 1)$ and $(-2/5, -2/5)$ The equation of line BC is



Or $7x+3y+4=0$

61. Vertex B is the point of intersection of $7x+3y+4=0$ and $x+y=2$, ie $B = \left(-5/2, \frac{9}{2}\right)$

62. Line AB is $y-3 = \frac{3-9/2}{1+5/2}(x-1)$ or $3x+7y=24$

63. Each family has parallel lines having the distance between them as $\frac{1}{\sqrt{2}}$ unit. Both the families are perpendicular to each other. So, to form a square of diagonal 2 units, lines of alternate pair to be chosen. Both the families have three such pairs. So, the number of squares possible is $3 \times 3 = 9$

66. position of a point with respect to align

67. Use the formula $|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$

68. Conceptual

69. Conceptual

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr.Super60-I

2014_P2

Date: 05-08-17

Time: 07:30 AM to 10:30 AM

WAT-13

Max. Marks: 180

Name of the Student: _____

H.T. NO:

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PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as :
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

05-08-17_Jr.Super60-I_WAT-13_SYLLABUS

MATHS: SYLLABUS COVERED FROM 31-07-17 TO 04-08-17 (WAT-13)

PHYSICS: SYLLABUS COVERD FROM 31-07-17 TO 04-08-17(WAT)-13

CHEMISTRY: Classification of Elements and Periodicity in Properties : Modern periodic law and present form of the periodic table, s, p, d and f block elements Periodic trends in properties of elements atomic and ionic radii, ionization enthalpy, electron gain enthalpy

JEE-ADVANCED-2014-P2-Model

Time: 7.30 AM to 10.30 PM

IMPORTANT INSTRUCTIONS

Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 11 – 16)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec – III(Q.N : 17 – 20)	Matrix Matching Type	3	-1	4	12
Total				20	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 31 – 36)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec – III(Q.N : 37 – 40)	Matrix Matching Type	3	-1	4	12
Total				20	60

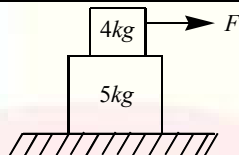
MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No. of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 51 – 56)	Questions with Comprehension Type (3 Comprehensions – 2 +2+2 = 6Q)	3	-1	6	18
Sec – III(Q.N : 57 – 60)	Matrix Matching Type	3	-1	4	12
Total				20	60

Section-1**(Only one Option correct Type)**

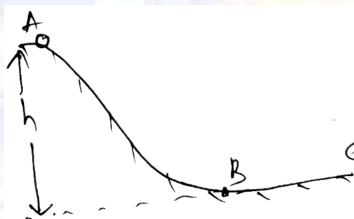
This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct

1. A body of mass 'm' is hauled from the earth's surface by applying a variable force F varying with height 'h' as $\vec{F} = 2(ah-1)m\vec{g}$ where a is a +ve constant. For first half of the height ascended, choose the correct statement: (Neglect variations of g with height)
- A) Work done by F is $\frac{5mg}{2a}$
- B) Increase in gravitational potential energy is $\frac{mg}{2a}$
- C) Data insufficient
- D) Kinetic energy of the body is $\frac{3mg}{4a}$
2. A large slab of mass 5kg lies on a smooth horizontal surface with a block of mass 4kg lying on top of it, μ between block and slab is 0.25. If the block is pulled horizontally by a force of F=6N. The work-done by a force of friction on the slab, between the instants t=2s and t=3s is ($g = 10ms^{-2}$) (Initial speeds is 0)

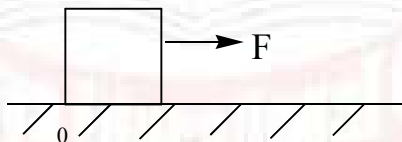


- A) 2.4 J B) 5.55 J C) 4.44 J D) 10 J

3. A small object of mass m slides down very slowly along a rough uneven hilly plane from a height h and stops after travelling a certain distance BC on a horizontal surface. The work needed to be done to return the object to its initial position along the same path very slowly would be



- A) zero B) mgh C) $2mgh$ D) cannot be found
4. A block of mass ' m ' is kept at point O on a rough horizontal surface whose coefficient of friction varies from O as $\mu = 0.5 + \frac{1}{5-x} [\forall x < 3m]$



If the block is moved by an external force F , the work done by frictional forces during the first 3m is (neglect the size of the block)

A) $-mg \left[\frac{3}{2} - \ln \left(\frac{5}{2} \right) \right]$

B) $-mg \left[\frac{2}{3} - \ln \left(\frac{2}{5} \right) \right]$

C) $-mg \left[\frac{3}{2} - \ln \left(\frac{2}{5} \right) \right]$

D) $-mg \left[\frac{3}{2} - \ln \left(\frac{2}{3} \right) \right]$

5. A book of mass m is projected with a speed V across a horizontal surface. The book slides until it stops due to frictional force between book and surface. The surface is now tilted 30° and the book is projected up the surface with the same initial speed V . When the book has come to rest, how does the decrease in mechanical energy of the book-earth system compare to that when the book slide over the horizontal surface?
- A) It is the same B) It is larger
C) It is smaller D) The relationship can't be determine
6. The force exerted by a compression device is given by $F(x) = kx(x-l)$ for $0 \leq x \leq l$ where l is maximum possible compression and k is a constant. The work required to compress the device a distance d is maximum when

A) $d = l/4$

B) $d = \frac{l}{\sqrt{2}}$

C) $d = \frac{l}{2}$

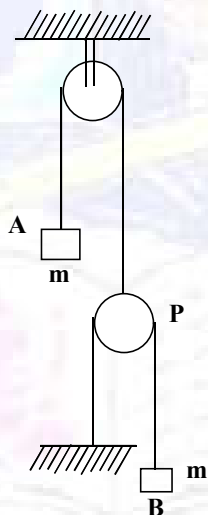
D) $d = l$

7. The potential energy of a 1 kg particle free to move along the X-axis is given by

$v(x) = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) J$. The total mechanical energy of the particle is 2J. Then, the maximum speed (in ms^{-1}) is

- A) $\frac{1}{\sqrt{2}}$ B) $\sqrt{2}$ C) $\frac{3}{\sqrt{2}}$ D) 2

8. In the figure shown the system is released from rest. Find the velocity of block A when block B has fallen a distance l . Assume all pulleys to be mass less and frictionless.



- A) $\sqrt{\frac{gl}{2}}$ B) $\sqrt{\frac{gl}{3}}$ C) $\sqrt{\frac{4gl}{5}}$ D) $\sqrt{\frac{gl}{5}}$

9. Consider the following statements regarding potential energy.
- (i) Potential energy can be positive or negative.
 - (ii) A moving body may or may not have potential energy.
 - (iii) Potential energy can be defined for any type of force.
 - (iv) Potential energy depends on frame of reference.
 - (v) Potential energy corresponding to an internal conservative force will increase when the particle moves opposite to the direction of that conservative force
 - (vi) Potential energy can be defined as a scalar function of position whose negative derivative (gradient) gives force

No. of correct statements

- A) 3 B) 5 C) 4 D) 6

10. Which of the following forces is a conservative force.

- A) $\vec{F} = (xyz)\hat{i} + (x^2z)\hat{j} + (x^2y)\hat{k} \text{ N}$ B) $\vec{F} = (2xyz)\hat{i} + (x^2z)\hat{j} + (x^2y)\hat{k} \text{ N}$
C) $\vec{F} = (xyz)\hat{i} + (xz^2)\hat{j} + (x^2y)\hat{k} \text{ N}$ D) $\vec{F} = (xyz)\hat{i} + (xz^2)\hat{j} + (xy^2)\hat{k} \text{ N}$

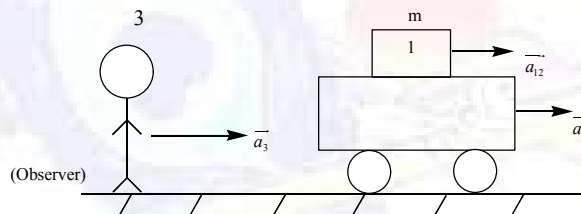
Section-2

(Paragraph Type)

This section contains 3 paragraphs each describing theory, experiment, data etc. Six questions relate to three paragraphs with two questions on each paragraph. Each question pertaining to a particular paragraph should have **only one correct answer** among the four choices A, B, C and D.

PASSAGE: 01

A block of mass m_1 moves with an acceleration a_{12} relative to the plank 2 as shown in figure. Assume zero initial velocities of the bodies and observer (assume all accelerations are along the same direction)



11. What is the work done by the pseudo force as observed by the man on the block during time t ?

A) $\frac{+ma_3(a_{12} + a_2 - a_3)t^2}{2}$

C) $\frac{-ma_3(a_{12} + a_2 - a_3)t^2}{2}$

B) $\frac{-ma_3(a_{12} - a_2 - a_3)t^2}{2}$

D) $\frac{+ma_3(a_{12} + a_2 + a_3)t^2}{2}$
12. The work done by the inertial force on the block relative to the observer on the ground during time t .

A) $-\frac{1}{2}ma_3t^2$

B) $-\frac{1}{2}ma_3a_{12}t^2$

C) 0

D) $\frac{1}{2}m(a_{12} + a_2)t^2$

PASSAGE : 02

A particle A is fixed at origin of a fixed coordinate system. Another particle B which is free to move experiences an force $\vec{F} = \left(\frac{-2\alpha}{r^3} + \frac{\beta}{r^2} \right) \hat{r}$ due to particle A where $\vec{r}$ is the position vector of particle relative to A. It is given that the force is conservative in nature and potential energy at ∞ is zero. If has to be removed from the influence of A, energy has to be supplied for such a process. The ionisation energy E_0 is work that has to be done by an external agent to move the particle from a distance r_0 very slowly. Here r_0 is the equilibrium of the particle

13. The potential energy of particle as a function of r

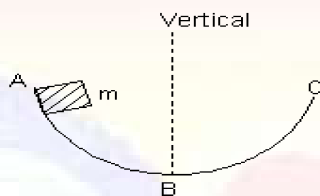
- A) $\frac{\alpha}{r^2} - \frac{\beta}{r}$ B) $\frac{-\alpha}{r^2} + \frac{\beta}{r}$ C) $\frac{-\alpha}{r^2} - \frac{\beta}{r}$ D) $\frac{\alpha}{r^2} + \frac{\beta}{r}$

14. The ionisation energy E_0 of the particle B (magnitude)

- A) $\frac{\beta^2}{2\alpha}$ B) $\frac{2\beta^2}{\alpha}$ C) $\frac{\beta^2}{4\alpha}$ D) $\frac{\beta^2}{\alpha}$

PASSAGE : 03

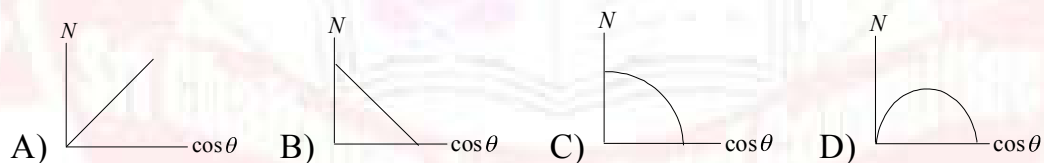
A fixed hemispherical bowl is arranged as shown. A and C are points at the same horizontal level. B is the bottom most point. A small block of mass m is initially placed at A. R is the radius of the bowl. (If necessary take $g = 10\text{m/s}^2$)



15. If bowl is considered as rough and when block is released from A, it comes to rest at B. The work done by the friction as it moves from A to B is

A) $-mgR$ B) $\frac{mgR}{2}$ C) $-2mgR$ D) $-\frac{mgR}{2}$

16. Now consider that bowl is completely smooth and block is released from A. If N is the normal reaction and θ is the angle made by the radius vector at any instant with the vertical axis, which of the following graphs represent the graph of N versus $\cos \theta$, as it moves from A to B ?



Matching - 1

17. In column-I a force $\vec{F}$ is given and in column-II work done by that force between two specified points A and B is given. Match the appropriate column.

Column - I	Column-II
A) $\vec{F} = (3x^2y)\hat{i} + (x^3)\hat{j}$ From A(0,0,0) to B(1,1,1)	P) $2\pi Rk$
B) $\vec{F} = -k(y\hat{i} + x\hat{j})$ From A(0,0) to B(a, a)	Q) -9
C) $\vec{F} = k\left(\frac{-y\hat{i}}{\sqrt{x^2+y^2}} + \frac{x\hat{j}}{\sqrt{x^2+y^2}}\right)$ moves from A(0,0) to B(0,0) along a circular path of radius R in xy plane centered at origin in anti clockwise direction.	R) $-ka^2$
D) $\vec{F} = k\left(\frac{x\hat{i}}{(x^2+y^2)^{3/2}} + \frac{y\hat{j}}{(x^2+y^2)^{3/2}}\right)$ A(0,0) to B(0,a) along a circular arc of radius R centered at origin.	R) zero

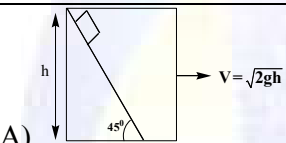
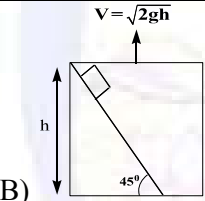
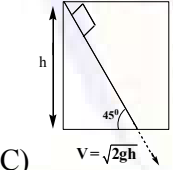
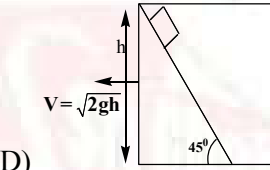
A) A-S, B-P, C-Q, D-R

B) A-Q, B-R, C-P, D-S

C) A-R, B-P, C-S, D-Q

D) A-P, B-R, C-Q, D-S

18. Figure shows four situations in which a small block of mass 'm' is released from rest (with respect to smooth fixed wedge) as shown in figure. Column –II shows work done by normal reaction on the block (with respect to an observer who is stationary on ground) till block reaches at the bottom of inclined wedge, match the appropriate column.

Column – I	Column-II
A) 	P) positive
B) 	Q) negative
C) 	R) equal to mgh in magnitude
D) 	S) equal to zero

A) A-P; B-P; C-Q; D-S

B) A-P; B-P; C-S; D-Q

C) A-P; B-Q; C-Q; D-S

D) A-P; B-P; C-P; D

19. A chain of length l and mass m lies on the surface of a smooth sphere of radius $R > l$ with one end tied to the top of the sphere

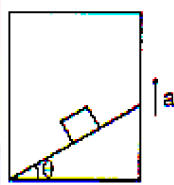
Column I		Column II	
A)	Gravitational potential energy of the chain with reference level at the centre of the sphere	P)	$\frac{Rg}{l} \left(1 - \cos \left(\frac{l}{R} \right) \right)$
B)	The chain is released and slides down, its KE when it has slid by an angle θ	Q)	$\frac{2Rg}{l} \left(\sin \left(\frac{l}{R} \right) + \sin \theta - \sin \left(\theta + \frac{l}{R} \right) \right)$
C)	The initial tangential acceleration (just after it is released)	R)	$\frac{mR^2 g}{l} \sin \left(\frac{l}{R} \right)$
D)	The magnitude of radial acceleration a_r of any element of the chain when it had slid down by angle θ	S)	$\frac{mR^2 g}{l} \left(\sin \left(\frac{l}{R} \right) + \sin \theta - \sin \left(\theta + \frac{l}{R} \right) \right)$

- A) A – S, B – R, C – P, D – Q B) A – Q, B – S, C – P, D – R
 C) A – R, B – S, C – P, D – Q D) A – R, B – S, C – Q, D – P

20. A block of mass m is stationary with respect to a rough wedge as shown in figure.

Starting from rest work done on block in time t :

If $m = 1\text{ kg}$, $\theta = 30^\circ$, $a = 2\text{ m/s}^2$, $t = 4\text{ s}$ and $g = 10\text{ m/s}^2$



Column - I	Column - II
A) By gravity	P) 144 J
B) By normal reaction	Q) 32 J
C) By friction	R) 56 J
D) By all the forces	S) 48 J
	T) none

A) A - T; B - P; C - Q; D - S

B) A - Q; B - P; C - S; D - T

C) A - T; B - Q; C - S; D - P

D) A - T; B - P; C - S; D - Q

Section-1

(One or more options correct type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct

21. The statement that is not true for the long form of the periodic table:
- A) It reflects the sequence of filling electrons in the order of sublevels s, p, d and f.
 - B) It helps to predict the stable valence states of elements
 - C) It reflects trends in physical and chemical properties of elements
 - D) It helps to predict the relative ionicity of bond between any two elements
22. Which of the following represent correct order of first ionization enthalpy for Ca, Ba, Se, S and Ar?
- A) $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$
 - B) $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$
 - C) $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$
 - D) $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$
23. If $1.96 \times 10^4 \text{ kJ}$ energy is supplied to 1 mole $\text{Li}_{(g)}$ it ionises to Li^{+3} . If IE_1 for Li is 520 kJ mol^{-1} and IE for H is $2.18 \times 10^{-18} \text{ J atom}^{-1}$. The IE_2 for Li will be
- A) 7270 kJ mol^{-1}
 - B) $11.81 \times 10^3 \text{ kJ mol}^{-1}$
 - C) $6.54 \times 10^{-3} \text{ kJ mol}^{-1}$
 - D) $3.32 \times 10^4 \text{ kJ mol}^{-1}$

24. The correct order of decreasing electronegativities values among Be(I), oxygen (II), Nitrogen(III) and magnesium (IV) is:
- A) II > III > I > IV B) III > I > V > II > I
C) II > III > IV > I D) II > I > III > IV
25. The first and second ionization potentials of an element M (atomic weight = 25) are 800 and 1500 kJ mol⁻¹ respectively. Calculate the percentage of M⁺² (g) ions formed if 5 g of M(g) absorbs 250 kJ of energy.
- A) 70 B) 40 C) 60 D) 30
26. Which of the following order is correct regarding the atomic radii?
- A) Cr > Mn > Fe B) V < Nb < Ta
C) Sc < Y < La D) Ce < Eu < Lu
27. The correct order of size is:
- A) Cr⁺³ < Cr⁺⁴ < Cr⁺⁵ B) C⁴⁻ < N³⁻ < O²⁻
C) Fe > Fe⁺² > Fe⁺³ D) He⁺ < Li⁺² < Be⁺³

28. Given that
- i) IP_2 order of $C < N < F < O$
 - ii) First negative electron gain enthalpy of $S > Se > Te > O$
 - iii) First ionisation enthalpy of $Sc > Y > La$
 - iv) Electron gain enthalpy with sign $Ne > Ar > Kr > Xe$
- Number of correct statements are
- A) 1 B) 2 C) 3 D) 4
29. In which of the following the order is not correct, according to the property indicated against it?
- A) $Na > Mg > Al$ – Second ionization potential
 - B) $P < S < Cl$ non metallic character
 - C) $F_2 > Cl_2 > Br_2 > I_2$ Oxidising power
 - D) $P_2O_5 < SO_3 < Cl_2O_7 \rightarrow$ Acidic nature
30. Linus Pauling wanted to determine the electron gain enthalpy (electron affinity) of an unknown element **X**. He took one mole of atoms of **X** and found that on absorption of 410 kJ of energy half of **X**-atoms transfer one electron to the other half. To convert all the resulting X^- ions to X^+ ions, an additional energy of 735 kJ was required. What was the electron gain enthalpy of **X** as obtained by Linus Pauling?
- A) -270 kJ mol^{-1} B) -325 kJ mol^{-1} C) -540 kJ mol^{-1} D) -650 kJ mol^{-1}

Section-2

(Paragraph Type)

This section contains 3 paragraphs each describing theory, experiment, data etc. Six questions relate to three paragraphs with two questions on each paragraph. Each question pertaining to a particular paragraph should have **only one correct answer** among the four choices A, B, C and D.

Comprehension: 1

The first ($\Delta_i H_1$) and the second ($\Delta_i H_2$) ionisation enthalpies in kJ mol^{-1} and the electron gain enthalpy in kJ mol^{-1} of a few elements (two metals of same group and two non-metals of same group along with a another non metal) are given below:

Elements	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{eg} H$
(P)	520	7300	-60
(Q)	419	3051	-48
(R)	1681	3374	-333
(S)	1008	1846	-295
(T)	2372	5251	+48

Answer the following questions.

31. Which one of the above elements is least reactive non metal and most reactive metal respectively?
 A) (R) and (Q) B) (S) and (T) C) (T) and (Q) D) (T) and (P)
32. Which one of the above elements is most reactive non metal and least reactive metal respectively?
 A) (P) and (Q) B) (R) and (P) C) (R) and (T) D) (S) and (P)

Comprehension: 2

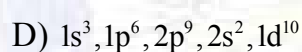
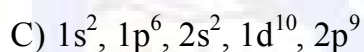
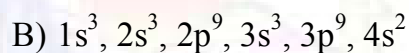
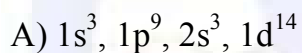
E.N is the tendency of an atom to attract the shared electron pair towards it in a covalent bond. Since it is a relative property it has no units. Hence various scales are proposed such as Mullikan scale, Pauling scale, Allred-Rochowe scale. The most accepted is of Pauling based on bond energy data and more particularly on Ionic-covalent bond resonance energy. While Mulliken scale is applicable to mono valent elements says that E.N is the average of I.P and E.A. Electro negativity on Mulliken scale is 2.8 times more than that of calculated from Pauling scale.

33. H – H, X – X, and H – X bond energies are 104 kcal/mol 38 kcal/mol and 135 kcal/mol respectively. Electro negativity of 'X' is.
- A) 3.76 B) 3.21 C) 2.85 D) 1.72
34. If electron affinity of 'X' is 333 kJ/mol then Ionization energy of 'X' is:
- A) 1712 kJ/mol B) 874 kJ/mol
- C) 1012 kJ/mol D) 2045 kJ/mol

Comprehension: 3

In an universe other than earth all the quantum numbers have same value as on earth except ' ℓ ' and ' s '. Azimuthal quantum number can have values from 0 to $(n + 1)$ in integral step and s (spin quantum no.) has values of $\left(-\frac{1}{2}, 0 \& +\frac{1}{2}\right)$. If all rules like Aufbau rule, Hund's rule and Paulis exclusion rule follow in other universe also.

35. Configuration of element with atomic number 29 will be



36. No. of elements in third period of another universe will be

A) 27

B) 12

C) 36

D) 42

Section-3**(Matching List Type)**

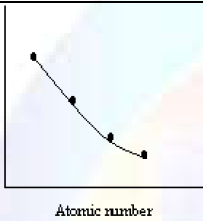
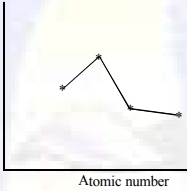
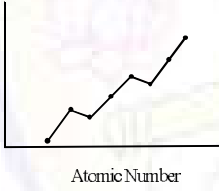
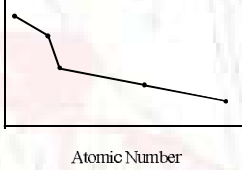
This section contains four questions, each having two matching lists (**List-I** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

37. Consider Slater rules for given code type question

	Column -1		Column -2
P)	Screening constant for Na	1)	8.8
Q)	Screening constant for Na^+	2)	2.2
R)	Z_{eff} for Na	3)	4.15
S)	Z_{eff} for Na^+	4)	6.85

P	Q	R	S
A) 1	3	2	4
B) 3	1	2	4
C) 4	1	2	3
D) 1	3	4	2

38. Match the column -1 with column-2

Column-1		Column-2	
P		1	Magnitude of Electron gain enthalpy for halogens
Q		2	First ionisation enthalpy order for second period elements of long form of periodic table.
R		3	Electro negativity order for halogens
S		4	First ionisation enthalpy of alkali metals

(P.P=Periodic property, Graphs are not up to the scale, use them to predict order)

P	Q	R	S
A) 2	4	1	3
B) 3	1	2	4
C) 3	1	4	2
D) 1	3	2	4

39.

	Column-1		Column-2
P	(Xe) $4f^2 5d^0 6s^2$	1	d-block 6 th group
Q	(Rn) $5f^3 6d^1 7s^2$	2	f-block-6 th period
R	(Kr) $4d^{10} 5s^2$	3	Radioactive element
S	(Ar) $3d^5 4s^1$	4	d-block- 5 th period

P	Q	R	S
A) 3	1	4	2
B) 2	3	1	4
C) 2	3	4	1
D) 4	3	2	1

40. Match the elements with respective periodic property given in the column-1 with their gradation given in column-2

	Coloumn-I		Coloumn-II
P	$Li \xrightarrow{(i) \text{ Basic nature of their oxides }} Be$ $\xrightarrow{(ii) \text{ Metallic character }} B \xrightarrow{(iii) \text{ atomic radius }} C$	1	(i)Increases (ii)Increases (iii)Decreases
Q	$Na \xrightarrow{(i) \text{ Electro positive character }} Mg$ $\xrightarrow{(ii) IP_1} Al \xrightarrow{\text{non metallic character}} Si$	2	(i)Decreases (ii)Decreases (iii)Increases
R	$F \xrightarrow{(i) \text{ magnitude of electron gain enthalpy }} Cl$ $\xrightarrow{(ii) \text{ Atomic size }} Br \xrightarrow{(iii) \text{ oxidising power }} I$	3	(i)Increases (ii)Decreases (iii)Decreases
S	$O \xrightarrow{(i) \text{ magnitude of electron gain enthalpy }} S$ $\xrightarrow{(i) \text{ Acidic nature of oxides }} Se \xrightarrow{(iii) \text{ Electro negetivity }} Te$	4	(i)Decreases (ii)Decreases (iii)Decreases

P	Q	R	S
A) 4	1	2	3
B) 2	1	4	3
C) 3	2	1	4
D) 4	2	1	3

Section-1

(Only one Option correct Type)

This section contains 10 Multiple Choice questions. Each Question has Four choices (A), (B), (C) and (D). Out of Which Only One is correct

41. The number of value(s) of 'a' for which the function $f(x) = \sin(ax) + \tan(\sqrt{2}ax)$ will not be a periodic will be ____
A) 0 B) 1 C) 2 D) None of These
42. If fractional part of $\frac{1}{x}$ and x^2 for some $x \in (\sqrt{2}, \sqrt{3})$ are equal then the value of $x^4 - \frac{3}{x}$ is
A) 3 B) 7 C) 5 D) 0
43. The function $f(x) = x^{\frac{1}{\ln x}}$ is
A) a constant function in its domain
B) has domain $(0, \infty)$
C) a periodic function in its domain
D) All of the above
44. The fundamental period of $f(x) = \sin^9 x + \cos^9 x$ is
A) $\frac{\pi}{2}$ B) 2π C) π D) None of these

45. If $f(x) = \frac{xe^x}{e^x - 1} - \frac{x}{2}$, then $f(x)$ is
A) an even function B) an odd function
C) neither odd nor even function D) both odd and even function
46. If domain of $f(x)$ is $[1, 3]$, then the domain of $f(\log_2(x^2 + 3x - 2))$ is
A) $[-5, -4] \cup [1, 2]$ B) $[-13, -2] \cup \left[\frac{3}{5}, 5\right]$
C) $[-4, 1] \cup [2, 7]$ D) $[-3, 2]$
47. The fundamental period of $f(x) = (\sin x)^0 + (\cos x)^0 + (\tan x)^0$ is
A) 2π B) π C) $\pi/2$ D) Indeterminate
48. If $f(x) = \sqrt{-\sqrt{-x}}$, then domain of $f(x)$ is
A) $[0, \infty)$ B) $(-\infty, 0]$ C) $\{0\}$ D) None of these
49. The domain of $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is
A) $(1, 2)$ A) $(-1, 0) \cup (1, 2)$ C) $(-1, 0) \cup (2, \infty)$ D) $(-1, 0) \cup (1, 2) \cup (2, \infty)$
50. The domain of the function $f(x) = \log_{10}\{1 - \log_{10}(x^2 - 5x + 10)\}$ is
A) $(0, \infty)$ B) $(0, 5)$ C) $(-\infty, 0)$ D) None of these

Section-2

(Paragraph Type)

This section contains 3 paragraphs each describing theory, experiment, data etc. Six questions relate to three paragraphs with two questions on each paragraph. Each question pertaining to a particular paragraph should have **only one correct answer** among the four choices A, B, C and D.

Paragraph for Question Nos. 51 & 52

Let us define a new function of $R_g(x)$ or $R_g(x) = \begin{cases} x, & \text{if } x < 2013 \\ 4026 - x, & \text{if } x \geq 2013 \end{cases}$, then

51. The solution of the equation $R_g(x - 2013) = 2014x + 2013$ is
 A) 0 B) -2 C) 5 D) None of These
52. The maximum value of $R_g(x)$ is
 A) 0 B) 2013 C) 1 D) Can't Say

Paragraph for Question Nos. 53 & 54

Any function $f(x)$ is said to be an EVEN FUNCTION, if $f(-x) = f(x)$.

Any function $f(x)$ is said to be an ODD FUNCTION, if $f(-x) = -f(x)$.

Any function $f(x)$ can be expressed as the sum of an even and an odd function

$$\text{as } f(x) = \left(\frac{f(x) + f(-x)}{2} \right) + \left(\frac{f(x) - f(-x)}{2} \right),$$

Where $\left(\frac{f(x) + f(-x)}{2} \right)$ is EVEN and $\left(\frac{f(x) - f(-x)}{2} \right)$ is ODD.

53. If $f(x) = ae^x + be^{-x} + c$ is an even function, then the best possible relation between a, b, c can be

- A) $a + b = 0$ B) $a = b$ C) $a = b + c (c \neq 0)$ D) $b = a + c (c \neq 0)$

54. The function $f(x) = x \left(\frac{2018^x - 1}{2018^x + 1} \right)$ is

- A) Odd B) Even C) Both (A) & (B) D) Neither (A) nor (B)

Paragraph for Question Nos. 55 & 56

An even periodic function $f: R \rightarrow R$ with period 4 is such that

$$f(x) = \begin{cases} \max(|x|, x^2) & 0 \leq x < 1 \\ x & 1 \leq x \leq 2 \end{cases}$$

55. The value of $\{f(5.12)\}$ is (where $\{.\}$ denotes fractional part of x) is:

- A) $\{f(3.26)\}$ B) $\{f(7.88)\}$ C) $\{f(2.12)\}$ D) $\{f(5.88)\}$

56. The number of solutions of $f(x) = |3 \sin x|$ for $x \in (-6, 6)$ are

- A) 5 B) 3 C) 7 D) 9

Section-3

(Matching List Type)

This section contains four questions, each having two matching lists (**List-I** & **List-II**). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

57. Here $[]$ represents Greatest Integer Function, $\{ \}$ denote fractional part Function

Column – I (Function)		Column – II (fundamental Period)	
P	$f(x) = e^{\cos^4 \pi x} + x - [x] + \cos^2 \pi x$	1	$\frac{1}{3}$
Q	$f(x) = \cos(2\pi\{2x\}) + \sin(2\pi\{2x\})$	2	$\frac{1}{4}$
R	$f(x) = \sin(3\pi\{x\}) + \tan(\pi[x])$	3	$\frac{1}{2}$
S	$f(x) = 3x - [3x + a] - b$ where $a, b \in \mathbb{R}^+$	4	1

	P	Q	R	S
A)	4	3	4	1
B)	4	3	1	3
C)	2	1	4	3
D)	2	1	1	4

58. Match the following

Here $[]$ represents Greatest Integer Function

	COLUMN – I		COLUMN - II
P	Odd function	1	$x - [x] \forall x \in R$
Q	Even function	2	$\log(x + \sqrt{1+x^2}) \forall x \in R$
R	Neither even nor odd	3	$x \log\left(\frac{1+x}{1-x}\right) \forall x \in (-1,1)$
S	Periodic function	4	$\frac{2^{x/2}}{1+2^{x/2}} \forall x \in R$

	P	Q	R	S
A)	2	3	1, 4	1
B)	2	3	1	3
C)	4	1	1, 4	3
D)	2	1	1	1

59. Column-B discusses about the nature of functions given in Column-A.

Column-A	Column-B
P) $f: R \rightarrow R, f(x) = \frac{e^{2x} + 1}{e^x}$	1) Periodic Function
Q) $f: R \rightarrow R, f(x) = x + \sin x$	2) Even Function
R) $f: R \rightarrow R, f(x) = (\sin x + \cos x)$	3) Odd Function
S) $f: R \rightarrow R, f(x) = \sin^6 x + \cos^6 x$	4) Neither even nor odd function

	P	Q	R	S
A)	4	1,3	1	3
B)	2	3	1,4	1,2
C)	2	1,3	1,4	1,2
D)	4	3	1	3,4

60. Column-B consists of period of functions given in Column-A.

Column-A	Column-B
P) $f(x) = \sin^{2017}(2016x) + \cos^{2017}(2016x)$	1) $\frac{\pi}{2}$
Q) $f(x) = \sin^{2016}(2017x) + \cos^{2017}(2016x)$	2) π
R) $f(x) = \sin^{2017}(2016x) + \cos^{2016}(2017x)$	3) 2π
S) $f(x) = \sin^{2016}(2017x) + \cos^{2016}(2017x)$	4) $\frac{\pi}{3}$

	P	Q	R	S
A)	1,2,3,4	2,3	1,2,3	1,2,3
B)	1,2,3,4	1,2,3	1,2,3	1,2,3,4
C)	2,3,4	2,3	1,2,3	2,3
D)	1,2,3,4	2,3	2,3	2,3



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

WAT-13

Date: 05-08-17

Time: 07:30 AM to 10:30 AM

2014_P2

Max.Marks:180

KEY SHEET

PHYSICS

1	B	2	B	3	C	4	C	5	C
6	D	7	C	8	D	9	B	10	B
11	C	12	C	13	B	14	C	15	A
16	A	17	B	18	B	19	C	20	D

CHEMISTRY

21	A	22	D	23	A	24	A	25	D
26	C	27	C	28	D	29	A	30	B
31	C	32	B	33	A	34	A	35	A
36	B	37	A	38	B	39	C	40	D

MATHS

41	B	42	C	43	A	44	A	45	B
46	A	47	B	48	C	49	D	50	B
51	B	52	B	53	B	54	B	55	B
56	C	57	A	58	A	59	B	60	D

SOLUTIONS**PHYSICS**

1. $W_c + W_{nc} + W_{ext} = \Delta KE$

2. with $F=6N$ both blocks move together, $W_f = f.S$

3. from A to C, $W_f = -mgh$

From C to A, $-mgh + W_f + W_F = 0$

4. $W_f = -fdr$

6. $\frac{dW}{dx} = 0$

7. Work done by conservative force is independent of path followed

10. Apply W-E theorem

12. Apply W_{pseudo} concept

14. $du = -\vec{F} \cdot \vec{dr}$, Apply work energy theorem

15. When it moves from A to B and surface is rough from W - E theorem

$$W_{net} = \Delta KE$$

$$W_f + W_g = W_{net}$$

$$W_f + mgR = 0 \Rightarrow W_f = -mgR$$

16. A to B surface is rough, B to C smooth let it reach a point P above C, when pushed from A. So from W - E theorem $W_{net} = \Delta KE = KE_f - KE_i$

motion from A to point (beyond C)

$$W_{f(AB)} + W_g = KE_P - KE_A = 0 - \frac{1}{2}mv_A^2$$

Hence $W_{f(AB)} = -mgR$

Given $\frac{1}{2}mv_A^2 = mgR$, let y be the position of point P above horizontal level A_C , where the body finally reaches

$$\Rightarrow -mgR + mgy = -mgR$$

$$\Rightarrow y = 0$$

$\therefore$ it reaches to a height of R above B

20. apply work energy theorem

A - T; B - P; C - S; D - Q

CHEMISTRY

23. Solution:



also $b = E_1$ for $\text{Li}^{+2} = E_1$ for $H \times Z^2$

$$= 2.18 \times 10^{-18} \times 3^2 \times 6.023 \times 10^{23} \text{ J mol}^{-1}$$

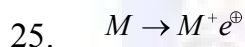
$$= 11.81 \times 10^6 \text{ J mol}^{-1} = 11.81 \times 10^3 \text{ kJ mol}^{-1}$$

For the above eq

$$1.96 \times 10^4 = 520 + a + 11.81 \times 10^3$$

$$a = 7270 \text{ kJ mol}^{-1}$$

24. Conceptual



number of moles $\frac{5}{25} = 0.2$

Energy required to form M^{+} ions

$$= 0.2 \times 800$$

$$= 160 \text{ KJ mol}^{-1}$$

Remaining energy = 90 KJ moles

This is used to convert M^{+} to M^{+2}

Number of moles of M^{+2} formed

$$\frac{90}{1500} = 0.06$$

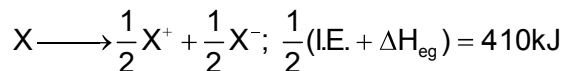
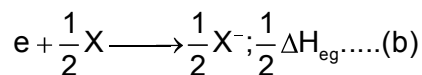
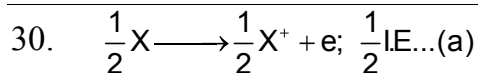
$$\% M^{+2} = \frac{0.06}{0.2} \times 100 = 30\%$$

26. Conceptual

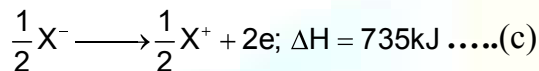
27. Conceptual

28. Conceptual

29. $\text{Ne} > \text{Ar} > \text{Kr} > \text{Xe} > \text{Rn} > \text{Ne}$ (NCERT)



$$I.E. + \Delta H_{eg} = 820 \text{ kJ}$$



$$(a)-(b) = (c)$$

$$\frac{1}{2}I.E. - \frac{1}{2}\Delta H_{eg} = 735$$

$$\Rightarrow I.E. - \Delta H_{eg} = 1470$$

$$I.E. + \Delta H_{eg} = 820 \text{ kJ}$$

$$(-) - 2\Delta H_{eg} = 650 \Rightarrow \Delta H_{eg} = -325 \text{ kJ mol}^{-1}.$$

32. **Hint: (1)** The element (E) having very high ionisation enthalpy and positive electron gain enthalpy would be a noble gas, i.e., least reactive.

(2) The element (B) would be most reactive metal as it has low ionisation enthalpy and low negative electron gain enthalpy. Probably it is an alkali metal.

(3) The element (C) would be most reactive non-metal as it has high negative value of electron gain enthalpy. Probably it is a halogen.

(4) The element (D) would be least reactive non-metal.

(5) The element (A) has low first ionisation enthalpy but very high second ionisation enthalpy. It would be least reactive alkali metal, i.e., lithium which forms covalent MX]

$$33. \quad E_{H-X}^1 = \frac{1}{2}(B.E. H_2 + B.E. \text{ of } X_2)$$

$$= \frac{104 + 38}{2} = 71 \text{ k.mol}$$

$$\Delta = E_{H-X} - E_{HX}^1$$

$$= 135 - 71 = 64 \text{ k. mol}$$

$$X_A - X_B = 0.208\sqrt{64} \times 8$$

$$X_A = 0.208 \times 8 + 2.1 = 3.764$$

34. $3.76 = \frac{333 + x}{544}$

$$x = 1712 \text{ kJ}$$

35 and 36.

n = 1	n = 2	N = 3	n = 4
$\ell = 0 \rightarrow 1s$	$\ell = 0 \rightarrow 2s$	$\ell = 0 \rightarrow 3s$	$\ell = 0 \rightarrow 4s$
$\ell = 1 \rightarrow 1p$	$\ell = 1 \rightarrow 2p$	$\ell = 1 \rightarrow 3p$	$\ell = 1 \rightarrow 4p$
$\ell = 2 \rightarrow 1d$	$\ell = 2 \rightarrow 2d$	$\ell = 2 \rightarrow 3d$	$\ell = 2 \rightarrow 4d$
	$\ell = 3 \rightarrow 2f$	$\ell = 3 \rightarrow 3f$	$\ell = 3 \rightarrow 4f$
		$\ell = 4 \rightarrow 3g$	$\ell = 4 \rightarrow 4g$
			$\ell = 5 \rightarrow 4h$

Electronic configuration $\rightarrow 1s, 1p, 2s, 1d, 2p, 3s, 3d, 3p, 4s, 3d$

For third period $3s^3, 3d^{15}, 3p^9$

number of element = 27

(a)

$1s^3, 1p^9, 2s^3, 1d^{14}$

37. $Z_{\text{eff}} \text{ for Na} = 11 - (8 \times 0.85 + 2 \times 1)$

$$Z_{\text{eff}} \text{ for Na}^+ = 11 - (7 \times 0.35 + 0.85 \times 2)$$

38. Conceptual

39. P = Ce_{58} Q) Pa_{91} R) Cd_{48} S) Cr_{24}

40. Conceptual

MATHS

42. $x = -1, 0, 1$

43. $f(x) = x^{\log_e e} = e$ Domain is $(0, 1) \cup (1, \infty)$

44. Sol: For odd values of n , fundamental period is 2π

For $n = 2$, fundamental period is indeterminate

For even values of $n (\neq 2)$, fundamental period is $\frac{\pi}{2}$

46. $f(x) = c$

47. 0^0 is not defined

49. $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is defined $\Rightarrow 4-x^2 \neq 0, x^3 - x > 0 \Rightarrow x \neq \pm 2, (x+1)x(x-1) > 0$

53. Sol: $f(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2}$

$$f(x) = \left[\left(\frac{a+b}{2} \right) (e^x + e^{-x}) + c \right] + \left(\frac{a-b}{2} \right) (e^x - e^{-x})$$

56. Conceptual

57. Conceptual

58. $x - [x]$ is neither even nor odd function. It is a periodic function with period '1'

$\log(x - \sqrt{1+x^2})$ is odd function

$x \log \left(\frac{1+x}{1-x} \right), \frac{2^{\frac{x}{2}}}{1+2^{\frac{x}{2}}}$ are even functions

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr.Super60-I

2012_P1

Date: 12-08-17

Time: 07:30 AM to 10:30 AM

WAT-14

Max. Marks: 210

Name of the Student: _____

H.T. NO:

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PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as :
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or

"un-darken" a darkened bubble.

12-08-17_Jr.Super60-I_WAT-14_SYLLABUS

MATHS: Syllabus Covered From 07-08-17 TO 11-08-17 (WAT-14)

PHYSICS: Syllabus Covered From 07-08-17 TO 11-08-17(WAT)-14

CHEMISTRY: Chemical Bonding and Molecular Structure : Kossel – Lewis approach to chemical bond formation, concept of ionic and covalent bonds, Lewis structures, Formal Charge, Ionic Bonding and lattice enthalpy, Fajan's rules

IIT-JEE-2012-P1-Model

Time: 07:30 A.M to 10:30A.M

IMPORTANT INSTRUCTIONS

Max Marks: 210

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 11 – 15)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 16 – 20)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 31 – 35)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 36 – 40)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

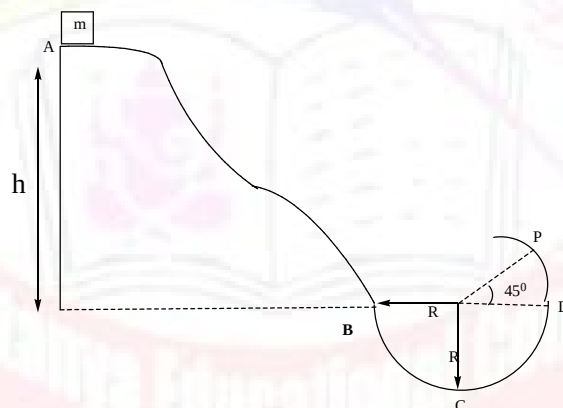
MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	3	-1	10	30
Sec – II(Q.N : 51 – 55)	Questions with Multiple Correct Choice	4	0	5	20
Sec – III(Q.N : 56 – 60)	Questions with Integer Answer Type	4	0	5	20
Total				20	70

PHYSICS:**Max.Marks : 70****SECTION I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A machine delivers power to a body which is proportional to the instantaneous velocity 'v' of the body. If the body starts with a velocity which is almost negligible, then the distance covered by the body is proportional to:
 A) $\sqrt{v}$ B) $\sqrt[3]{\frac{v}{2}}$ C) $v^{\frac{5}{3}}$ D) v^2
2. A particle of mass m slides on a frictionless surface ABCD, starting from rest as shown in the figure. The part BCD is a circular arc. If it loses contact at point P, find the maximum height attained by the particle from point C.



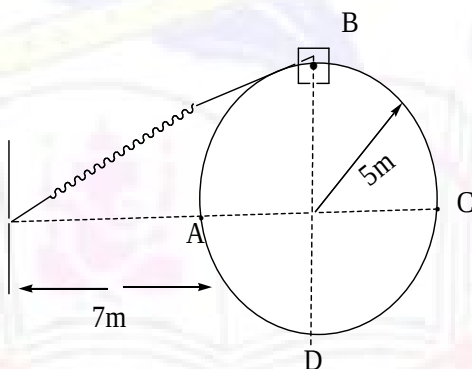
A) $R \left[2 + \frac{1}{2\sqrt{2}} \right] + \frac{h}{2}$

B) $R \left[2 - \frac{1}{2\sqrt{2}} \right] + \frac{h}{2}$

C) $3R + \frac{h}{2}$

D) none of these

3. A collar B of mass 2 kg is constrained to move along a horizontal smooth and fixed circular track of radius 5m. The spring lying in the plane of the circular track and having spring constant 200 N/m, is undeformed when the collar is at A. If the collar starts from rest at B, the normal reaction exerted by the track on the collar when it passes through A is:



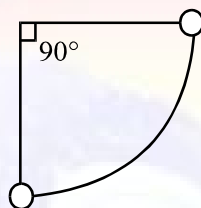
A) 360 N

B) 720 N

C) 1440 N

D) 2880 N

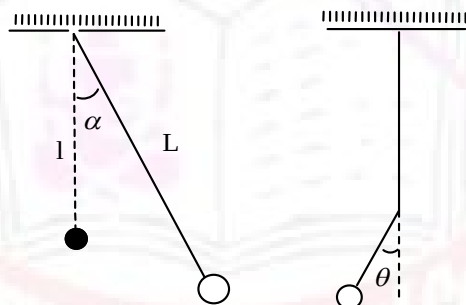
4. A simple pendulum is vibrating with an angular amplitude of 90° as shown in fig. For what value of α (angle between string and vertical) during its motion, the total acceleration is directed horizontally?



A) 0

B) 90° C) $\cos^{-1}(1/\sqrt{3})$ D) $\sin^{-1}(1/\sqrt{3})$

5. A simple pendulum consisting of a mass M attached to a string of length L is released from rest at an angle α . A pin is located at a distance l below the pivot point. When the pendulum swings down, the string hits the pin as shown in figure. The maximum angle which the string makes with the vertical after hitting the pin is:



A) $\cos^{-1}\left(\frac{L \cos \alpha + l}{L + l}\right)$

B) $\cos^{-1}\left(\frac{L \cos \alpha + l}{L - l}\right)$

C) $\cos^{-1}\left(\frac{L \cos \alpha - l}{L - l}\right)$

D) $\cos^{-1}\left(\frac{L \cos \alpha - l}{L + l}\right)$

6. A particle suspended from a fixed point, by a light inextensible thread of length L is projected horizontally from its lowest position with a velocity $\sqrt{\frac{7gL}{2}}$. The string will slack after swinging through an angle θ , the value of θ is:

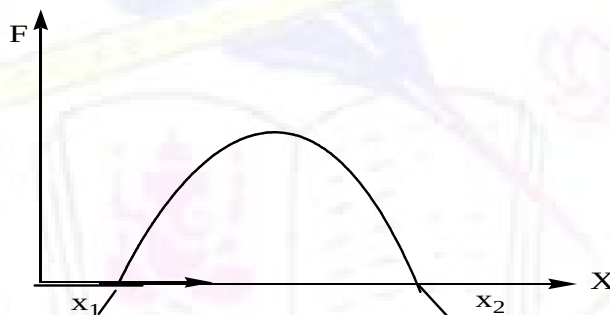
A) 30°

B) 135°

C) 120°

D) 150°

7. The force acting on a body moving along x-axis varies with the position of the particle as shown in figure. The body is in stable equilibrium at:



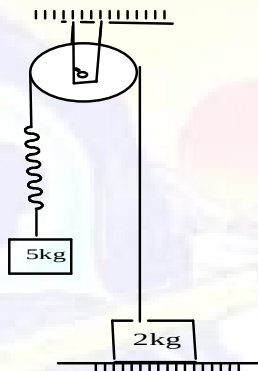
A) $x = x_1$

B) $x = x_2$

C) both x_1 and x_2

D) neither x_1 nor x_2

8. The system shown in figure is released from rest with spring in natural length and string just taut. The pulley and the springs are massless and friction is absent everywhere. The value of spring constant is $k = 40\text{N/m}$. The speed of 5kg block when 2kg block leaves contact with the ground is:



- A) $\sqrt{2}\text{m/s}$ B) $2\sqrt{2}\text{m/s}$ C) 2 m/s D) None of these
9. Consider the situation shown in figure. Initially the spring is unstretched when the system is released from rest. Assuming no friction in the pulley, find the maximum elongation of the spring?



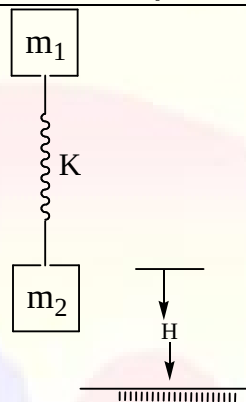
A) $\frac{mg}{k}$

B) $\frac{2mg}{k}$

C) $\frac{mg}{2k}$

D) $\frac{2mg}{3k}$

10. A spring mass system is held at rest at height H from the ground with the spring in relaxed state. Find the minimum value of ' H ' so that the m_2 has a tendency to rebound after hitting the ground during the subsequent motion. Assume that the lower block stops after hitting the ground. (Initial height of m_2 above the ground is H . Neglect the size of m_2)



A) $\frac{m_2 g}{K} \left(\frac{2m_1 + m_2}{2m_1} \right)$

B) $\frac{m_2 g}{2K} \left(\frac{2m_1 + m_2}{2m_1} \right)$

C) $\frac{m_2 g}{K}$

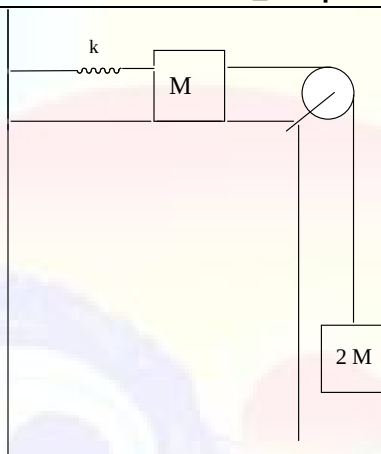
D) $\left(\frac{2m_1 + m_2}{2m_1^2} \right) \frac{g}{k}$

SECTION II

Multiple Correct Answer(s) Type

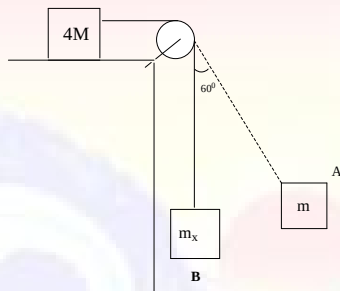
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. Two blocks of masses M and $2M$ are connected to a light spring of spring constant k that has one end fixed as shown in figure. The horizontal surface and pulley are frictionless. The blocks are released from rest when the spring is in its relaxed state.



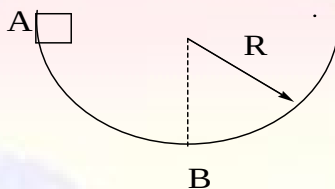
- A) Maximum extension in the string is $\frac{4Mg}{k}$
- B) Maximum kinetic energy of the system is $\frac{2M^2g^2}{k}$
- C) Maximum energy stored in the spring is four times that of maximum kinetic energy of the system
- D) When kinetic energy of the system is maximum, energy stored in the spring is $\frac{4M^2g^2}{k}$

12. In the system shown in figure the mass 'm' moves in a circular arc of angular amplitude 60° . Mass $4m$ is stationary, then:



- A) the minimum value of coefficient of friction between the mass $4m$ and the surface of the table is 0.5.
- B) the work done by the gravitational force on the block m is positive when it moves from A to B.
- C) the power delivered by tension when m moves from A to B is zero
- D) the kinetic energy of m in position B equals the work done by gravitational force on the block when it moves position A to B.

13. A small block of mass m is released from rest from position A inside a smooth hemispherical bowl of radius R as shown in figure. Choose the wrong option.



- A) Acceleration of block is constant throughout
- B) Acceleration of block is g at A
- C) Acceleration of block is $3g$ at B
- D) Acceleration of block is $2g$ at B.
14. The potential energy of the moving particle along x -axis is given by $U = 20 + 5\sin(4\pi x)$, Where U is in joule and ' x ' is in meter under the action of conservative force.
- A) If mechanical energy is $20J$, then at $x=7/8m$; particle is at equilibrium.
- B) If mechanical energy is $20J$, then at $x=7/8m$; particle is not at equilibrium.
- C) If mechanical energy is $20J$, then at $x=3/8m$; particle is at equilibrium.
- D) If mechanical energy is $20J$, then at $x=3/8m$; particle is not at equilibrium.

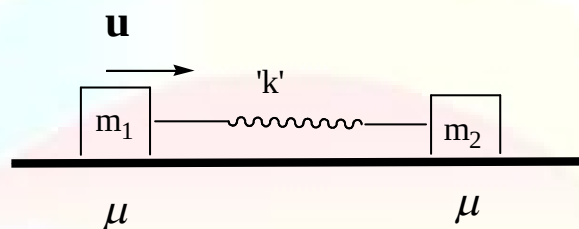
15. A small sphere of mass 'm' suspended by a thread is taken a side, so that thread forms the right angle with the vertical and then released, then which options are correct (Here ' θ ' $\Rightarrow$ is the angle of with vertical).
- A) The total acceleration of the sphere as a function of ' θ ' is " $g\sqrt{1+3\cos^2\theta}$ "
- B) The total tension in the thread as a function of ' θ ' is ' $3mg \cos\theta$ '
- C) The thread tension at the moment when the vertical component of the sphere's velocity is maximum will be $mg\sqrt{3}$
- D) The thread tension at the moment when the vertical component of the sphere's velocity is maximum will be ' mg '.

SECTION- III

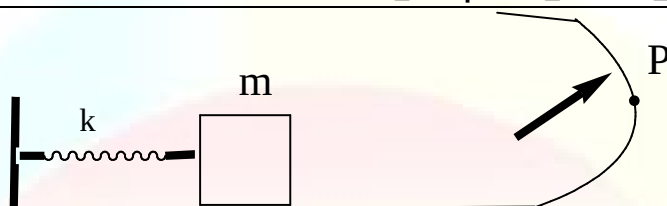
Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

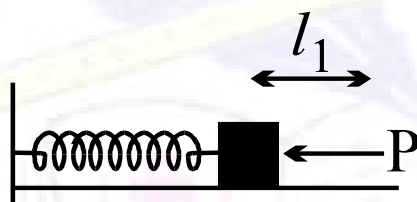
16. The blocks of masses $m_1 = 1\text{kg}$ and $m_2 = 2\text{kg}$ are connected by a spring and rest on a horizontal surface. The spring is unstretched initially. The spring constant of the spring is $k = 8\text{ N/m}$. The coefficient of friction between blocks and horizontal surface is $\mu = \frac{1}{2}$. Now the left block is imparted a velocity u towards right as shown in figure. Then what is the largest value of u such that the block mass m_2 never moves.



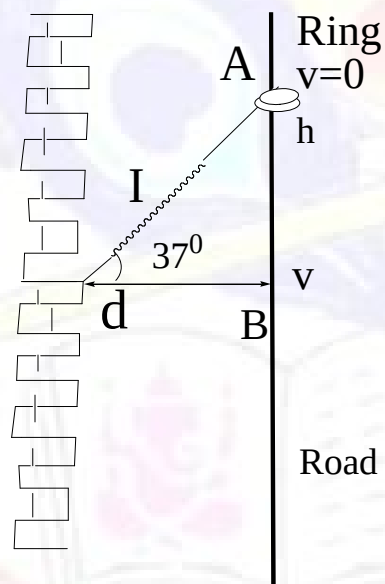
17. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constants and x is the distance between the atoms. If the dissociation energy of the molecule is $D = \left[U_{\text{at infinity}} - U_{\text{at equilibrium}} \right]$, then D is $\frac{b^2}{na}$. Find n ?
18. Figure shows a smooth track, a part of which is a circle of radius 10cm. A block of mass 3kg is pushed against a spring of spring constant 100 N/m fixed at the left end and is then released. If the initial compression of the spring is $10x$ cm, find the value of x so that the block pushes the track with a force of 30 N when it reaches point P, where the radius of the track is horizontal.



19. A block of mass m placed on a smooth horizontal surface is attached to a spring and is held at rest by a force P in equilibrium as shown (at equilibrium compression in spring is l_1). Suddenly the force P changes its direction opposite to the previous one. If the maximum extension of the spring during the subsequent motion is l_2 then find $l_2/l_1 = ?$



20. One end of a light spring of natural length d and spring constant $k=64\text{N/m}$ is fixed on a rigid wall and the other is attached to a smooth ring of mass $m=1\text{ kg}$ which can slide without friction on a vertical rod fixed at a distance $d=3\text{m}$ from the wall. Initially the spring makes an angle 37° with the horizontal as shown in fig. When the system is released from rest find the speed of the ring when the spring becomes horizontal (in m/s) [$\sin 37^\circ = 3/5$]



CHEMISTRY:**Max.Marks : 70****SECTION I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

21. Which of the following solubility order is not correct?
 A) $SrCrO_4 > BaCrO_4$ B) $BaC_2O_4 > SrC_2O_4$ C) $KHCO_3 > NaHCO_3$ D) $BaS_2O_3 > SrS_2O_3$
22. The types of bonds present in $CuSO_4 \cdot 5H_2O$ are only
 A) electrovalent and covalent
 B) electrovalent and co-ordinate covalent
 C) electrovalent, covalent and co-ordinate covalent
 D) covalent and co-ordinate covalent
23. Carbon suboxide (C_3O_2) has recently been shown as a component of the atmosphere of Venus. Which of the following formulation represents the correct ground state Lewis structure for carbon suboxide?
 A) $:O:C::C:C:O:$ B) $:O::C::C:C::O:$
 C) $:O::C::C::C::O:$ D) $:O:C:C:C:O:$

24. Thermodynamic decomposition temperature is maximum for

- A) MgCO_3 B) SrCO_3 C) BaCO_3 D) BeCO_3

25. Which of the following order is not correct?

A) $\text{AlCl}_3 > \text{MgCl}_2 > \text{NaCl}$ [covalent character]

B) $\text{AlF}_3 > \text{MgF}_2 > \text{NaF}$ [melting point]

C) $\text{AlCl}_3 > \text{MgCl}_2 > \text{NaCl}$ [melting point]

D) $\text{AlCl}_3 > \text{AlI}_3 > \text{AlBr}_3$ [melting point]

26. S_1 - size of cation increases due to polarization

S_2 - Al^{+3} has more polarizing power than Ga^{+3}

S_3 - S^{-2} has more polarisability than Cl^-

Which of the following given statements are correct:

- A) S_1, S_2, S_3 B) only S_3 C) S_1, S_3 D) S_2, S_3

27. Among the following, which compound will have the highest lattice energy?
- A) KF B) NaF C) CsF D) RbF.
28. Among LiCl, BeCl₂, BCl₃ and CCl₄, the covalent character follows the order?
- A) LiCl < BeCl₂ > BCl₃ > CCl₄ B) LiCl > BeCl₂ < BCl₃ < CCl₄
- C) LiCl < BeCl₂ < BCl₃ < CCl₄ D) LiCl > BeCl₂ > BCl₃ > CCl₄
29. Amongst LiCl, RbCl, BeCl₂ and MgCl₂, the compounds with the greatest and the least ionic character respectively are:
- A) LiCl and RbCl B) RbCl and BeCl₂
- C) RbCl and MgCl₂ D) MgCl₂ and BeCl₂
30. Choose the correct order of lattice energy:-
- A) CsF > CaF₂ B) AlF₃ > Al₂O₃ C) MgI₂ > KI D) MnBr₂ > MnF₂

SECTION II

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

31. Which of the following order is incorrect ?
- A) $Cd^{+2} > Hg^{+2}$ (polarizing power)
- B) $S^{-3} < P^{3-}$ (ionic radius)
- C) $F > Ne$ (covalent radius)
- D) $SnF_4 < SnCl_4$ (melting point)
32. Which of the following energy terms are associated with dissolution of $BaCl_2$ in water?
- A) Hydration energy of Ba^{2+} B) lattice energy of $BaCl_2$
- C) Sublimation energy of Ba D) Electron Affinity of Cl^-
33. The correct order of decreasing electron affinity of B, C, N and O is:
- A) $O > C > N > B$ B) $B > N > C > O$
- C) $O > C > B > N$ D) $O > B > C > N$

34. Which of the following orders is/are true?

A) $\text{AgF} > \text{AgCl} > \text{AgBr} > \text{AgI}$ [solubility in water]

B) $\text{ZnCl}_2 < \text{ZnBr}_2 < \text{ZnI}_2$ [covalent character]

C) $\text{LiClO}_4 > \text{NaClO}_4$ [thermal stability]

D) $\text{BeO} > \text{BeF}_2$ [melting Point]

35. Which of the following statements is/are true?

A) low solubility of LiF in water is due to high lattice energy

B) KNO_3 is water soluble due to high entropy change

C) CsI is less soluble in water due to smaller hydration energy

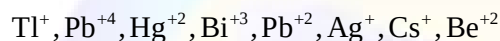
D) KI is more soluble in acetone than KCl

SECTION III

Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

36. How many of the followings cations have non-inert gas configuration

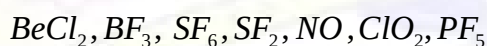


37. What is the value of $|x| + |y|$ where

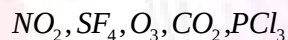
x = lowest Possible formal charge on 'N' atom in Lewis structure(s) of azide ion(N_3^{-1})

y = highest possible formal charge on 'N' atom in Lewis structure(s) of azide ion(N_3^{-1})

38. How many of the following molecule does not follow octet rule?



39. How many of the following molecules contain lone pair(s) on central atom in Lewis dot structures?



40. The number of total electrons shared between nitrogen atoms in N_2 is _____

MATHEMATICS:**Max.Marks : 70****SECTION I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

41. Let $f : [-1, 3] \rightarrow [-8, 72]$ be defined as $f(x) = 4x^3 - 12x$, then f is

- A) Injective but not surjective
- B) Injective as well as surjective
- C) Neither injective nor surjective
- D) surjective but not injective

42. The function $f : \mathbb{R} \rightarrow \{x \in \mathbb{R} : -1 < x < 1\}$ defined by $f(x) = \frac{x}{1+|x|}$ is

- A) one-one and into
- B) one-one and onto
- C) many-one and into
- D) many-one and onto

43. For the real number x , let $[x]$ denote the greatest integer less than or equal to x .

let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = 2x + [x] + \sin x \cdot \cos x$ then f is

- A) one-one but not onto
- B) onto but not one-one
- C) both one-one and onto
- D) neither one-one nor onto

44. If the following functions are defined from $\mathbb{R}$ to $\mathbb{R}$ then identify the function which is bijective?

A) $f(x) = \frac{e^x + e^{-x}}{2}$

B) $f(x) = x^4 - 3x^3 + 1$

C) $f(x) = 18x^3 - 21x^2 + 8x - 1$

D) $f(x) = x^3 - 4x^2 + 16x + 17$

45. If the function $f(x)$ and $g(x)$ are defined on $\mathbb{R} \rightarrow \mathbb{R}$ such

that $f(x) = \begin{cases} x+3, & x \in \text{rational} \\ 4x, & x \in \text{irrational} \end{cases}$ and

$g(x) = \begin{cases} x+\sqrt{5}, & x \in \text{irrational} \\ -x, & x \in \text{rational} \end{cases}$ then $(f-g)(x)$ is

A) one-one and onto

B) neither one-one nor onto

C) one-one but not onto

D) onto but not one-one

46. If $f(g(x)) = g(f(x)) = x$ for all real number x , and $f(2) = 5$ and $f(5) = 3$, then the value of $g(3) + g(f(2))$ is

A) 7

B) 5

C) 3

D) 2

47. Let $f(x) = ax+b$, Where **a** and **b** are integers. If $f(f(0)) = 0$ and $f(f(f(4))) = 9$, then the value of $f(f(f(f(10))))$ is equal to

A) 0

B) 4

C) 9

D) 10

48. If $g(x) = \left(4\cos^4(x) - 2\cos(2x) - \frac{1}{2}\cos(4x) - x^7 \right)^{\frac{1}{7}}$, then the value of $g(g(100))$ is equal to
 A)-1 B)0 C)1 D)100
49. Let $f : R \rightarrow R$ be defined as $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b$. the latest value of 'a' for which $f(x)$ is injective function is.
 A) $\frac{1}{4}$ B) $\frac{1}{8}$ C) $\frac{1}{2}$ D)1
50. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. then the set of all x satisfying $(f \circ g \circ f)(x) = (g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is
 A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
 C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$ D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

SECTION - II

Multiple Correct Answer(s) Type

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

51. The maximum value of the function defined by $f(x) = \min(e^x, 2 + e^x, 7)$ is α then integral value of x satisfying the inequality $\frac{x(x - [\alpha])}{x^2 - [\alpha]x + 12} < 0$ is
 A)1 B) 3 C) 5 D) 6

52. Let $f : R \rightarrow R$ defined by $f(x) = \min(|x|, 1 - |x|)$ Then which of the following hold(s) good?
- A) Rang of f is $(-\infty, 1]$ B) f is even function
 C) f Is function odd D) f is neither injective nor surjective.
53. Let: $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions and $g \circ f : A \rightarrow C$ is defined. Then which of the following statement(s) is (are) incorrect?
- A) If $g \circ f$ is onto then f must be onto
 B) If f is into and g is onto then $g \circ f$ must be onto function.
 C) If $g \circ f$ is one-one then g is necessarily one-one.
 D) If f is injective and g is surjective then $g \circ f$ must be bijective mapping.
54. Let f, g & h be three function defined as follows:
- $$f(x) = \frac{32}{4 + x^2 + x^4}, g(x) = 9 + x^2 \text{ and } h(x) = -x^2 - 3x + k$$
- Identify which of the following statement(s) is (are) correct?
- A) Number of integers in the range of $f(x)$ is 8.
 B) Number of integral value of k for which $h(f(x)) > 0$ and $h(g(x)) < 0 \forall x \in R$ is 20
 C) Number on integral value of k for which $h(f(x)) > 0$ and $h(g(x)) < 0 \forall x \in R$ is 19.
 D) Maximum value of $g(f(x))$ is 73.

55. Let $f(x)$ be a function defined on $[0, 1]$ such that

$$f(x) = \begin{cases} x & ; x \in \text{rational} \\ 1-x & ; x \in \text{irrational} \end{cases} \text{ If } f \circ f(x) = p + qx, \forall x \in [0, 1] \text{ Where } p, q \in I, \text{ then the value}$$

of $(p^2 + 2q)$ is greater than

A) 0

B) 1

C) 2

D) 3

SECTION III

Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

56. Let $f(x) = \left[\frac{1}{\cos\{x\}} \right]$ where $[y]$ and $\{y\}$ denote greatest integer and fractional part

function respectively and $g(x) = 2x^2 - 3x(k+1) + k(3k+1)$. if $g(f(x)) < 0 \forall x \in R$ then find

the number of integral value of k .

57. If $h(x) = Ax^5 + B \sin x + C \ln \left(\frac{1+x}{1-x} \right) + 7$, Where A, B, C are non-zero real constants and

$$h\left(\frac{-1}{2}\right) = 6, \text{ then find the value of } h\left(\frac{\operatorname{sgn}(e^{-x})}{2}\right).$$

58. Let $f : \mathbb{R} \rightarrow [0, \infty)$ be defined as $f(x) = \log_e(\sqrt{9x^2 - 12x + \lambda} + 1)$ be an onto function where λ is a real parameter belong to $(0, 10)$ Find the greatest possible value of λ .
59. Let a function f defined from $\mathbb{R} \rightarrow \mathbb{R}$ as $f(x) = \begin{cases} x+p^2, & \text{for } x \leq 2 \\ 2px+5, & \text{for } x > 2 \end{cases}$ If the function is surjective, then the sum of all possible integral value of p in $[-100, 100]$ is $\sum_{r=1}^{100} r - k$ then $(k-2) =$
60. Let the equation $(a-1)x^2 = x(2b+3)$ be satisfied by three distinct value of x , where $a, b \in \mathbb{R}$. if $f(x) = (a-1)x^3 + (2b+3)x^2 + 2x + 1$, and $f(g(x)) = 6x - 7$ where $g(x)$ is a linear function then find the value of $g(2)$ is equal to



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WAT-14

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2012_P1

Max.Marks:210

KEY SHEET

PHYSICS

1	D	2	A	3	C	4	C	5	C
6	C	7	B	8	B	9	B	10	A
11	ABC	12	ABCD	13	AC	14	AC	15	ABC
16	5	17	4	18	3	19	3	20	9

CHEMISTRY

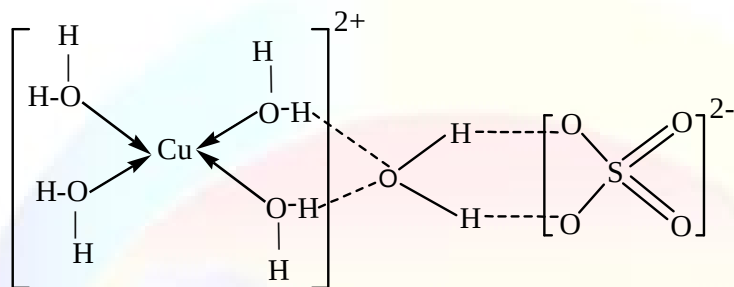
21	D	22	C	23	C	24	C	25	C
26	A	27	B	28	C	29	B	30	C
31	ABD	32	AB	33	C	34	ABD	35	ABCD
36	3	37	3	38	6	39	3	40	6

MATHS

41	D	42	B	43	A	44	D	45	B
46	A	47	D	48	D	49	A	50	A
51	ACD	52	BD	53	ABCD	54	ACD	55	AB
56	1	57	8	58	4	59	8	60	2

SOLUTIONS**CHEMISTRY**

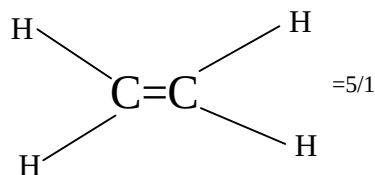
22. The bond between Cu^{2+} and SO_4^{2-} ion is ionic ; between S and O in SO_4^{2-} ions and between H and O atoms in H_2O are covalent ; those between Cu^{2+} and H_2O molecules are coordinate.



23. In (c), each C and O atom has octet of electrons.
24. A σ -bond is stronger than a π -bond hence option (a) is not correct.
- .
- .
27. For compounds containing ions of same charge, lattice energy increases as the size the ions decrease. Thus, NaF has highest lattice energy.
28. As we move from $\text{Li} \rightarrow \text{Be} \rightarrow \text{B} \rightarrow \text{C}$, the electronegativity (EN) increases and hence the EN difference between the element and Cl decreases and accordingly the covalent character increases.
- Thus option (c) i.e. $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$ is correct.
29. Electronegativity difference (EN) is highest in RbCl ($3 - 0.8 = 2.2$) and least in BeCl_2 ($3 - 1.5 = 1.5$) and hence option (b) is correct.

30. In $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}^*-\text{O}-\text{H}$ the asterisked carbon has a valency of 5 and hence this formula is not correct.
31. Because of the triple bond, the carbon –carbon bond distance in ethyne is shortest.
- .
33. Conceptual.

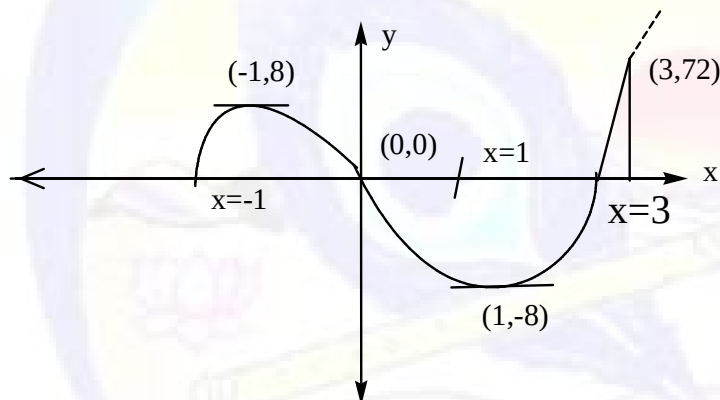
36.

38. BeCl_2 ; BF_3 ; NO ; ClO_2 ; PF_5 ; SF_6

39. Conceptual.

40. Each N atom in N_2 molecule shares three electrons, i.e., $:\text{N}::\text{N}:$ **MATHS**

41. $f(x) = 4x^3 - 12x$
 $f'(x) = 12(x^2 - 1) = 12(x+1)(x-1)$



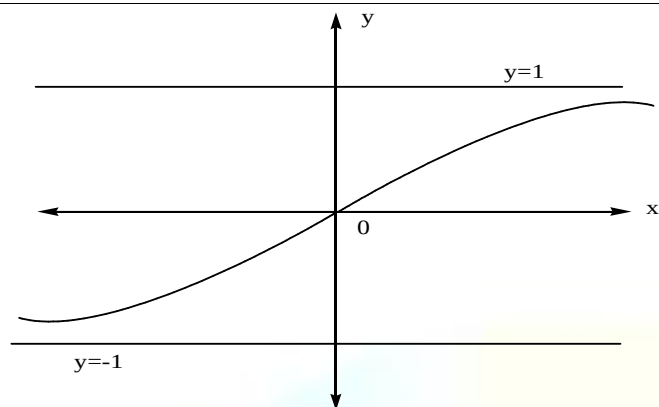
42.
$$f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ 0, & x = 0 \\ \frac{x}{1+x}, & x > 0 \end{cases}$$

$$\therefore f(0) = 0. \text{ Now, } y = \frac{x}{1-x}, x < 0$$

$$\Rightarrow x = \frac{y}{1+y} < 0 \Rightarrow y \in (-1, 0)$$

$$\text{Also, } y = \frac{x}{1+x}, x > 0 \Rightarrow x = \frac{y}{1-y} > 0 \Rightarrow y \in (0, 1)$$

So, range of function = $(-1, 1)$



43. If $x=a$, where 'a' is an integer then $f(a) = 2a + a + \frac{1}{2} \sin 2a$

But $\lim_{h \rightarrow 0} f(a-h) = 2a + (a-1) + \frac{1}{2} \sin 2a$

$\therefore$ Value between $\lim_{h \rightarrow 0} f(a-h)$ and $f(a)$ are never achieved.

Also, $f'(x) = 2 + \cos 2x > 0$, i.e., $f(x)$ is strictly increasing.

44. (A) $f(x) = \frac{e^x + e^{-x}}{2}$ is even function.

(B) $f(x) = x^4 - 3x^3 + 1$ is an even degree polynomial function.

(C) $f(x) = 18x^3 - 21x^2 + 8x - 1$

$f(x)$ is an odd degree polynomial function hence its range is $\mathbb{R}$.

$f'(x) = 3x^2 - 8x + 16 \Rightarrow D < 0$, hence $f(x)$ is one-one as well as onto.

45. We have $(f \circ g)(x) = (f(x) - g(x)) = (f - g)(x) = (f(x) - g(x)) = \begin{cases} 2x+3, & x \in \text{rational} \\ 4x, & x \in \text{irrational} \end{cases}$

As, $f\left(\frac{-3}{2}\right) = 0 = f\left(\frac{\sqrt{5}}{3}\right)$ and so on.

$\Rightarrow f(x)$ is many one function.

Also, $-\sqrt{5}$ does not belong to the range, because if $3x - \sqrt{5} = -\sqrt{5}$

$\therefore x = 0 \notin \mathbb{Q}$

$\Rightarrow f(x)$ is into function

46. Since g and f are inverses, $g(3)=5$ and so $g(3)+g(f(2))=5+2=7$.

47. $f(x) = ax + b$

$$f(0) = b$$

$$f(f(0)) = a.b + b = 0 \Rightarrow (a+1)b = 0 \Rightarrow a = -1 \text{ or } b = 0$$

$$f(f(f(4))) = 9$$

$$ff(f(4a+b)+b) = 9$$

$$\Rightarrow f(a(4a+b)+b) = 9$$

$$\Rightarrow f(4a^2 + ab + b) = 9$$

$$\Rightarrow a(4a^2 + ab + b) + b = 9$$

$$\Rightarrow 4a^3 + a^2b + ab + b = 9$$

$$\text{when } b = 0, 4a^3 = 9 \Rightarrow a^3 = \frac{9}{4} \Rightarrow a \text{ is not an integer}$$

$$\text{When } a = -1, -4 + b - b + b = 9 \Rightarrow b = 13$$

$$\therefore f(x) = -x + 13$$

$$f(f(x)) = -(-x + 13) + 13 = x$$

$$\therefore f(f(f(f(10)))) = f(f(10)) = 10$$

48. We have

$$4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7$$

$$= 4\cos^4 x - 2(2\cos^2 x - 1) - \frac{1}{2}(2\cos^2 2x - 1) - x^7$$

$$\therefore \text{We get } g(x) = \left(\frac{3}{2} - x^7\right)^{\frac{1}{7}}$$

$$\Rightarrow g(g(x)) = \left(\frac{3}{2} - (g(x))^7\right)^{\frac{1}{7}} = \left(\frac{3}{2} - \left(\frac{3}{2} - x^7\right)\right)^{\frac{1}{7}} = x$$

49. If $f(x)$ is one-one then $f(x)$ must be monotonic.

$$\text{Now, } f'(x) = x^2 + x + a \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow D \leq 0 \text{ i.e., } 1 - 4a \leq 0 \Rightarrow a \geq \frac{1}{4}$$

$$\text{So, } a_{\min} = \frac{1}{4}$$

50.

$$f(g(g(f(x)))) = g(g(f(x))) \Rightarrow \sin^2(\sin x^2) = \sin(\sin x^2)$$

$$\Rightarrow \sin(\sin x^2) = 0, 1$$

$$\Rightarrow \sin x^2 = m\pi, (4k+1)\frac{\pi}{2}, \text{ where } m, k \in I.$$

$$\text{But } -1 \leq \sin x^2 \leq 1$$

$$\Rightarrow \sin x^2 = 0$$

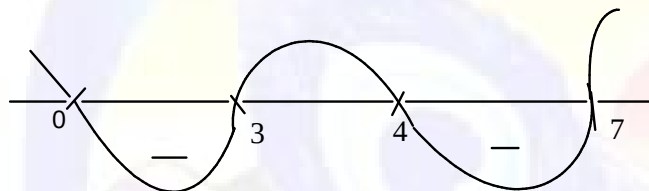
$$\Rightarrow x^2 = n\pi, n = 0, 1, 2, \dots$$

$$\Rightarrow x = \pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$$

51. $f(x) = \min(e^x, 2 + e^2, 8)$ From the graph it is clear that maximum value of $f(x)$ is, $\alpha = e^2$

$$\therefore [\alpha] = [e^2] = 7$$

$$\frac{x(x-7)}{x^2-7x+12} < 0 \Rightarrow \frac{x(x-7)}{(x-3)(x-4)} < 0$$



52. $f(x) = \min(|x|, 1 - |x|)$

$$D_f = R$$

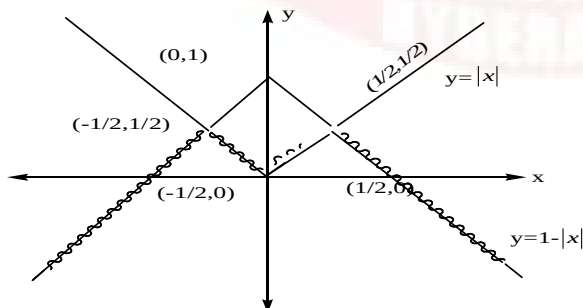
$$f(-x) = f(x)$$

$\Rightarrow f$ is even function.

Put

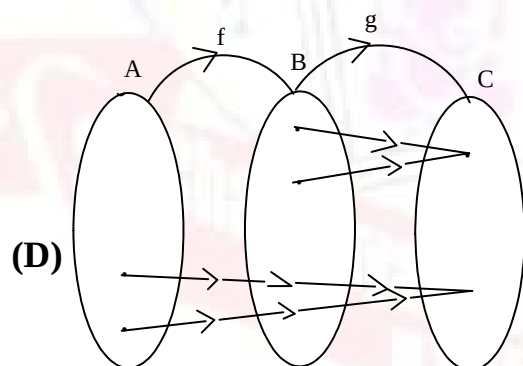
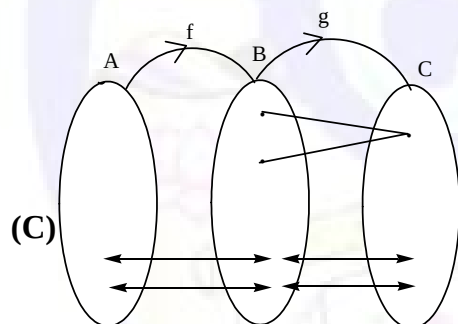
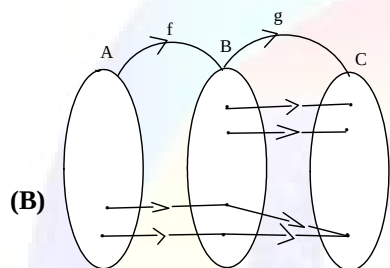
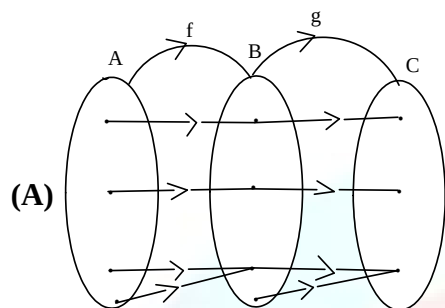
$$|x| = 1 - |x|$$

$$\therefore x = \pm \frac{1}{2}$$



53. (A) We have $f : A \rightarrow B$

$g : B \rightarrow C$ and $g \circ f : A \rightarrow C$



54. $f(x) = \frac{32}{4+x^2+x^4}$; $g(x) = 9+x^2$; $h(x) = -x^2-3x+k$

(A) Range of f is $[0,8]$

(B) $h(f(x)) > 0$ and $h(g(x)) < 0$

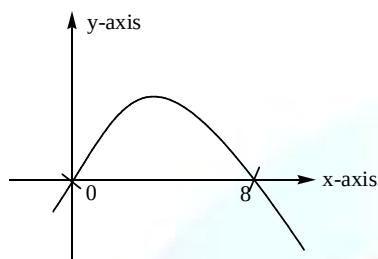
$\Rightarrow h(0) \geq 0 \Rightarrow k \geq 0$

$$h(8) > 0 \Rightarrow -64 - 24 + k > 0 \Rightarrow k > 88$$

$$\text{Also, } h(9) < 0 \Rightarrow -81 - 27 + k < 0 \Rightarrow k < 108$$

$\Rightarrow$ Number of integral value of k is 19.

(D) Maximum value of $g(f(x))$ is $g(8)=64+9=73$.



55. Given, $f(x) = \begin{cases} x & ; x \in Q \\ 1-x & ; x \notin Q \end{cases}$

$$f \circ f(x) = \begin{cases} x & ; x \in Q \\ 1-x & ; x \notin Q \end{cases}$$

$$\text{So, } f \circ f(x) = \begin{cases} x & ; x \in Q \\ 1-x & ; x \notin Q \end{cases}$$

$$\therefore f(f(x)) = x, \forall x \in [0, 1]$$

$$\Rightarrow p = 0 \text{ and } q = 1$$

$$\text{hence, } (p^2 + 2q) = 2$$

56. $f(x) = 1 \forall x \in R$

$$\Rightarrow g(1) < 0 \forall x \in$$

$$\Rightarrow 2 - 3(k+1) + 3k^2 + k < 0$$

$$\Rightarrow 3k^2 - 2k - 1 < 0$$

$$\Rightarrow (3k+1)(k-1) < 0$$

$$\Rightarrow k \in \left(\frac{-1}{3}, 1 \right)$$

57. Given $h\left(\frac{-1}{2}\right) = 6$

$$h(x) = Ax^5 + B \sin x + c \ln \left(\frac{1-x}{1+x} \right) + 7$$

$$\Rightarrow h(-x) = -Ax^5 - B \sin x + C \ln \left(\frac{1-x}{1+x} \right) + 7$$

$$\therefore h(x) + h(-x) = 14$$

$$\Rightarrow h\left(\frac{1}{2}\right) + h\left(-\frac{1}{2}\right) = 14$$

$$\therefore h\left(\frac{1}{2}\right) + 6 = 14$$

$$\Rightarrow h\left(\frac{1}{2}\right) = 8$$

58. Given $r9x) = \log(\sqrt{9x^2 - 12x + \lambda} + 1)$

$D_f = R$ & $R_f = [0, \infty]$ So, it is possible when $9x^2 - 12x + \lambda = 0$ hence equal roots

$$\Rightarrow \text{Disc.} = 0$$

$$\text{So, } 144 = 36\lambda$$

$$\Rightarrow \lambda = 4$$

59. For f to be surjective range = Co-domain $f(x) = \begin{cases} x+p^2, & \text{for } x \leq 2 \\ 2px+5, & \text{for } x > 2 \end{cases}$

For $x \leq 2$, range of f is $(-\infty, 2+p^2]$ and

For $x > 2$, range of f is $(4p+5, \infty)$. [Note that $p > 0$]

For range of f to be R , $4p+5 \leq p^2+2$

$$p^2 - 4p - 3 \geq 0 \Rightarrow (p-3)(p+1) \geq 0 \Rightarrow p \in (-\infty, -1) \cup [3, \infty] \text{ but } p > 0.$$

Hence, $p \in [3, \infty)$

$$\Rightarrow s = 3 + 4 + \dots + 100 = 5050 - 3 = 5047$$

60. $(a-1)x^2 = x(2b+3)$

The above equation is satisfied by three distinct values of x therefore it is an identity.

$$\therefore 2-2a=0 \Rightarrow a=1 \text{ and } 2b+3=0 \Rightarrow b=-\frac{3}{2}$$

Now, $f(x) = 2x+1$

$$\text{Let } g(x) = px+q \Rightarrow g'(x) = p$$

$$f(g(x)) = 6x-7 \Rightarrow 2(px+q)+1 = 6x-7$$

$$\Rightarrow 2px+2q+1 = 6x-7$$

$$\Rightarrow 2p=6 \Rightarrow p=3 \text{ and } q=-4$$

$$\therefore g'(2012) = 3$$



Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Jr.Super60-I

2016_P2

Date: 19-08-17

Time: 07:30 AM to 10:30 AM

WAT-15

Max. Marks: 186

Name of the Student: _____

H.T. NO:

PAPER-II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as :
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

19-08-17_Jr.Super60-I_WAT-15_SYLLABUS

MATHS: SYLLABUS COVERED FROM 14-08-17 TO 18-08-17 (WAT-15)

PHYSICS: SYLLABUS COVERD FROM 14-08-17 TO 18-08-2017(WAT-15)

CHEMISTRY: VSEPR Theory & shapes of simple molecules, VBT and Hybridization

Sri Chaitanya IIT Academy

19-08-17_Jr.Super60-I_JEE-ADV_(2016_P2)_WAT-15_Q.P

JEE-ADVANCE-2016-P2-Model

Time: 07:30AM to 10:30 AM

IMPORTANT INSTRUCTIONS

Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 7 – 14)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 25 – 32)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions with Single Correct Choice	3	-1	6	18
Sec – II(Q.N : 43 – 50)	Questions with Multiple Correct Choice (Partial Marking +1)	4	-2	8	32
Sec – III(Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	3	0	4	12
Total				18	62

Jr.Super60-I

space for rough work

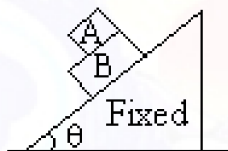
Page 2

PHYSICS**Max Marks : 62****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. A block A placed over another block B which is placed over a smooth inclined plane as shown in figure. The coefficient of friction between the two blocks A and B is μ . Mass of block B is two times the mass of block A. Now the blocks are released. The acceleration of the centre of mass of two blocks is



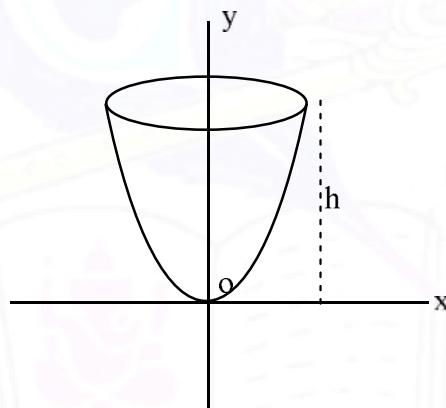
- A) $g \sin \theta$ B) $\frac{g \sin \theta - \mu g \cos \theta}{3}$ C) $\frac{g \sin \theta}{3}$ D) $\frac{2g \sin \theta - \mu g \cos \theta}{3}$
2. Two blocks of equal mass m are connected by an unstretched spring and the system is kept at rest on a frictionless horizontal surface. At $t=0$, a constant force F is applied on the first block pulling it away from the other as shown in figure.



If the extension of the spring is x_0 at time t , then the displacement of the first block in time t is

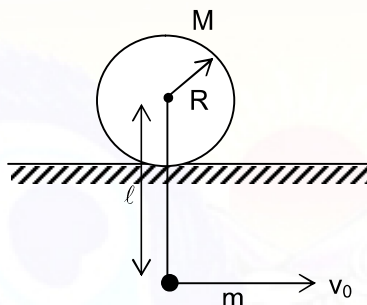
- A) $\frac{1}{2} \left(\frac{Ft^2}{2m} + x_0 \right)$ B) $-\frac{1}{2} \left(\frac{Ft^2}{4m} + x_0 \right)$ C) $\frac{1}{2} \left(\frac{Ft^2}{2m} - x_0 \right)$ D) $\left(\frac{Ft^2}{2m} + x_0 \right)$

3. In a boat of mass $4M$ and length ' L ' on a frictionless water surface. Two men A (mass = M) B (mass = $2M$) are standing on the opposite ends. Now A travels a distance $L/4$ relative to boat towards its centre and B moves a distance $3L/4$ relative to boat and meet A. The displacement of the boat on water till A and B meet is
- A) $\frac{5L}{28}$ B) zero C) $\frac{L}{2}$ D) $\frac{23L}{2}$
4. A paraboloid shaped solid object is formed by rotating on parabola $y = 2x^2$ about y-axis as shown in figure. If the height of the body is ' h ' then the position of centre of mass from origin. (Assume density to be uniform throughout).

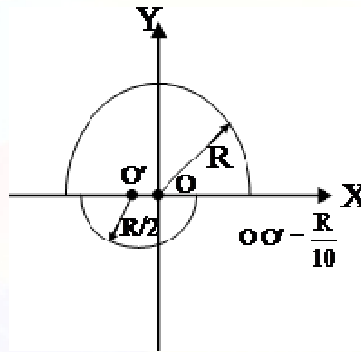


- A) $\frac{h}{3}$ B) $\frac{h}{4}$ C) $\frac{2h}{3}$ D) $\frac{2h}{4}$

5. A disc of radius R and mass M is placed on smooth horizontal surface as shown in figure. A light rod of length ℓ is hanged from the centre of disc and a small mass m is attached at the end as shown in figure. Now a velocity V_0 is given to mass m . Find the maximum height which mass m can attain. (Assume rod can rotate $< 90^\circ$)



- A) $\frac{1}{2} \frac{V_0^2}{g} \frac{m}{M+m}$ B) $\frac{1}{2} \frac{V_0^2}{g} \frac{M}{2m}$ C) $\frac{1}{2} \frac{V_0^2}{g} \frac{m}{2M}$ D) $\frac{1}{2} \frac{V_0^2}{g} \frac{M}{M+m}$
6. Two solid hemispheres of radii R and $R/2$ with centers O and O' respectively as shown in figure. The density of bigger hemisphere is ρ and that of smaller hemisphere is 2ρ . Taking center of bigger hemisphere is at origin and the distance between centers of two hemispheres OO' is $R/10$, find co-ordinates of center of mass of the system.



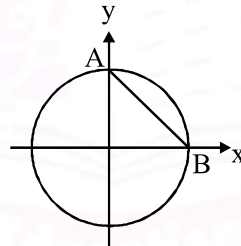
- A) $\left(\frac{-R}{50}, \frac{21R}{80}\right)$ B) $\left(\frac{-R}{30}, \frac{21R}{80}\right)$ C) $\left(\frac{-R}{50}, \frac{7R}{16}\right)$ D) $\left(\frac{-R}{30}, \frac{7R}{16}\right)$

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

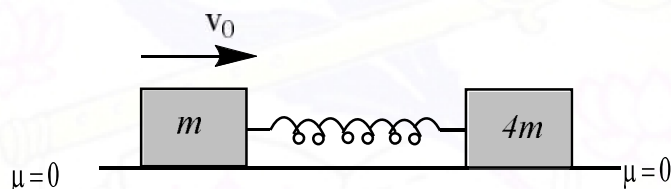
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

7. An object comprises of a uniform ring of radius R and a uniform chord AB (not necessarily made of the same material) as shown. Which of the following can not be the center of mass of system



- A) $\left(\frac{R}{3}, \frac{R}{3}\right)$ B) $\left(\frac{R}{3}, \frac{R}{4}\right)$ C) $\left(\frac{R}{2}, \frac{R}{2}\right)$ D) $\left(\frac{R}{\sqrt{2}}, \frac{R}{\sqrt{2}}\right)$

8. Which of the following is/are correct?
- A) If the center of mass of three particles is at rest and it is known that two of them are moving along different non-collinear lines, then the third particle must also be moving.
- B) If the center of mass of a system remains at rest, then the net work done by all the forces acting on the system must be zero.
- C) If the velocity of center of mass of a system remains zero, then the net external force acting on the system must be zero.
- D) If the speed of center of mass is changing, then there must be some work being done by the internal forces on the system.
9. Two blocks of masses m and $4m$ lie on a smooth horizontal surface connected with a spring in its natural length. Mass m is given initial velocity V_0 as shown in figure. Which of the following is true about subsequent motion?



- A) Kinetic energy of mass m is maximum in ground frame and centre of mass (CM) frame simultaneously.
- B) Value of maximum and minimum kinetic energy is same in CM and ground frame.
- C) Minimum kinetic energy is zero in CM frame but non-zero in ground frame.
- D) Maximum and minimum kinetic energy of m in ground frame is, respectively, $\frac{1}{2}mv_0^2$ and zero.

10. Two identical buggies move one after other due to inertia (without friction) with the same velocity v_0 . A man of mass m rides the rear buggy. At a certain moment the man jumps into the front buggy with a velocity u relative to his buggy. If mass of each buggy is equal to M and velocity of buggies after jumping of man are v_{rear} and v_{front} . Then

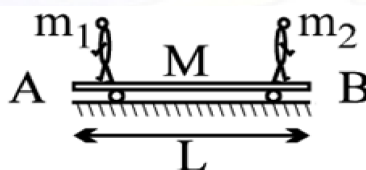
A) $v_{rear} = v_0 + \frac{m}{m+M}u$

B) $v_{rear} = v_0 - \frac{m}{m+M}u$

C) $v_{front} = v_0 + \frac{m}{(m+M)^2}u$

D) $v_{front} = v_0 - \frac{m}{(m+M)^2}u$

11. Two person of mass m_1 and m_2 are standing at the two ends A and B respectively, of a trolley of mass M as shown. When the person standing at A jumps from the trolley towards left with u_{rel} with respect to the trolley, then (Ground is frictionless)



A) the trolley moves towards right

B) the trolley rebounds with velocity $\frac{m_1 u_{rel}}{m_1 + m_2 + M}$

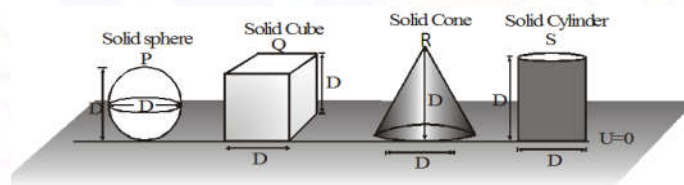
C) the trolley rebounds with velocity $\frac{m_1 u_{rel}}{m_2 + M}$

D) the centre of mass of the system consisting of two persons and trolley remains stationary

12. Choose the correct statement(s) regarding center of mass frame

- A) If $\overline{F}_{ext} = 0$ center of mass frame is inertial frame
- B) Centre of mass frame is a zero momentum frame
- C) Kinetic energy of a system is minimum in centre of mass frame
- D) None of these

13. Assuming potential energy 'U' at ground level to be zero.



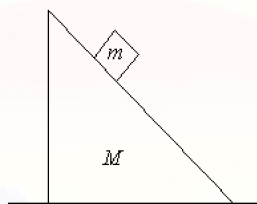
All objects are made up of same material.

U_P = Potential energy of solid sphere U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone U_S = Potential energy of solid cylinder

- A) $U_S > U_P$ B) $U_Q > U_S$ C) $U_P > U_Q$ D) $U_S > U_R$

14. A block of mass m is placed at rest on a smooth wedge of mass M placed at rest on a smooth horizontal surface. As the system is released



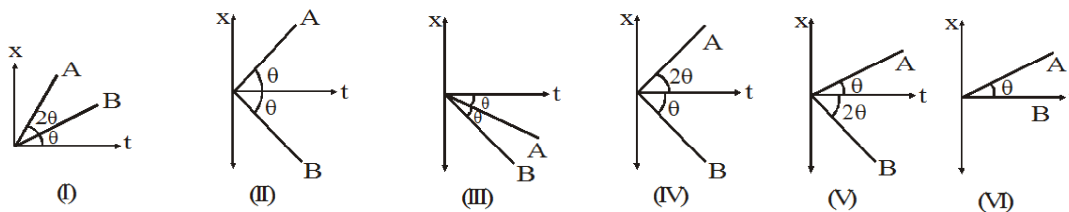
- A) The COM of the system remains stationary
- B) The COM of the system has an acceleration g vertically downward
- C) Momentum of the system is conserved along the horizontal direction
- D) Acceleration of COM vertically downward is $a < g$

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16:

An initially stationary box on a frictionless floor explodes into two pieces, piece A with mass m_A and piece B with mass m_B . Two pieces then move across the floor along x -axis. Graph of position versus time for the two pieces are given.



15. Based on the above question, Match column A with the column B.

Column A

P) $m_A = m_B$

Q) $m_A > m_B$

R) $m_A < m_B$

Column B

(Graph number)

I

II

III

IV

V

VI

A) P – VI, Q – III, R – I

B) P – II, Q – V, R – IV

C) P – II, Q – IV, R – V

D) P – VI, Q – II, R – IV

16. If all the graphs are possible then, in which of the following cases external force must be acting on the box :-

A) I & II

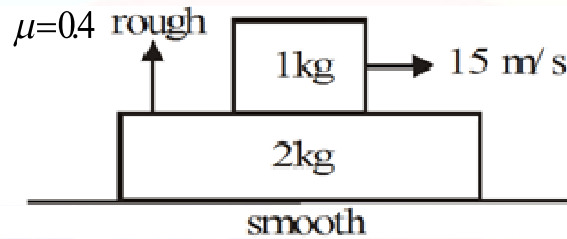
B) II & III

C) I, III & VI

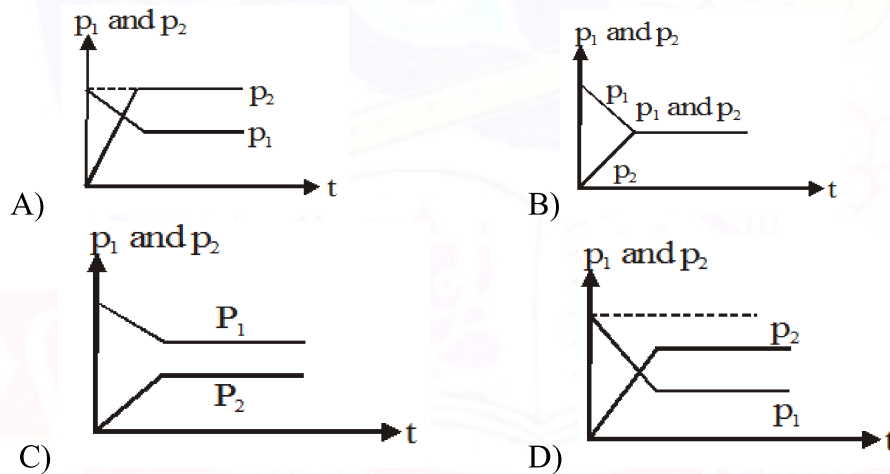
D) I, III, V

Paragraph For Questions 17 and 18:

A 1 kg block is given a velocity of 15 m/s towards right over a very long rough plank of mass 2 kg as shown in figure.



17. The correct graph showing linear momentum of 1 kg (i.e. p_1) and of 2 kg (i.e. p_2) versus time is :



18. Momentum of both the blocks are equal at time $t = \dots\dots\dots$ seconds:

- A) 1.75 B) 1.875 C) 2.5 D) 1.25

CHEMISTRY

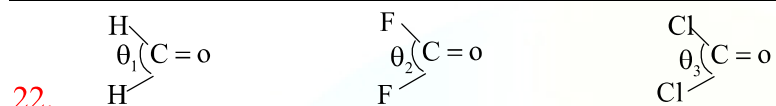
Max Marks: 62

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

19. Which of the following species will have the lone pair effects cancelled
- A) ICl_2^- B) CIF_3 C) PCl_3 D) BrF_5
20. SH_6 does not exist because
- A) Hydrogen is too small to overlap with large orbital of sulphur atom
B) Hydrogen cannot cause contraction of d-orbital of sulphur as it cannot induce a high positive formal charge on sulphur
C) Sulphur cannot expand its octet
D) Sulphur is unreactive towards Hydrogen
21. What hybridization is expected on the central atom of each of the following molecules.
- i) BeH_2 ii) CH_2Br_2
iii) PF_6^- iv) BF_3
A) sp^2, sp, sp^3, sp^2 B) $sp, sp^3, sp^3 d, sp^2$
C) $sp, sp^3, sp^3 d^2, sp^2$ D) sp^2, sp, sp^2, sp^3



Identify the correct order of bond angle

- A) $\theta_1 > \theta_2 > \theta_3$ B) $\theta_1 > \theta_3 > \theta_2$ C) $\theta_3 > \theta_2 > \theta_1$ D) $\theta_1 = \theta_2 = \theta_3$

23. Which is correct statement

As the % S-character of a hybrid orbital decreases

I) The bond angle decreases

II) Size of orbital decreases

III) The bond length increase

IV) Size of orbital increase

- A) I,III,IV B) II,III,IV C) I & II D) All are correct

24. Shape of IF_7 molecule is

- A) Pentagonal bipyramidal B) Trigonal pyramidal
C) Tetrahedral D) Square planar

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

25. In the structure of H_2CSF_4 which of the following statement is /are correct.

- A) Two C-H bonds are in the same plane of axial S-F bonds.
- B) Two C-H bonds are in the same plane of equatorial S-F bonds
- C) Total six atoms are in the same plane
- D) Equatorial S-F bonds are perpendicular to plane of π -bond

26. Identify the correct statement in the following

- A) Hybridisation of carbon in C_3O_2 is SP
- B) $N(Me)_3$ and $N(SiMe_3)_3$ are iso structural
- C) In trigonal bipyramidal arrangement lone pair of electron are usually occupied at equatorial position.
- D) The shape of I_3^- is linear.

27. Which of the following statement(s) is/are a valid postulate(s) of valence bond theory?
- A) Overlapping orbitals have unpaired electron
 - B) Greater is the extent of overlapping higher is the bond strength
 - C) Greater is the orbital overlapping higher is the bond length
 - D) Lateral overlapping of atomic orbitals results in formation of π -bond
28. Which of the following statement is (are) correct?
- A) In ClF_3 , the axial $Cl-F$ bond length is longer than equatorial $Cl-F$ bond length.
 - B) In SF_4 , $F-S-F$ equatorial bond angle is 120°
 - C) In $[ICl_4]^-$, $Cl-I-Cl$ bond angle is 90°
 - D) There are two $p^\pi-d^\pi$ bonds in SO_3
29. There is a change in the hybridisation when
- A) NH_3 combines with H^+
 - B) AlH_3 combines with H^-
 - C) NH_3 forms NH_2^-
 - D) SiF_4 forms SiF_6^{2-}

30. Which of the following orbital can form δ -bond (Delta bond).

- A) $dx^2 - y^2$ orbital B) dz^2 orbital C) dxy Orbital D) P_z orbital

31. In which compound vacant orbitals take part in hybridisation

- A) Al_2Cl_6 B) C_2H_5Cl C) B_2H_6 D) H_3BO_3

32. Identify the incorrect statement(s) amongst the following.

- A) molecule having trigonal bipyramidal geometry and sp^3d hybridisation, the d-orbital involved in hybridization is $dx^2 - y^2$
- B) In sp^3d^2 hybridisation, $dx^2 - y^2, dz^2$ sets of d-orbital is involved
- C) I_3^+ and XeF_7^+ have same hybridization
- D) CO_3^{2-} and SO_3 are iso electronic and isostructural.

SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Questions 33 and 34:

The Approximate shape of a molecule can often be predicted by using what is called the valence shell Electron pair repulsion (VSEPR) model. Electrons in bonds and in lone

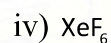
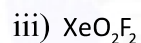
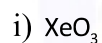
pairs can be thought of as charge cloud that repel one another and stay as far apart as possible, thus causing molecules to assume specific shapes.

The repulsive interactions of Electron pairs decrease in the order.

Lone pair- lone pair > lone pair – bond pair > bond pair – bond pair.

These repulsion effect result in deviations from idealised shapes and alteration in bond angles in molecule.

33. Among the following molecules



Those having different molecular geometry but same number of lone pairs on Xe are,

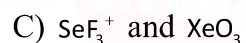
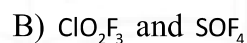
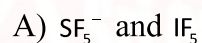
A) (i),(ii) and (iii) only

B) (i),(ii),(iv) only

C) (ii)(iii) and (iv) only

D) (i),(ii),(iii) and iv

34. which of the fallowing represent the isostructural pair?



D) All of the above

Paragraph for Questions 35 and 36:

Different types of bonds are present in the chemical compounds. These bonds have different strengths and bond energies associated with them. These bonds are formed in different environments associated with atoms and the compounds in which they are present.

35. which of the following will form only σ bond (generally).

- A) s-s overlapping B) Hybrid orbital overlapping
C) s-p overlapping D) All of these

36. Shape of the molecule is decided by

- A) σ bond B) π bond
C) both σ and π bond D) neither σ nor π -band

MATHEMATICS**Max Marks : 62****SECTION – I**
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

37. The number of solutions of $\sin x = \frac{x}{10}$ is
- A) 6 B) 7 C) 10 D) 9
38. The number of solutions of $2^x + 3^x + 4^x - 5^x = 0$ is
- A) 2 B) 1 C) 3 D) 0
39. If $f: R \rightarrow R$ is a function satisfying the property $f(x+1) + f(x+3) = 2$ for $x \in R$, then the period of $f(x)$ is
- A) 3 B) 2 C) 4 D) 5
40. Let $f(x) = 2x(2-x); 0 \leq x \leq 2$, then the number of solution of $f(f(f(x))) = \frac{x}{2}$ is
- A) 2 B) 4 C) 8 D) 12

41. The number of ordered pairs (a, b) from the set $A = \{1, 2, 3, 4, 5\}$ so that the function

$$f(x) = \frac{x^3}{3} + \frac{ax^2}{2} + bx + 10 \text{ is an injective mapping } \forall x \in R, \text{ is}$$

- A) 13 B) 14 C) 15 D) 16

42. $f(x)$ is a function satisfying the equation $f(1-x) + 2.f(x) = 3x, \forall x \in R$ then $f(0) =$

- A) -2 B) -1 C) 0 D) 1

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

43. Let $f(x)$ be invertible function and let $f^{-1}(x)$ be its inverse, let equation

$$f(f^{-1}(x)) = f^{-1}(x) \text{ has two real roots } \alpha, \beta \text{ (within the domain of } f(x)) \text{ then}$$

- A) $f(x) = x$ also has same two real roots
B) $f^{-1}(x) = x$ also has same two real roots
C) $f(x) = f^{-1}(x)$ also has the same two real roots
D) Area of triangle formed by $(0,0), (\alpha, f(\alpha)), (\beta, f(\beta))$ is 1 unit

44. Let 'n' be a +ve integer with $f(n) = 1 + 2 + 3 + \dots + n$ and $p(x), q(x)$ be polynomials in

'x' such that $f(n+2) = p(n).f(n+1) + q(n).f(n)$ for all $n \geq 1$, then

A) $p(x) = x + 3$ B) $q(x) = -x - 2$ C) $p(x) = -x - 2$ D) $q(x) = -x + 3$

45. All the periodic functions are

A) one-one

B) Many one

C) Invertible

D) Invertible by restricting the domain

46. If $f: [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$ then $f^{-1}(x) =$

A) $\frac{x + \sqrt{x^2 - 4}}{2}$ B) $\frac{x - \sqrt{x^2 - 3}}{2}$ C) $\frac{x + \sqrt{x^2 - 3}}{2}$ D) $\frac{x - \sqrt{x^2 - 4}}{2}$

47. Let $f: (2, 4) \rightarrow (1, 3)$ be a function defined by $f(x) = x - \left[\frac{x}{2} \right]$ (where $[.]$ denotes

the greatest integer function), then which of the following is/are true?

A) $f^{-1}\left(\frac{5}{4}\right) = \frac{9}{4}$

B) $f^{-1}\left(\frac{7}{4}\right) = \frac{11}{4}$

C) $f^{-1}(2) = 3$

D) f is not bijective function

48. The function $f: X \rightarrow Y$ where $f(x) = \sin x + \cos x + 2\sqrt{2}$ is invertible if

- A) $X = \left[\frac{\pi}{4}, \frac{5\pi}{4}\right], Y = [\sqrt{2}, 3\sqrt{2}]$ B) $X = \left[\frac{\pi}{4}, \frac{\pi}{2}\right], Y = [1+\sqrt{2}, 3\sqrt{2}]$
 C) $X = \left[\frac{3\pi}{4}, \frac{5\pi}{4}\right], Y = [\sqrt{2}, 2\sqrt{2}]$ D) $X = \left[\frac{-\pi}{4}, \frac{3\pi}{4}\right], Y = [2\sqrt{2}, 3\sqrt{2}]$

49. For the function $f(x) = \log_{10}(3x^2 - 4x + 5)$

- A) domain is $(0, \infty)$ B) range is $\mathbb{R}$
 C) domain is $\mathbb{R}$ D) range is $\left[\log_{10}\left(\frac{11}{3}\right), \infty\right)$

50. The equation $\|x-1\| + a = 4$ $a \in \mathbb{R}$ has

- A) 3 distinct real roots for unique value of 'a'
 B) 4 distinct real roots for $a \in (-\infty, -4)$
 C) 2 distinct real roots for $|a| < 4$
 D) no real root for $a > 4$

SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.

Paragraph for Questions 51 and 52:

Let $f(x)$ be a real valued continuous function such that

$$f(0) = \frac{1}{2}; f(x+y) = f(x)f(a-y) + f(y)f(a-x) \forall x, y \in R \text{ then for some real values of 'a'}$$

51. $f(x)$ is a

- A) periodic function B) periodic function with no fundamental period
C) Not a periodic function D) increasing function

52. $f(x) =$

- A) $\frac{\cos x}{2}$ B) $\frac{-1}{2}$ C) $\frac{1}{2}$ D) $\sin x$

Paragraph for Questions 53 and 54:

Consider the function $f: R \rightarrow (0, \infty)$ defined by $f(x) = 2^x + 2^{|x|}$

53. $f(x)$ is

- A) One-One Onto B) One-One Into
C) Many-One Onto D) Many-One Into

54. The number of solutions of $f(x) = 2x + 3$ is

- A) 0 B) 1 C) 2 D) 4



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

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2016_P2

Max.Marks: 186

KEY SHEET PHYSICS

1	A	2	A	3	A	4	C	5	D
6	A	7	BCD	8	AC	9	ACD	10	BC
11	ABD	12	ABC	13	ABD	14	CD	15	B
16	C	17	D	18	C				

CHEMISTRY

19	A	20	B	21	C	22	B	23	A
24	A	25	AC	26	ACD	27	ABD	28	ACD
29	BD	30	AC	31	AC	32	ACD	33	D
34	D	35	D	36	A				

MATHS

37	B	38	B	39	C	40	C	41	C
42	B	43	ABC	44	AB	45	BD	46	A
47	ABC	48	ABC	49	CD	50	ABCD	51	B
52	C	53	D	54	C				

SOLUTIONS**PHYSICS**

01. There is no friction between the blocks and $a_{cm} = \sin \theta$

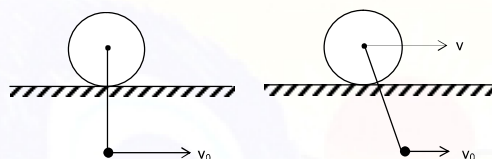
02. $\vec{S}_{\text{block wrt g}} = \vec{S}_{\text{wrt cm}} + \vec{S}_{\text{cm, g}}$

$$= \frac{x_0}{2} + \frac{1}{2} \times a_{cm} \times t^2$$

03. $2M \left(\frac{3L}{4} - x \right) = M \left(\frac{L}{4} + x \right) + 4Mx$

04. $h_{cm} = \frac{\int y dm}{\int dm} = \frac{\int y dV}{\int dV}$

$$dV = (\pi x^2) dy$$



05.

$$mV_0 = (M + m) V$$

$$V = \frac{mV_0}{M + m}$$

From energy conservation

$$\frac{1}{2} mV_0^2 + Mg\ell = mgh + Mg\ell + \frac{1}{2} (M+m)V^2$$

$$\frac{1}{2} mV_0^2 - \frac{1}{2} \frac{m^2 V_0^2}{(M+m)} = mgh$$

$$\frac{1}{2} \frac{mV_0^2 [M+m-m]}{[M+m]} = mgh$$

$$h = \frac{1}{2} \frac{V_0^2}{g} \frac{M}{(M+m)}$$

06.

$$x_{cm} = \frac{M_1 \left(\frac{R}{10} \right) - M_2 \left(\frac{R}{10} \right)}{M_1 + M_2}$$

$$y_{cm} = \frac{M_1 \left(\frac{3R}{8} \right) - M_2 \left(\frac{3R}{16} \right)}{M_1 + M_2}$$

07. Centre of mass always lies b/w centre of mass of segments and always lies on

axis of symmetry.

08. Conceptual

09. In both CM and ground frame, $K_{\max}$ is there, when x is zero in spring, which occurs simultaneously.

$$v_{CM} = \frac{m(v_0) + 0}{5m} = \frac{v_0}{5}$$

$$K_{\max CM} = \frac{1}{2}m\left(\frac{4v_0}{5}\right)^2 + \frac{1}{2}(4m)\left(\frac{v_0}{5}\right)^2 = \frac{2}{5}mv_0^2$$

$$K_{\max \text{ ground}} = \frac{1}{2}mv_0^2$$

$$K_{\min CM} = 0$$

$$K_{\min \text{ ground}} = \frac{1}{2}(m + 4m)v_{CM}^2 = \frac{mv_0^2}{10}$$

$$K_{\max m} = \frac{1}{2}mv_0^2 \text{ (ground frame)}$$

$K_{\min m} = 0$ (ground frame when energy is shared by spring and $4m$ and m will reverse direction of motion).

10. $(M + m)V_0 = MV + m(V + u)$

$$V = v_0 - \frac{mu}{M + m}$$

$$m(u + v) + MV_0 = (M + m)V^1$$

$$V^1 = v_0 + \frac{Mmu}{(M + m)^2}$$

11.

(A), (B), (D)

F_{ext} on the system $(m_1 + m_2 + M)$ in horizontal direction is zero.

$\therefore$ centre of mass of the system remains stationary.

Again $\vec{v}_{m_1g} = \vec{v}_{m_1i} + \vec{v}_{ig} = (-u_{\text{rel}} + v)\hat{i}$

$$\vec{v}_{(m_2+M)g} = v\hat{i}$$

Conserve linear momentum in horizontal direction

$$m_1(-u_{\text{rel}} + v)\hat{i} + (m_2 + M)v\hat{i} = 0$$

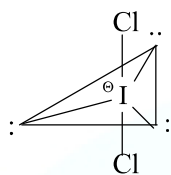
$$\therefore v = \left(\frac{m_1 u_{\text{rel}}}{m_1 + m_2 + M}\right)\hat{i}$$

13. Potential energy depends on both mass and centre of gravity.

CHEMISTRY

19. Ans: A

ICl_2^- have a Linear structure and lone pair of electrons occupied at equatorial positions



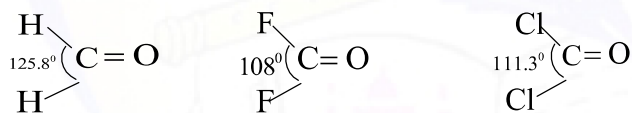
20. Ans: B

Conceptual

21. Ans : c

**SINGLE BOND**

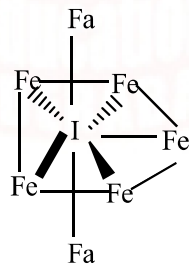
22. Ans: B



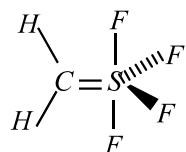
Higher the electronegativity of surrounding atom, lesser is the bond angle between the bonds.

23. Ans: A

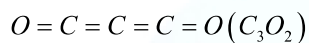
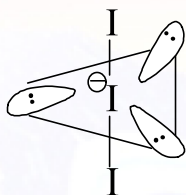
24. Ans: A



25. Sol: A & C

C- sp^2 HYBRIDISATIONS- sp^3d HYBRIDISATION

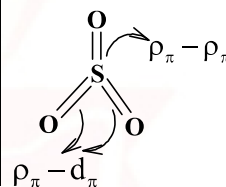
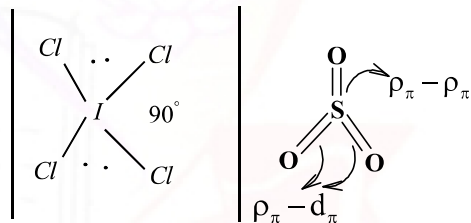
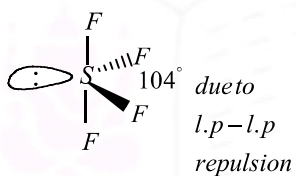
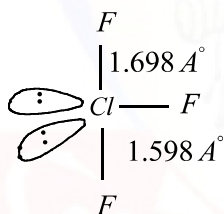
26. Sol: A,C,D

 $N(Me)_3$: pyramidal, $Ni(SiMe_3)_3$ -Trigonal planar

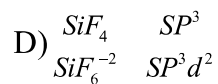
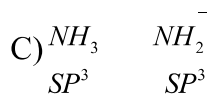
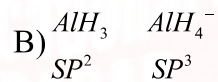
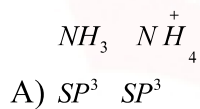
27. Sol: A,B,D

Conceptual

28. Sol: A,C, D



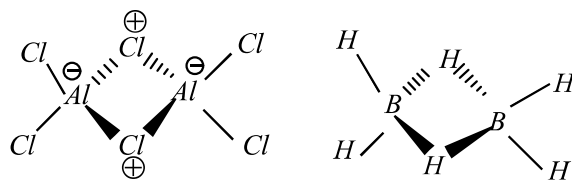
29. Sol: B,D



30. Sol: A & C

Conceptual

31. Sol: A & C

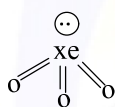


32. Sol: A, C, D

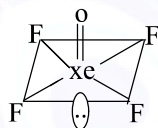
In sp^3d hybridization dz^2 orbital is used I_3^+ have sp^3 hybridisation, XeF_7^+ have sp^3d^3

Comprehension : I CO_3^{2-} and SO_3 are isostructural but not isoelectronic

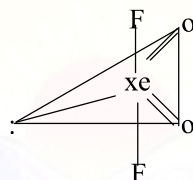
33. Ans :D



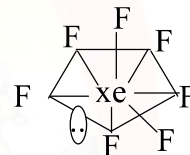
Pyramides



square pyramides



see-saw



capped octa hedron

34. Ans: d

All pairs having same no of sigmas bonds and lone pair of e^-

Comprehension : II

35. Ans:d

All the orbitals are overlap along the axis

36. Ans : a

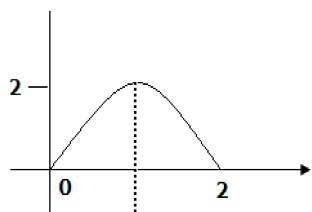
Only sigma bond decided the shape of molecule.

MATHS

37. Draw the graphs of $y = \sin x$; $y = \frac{x}{10}$ 38. Draw the graphs of $y = \left(\frac{2}{5}\right)^x + \left(\frac{3}{5}\right)^x + \left(\frac{4}{5}\right)^x$ & $y = 1$

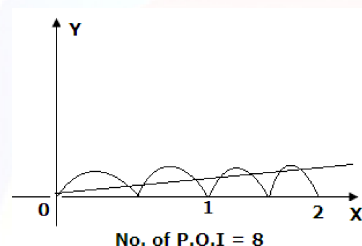
39. $f(x) = f(x+4) \Rightarrow \text{period} = 4$

40.



Since the graph of $y = f(x)$ is symmetrical about the line $x = 1$

$\Rightarrow$ graph of $y = f(f(x))$ and $y = f(f(x))$ are also symmetrical about the line $x = 1$



41. $f^{-1}(x) = x^2 + ax + b$ is injective if disc ≤ 0

$a = 1 \Rightarrow b = 1, 2, 3, 4, 5$

$= 2 \quad = 1, 2, 3, 4, 5$

$= 3 \quad = 3, 4, 5$

$= 4 \quad = 4, 5$

42. replace x by $\frac{1}{2} + x$ _____ (1)

now replace x by $\frac{1}{2} - x$ _____ (2)

$(1) + (2) \Rightarrow f\left(\frac{1}{2} + x\right) = \frac{1}{2} + 3x$

Put $x = -\frac{1}{2} \Rightarrow f(0) = -1$

Multi answer question

43. $f^{-1}(f(x)) = f(f^{-1}(x)) = x$ if $f(f^{-1}(x)) = f^{-1}(x) \Rightarrow x = f^{-1}(x)$

If $f(f^{-1}(x)) = f^{-1}(x) \Rightarrow f(f^{-1}(f(x))) = f^{-1}(f(x))$

$$\Rightarrow f'(x) = f^{-1}(f(x)) = x$$

44. $f(n+2) - f(n+1) = \underline{n+2}$

$$= (n+2)\underline{n+1}$$

$$= (n+2)[f(n+1) - f(n)]$$

$$\Rightarrow f(n+2) = (n+3)f(n+1) - (n+2)f(n)$$

$$\therefore p(x) = x+3; Q(x) = -x-2$$

45. Horizontal line test fails

46. $f^{-1}(x) = \frac{x \pm \sqrt{x^2 - 4}}{2}$

But range of $f^{-1}(x)$ is $[1, \infty)$, so ignore $\frac{x - \sqrt{x^2 - 4}}{2}$

47. Conceptual

48. $f(x) = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right) + 2\sqrt{2}$ (or) $\sqrt{2} \cos\left(x - \frac{\pi}{4}\right) + 2\sqrt{2}$

$$\frac{-\pi}{2} \leq x + \frac{\pi}{4} \leq \frac{\pi}{2} \Rightarrow X = \left[-\frac{3\pi}{4}, \frac{\pi}{4}\right]$$

$$\text{(or) } X = \left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$$

$$\text{Similarly } y = [\sqrt{2}, 3\sqrt{2}]$$

49. $3x^2 - 4x + 5 > 0$

$$\Rightarrow \Delta < 0 \Rightarrow 3x^2 - 4x + (5 - 10^y) = 0$$

$$X \text{ is real} \Rightarrow \Delta \geq 0$$

$$10^y \geq \frac{11}{3} \Rightarrow y \geq [\log_{10}(11/3), \infty]$$

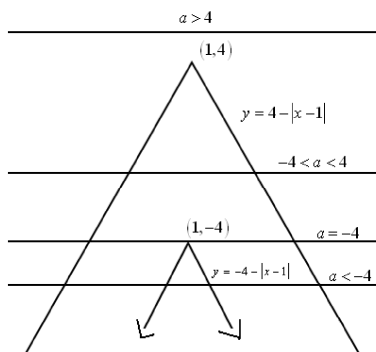
50. If $a > 0$

If $a < 0$

4 real roots $a \in (-\infty, -4)$

$-4 < a < 4$ 2 roots

$a > 4$ no root

**Passage type questions 51 & 52 hint**

$$f(x) = \frac{1}{2} \Rightarrow \text{periodic constant function}$$

Passage type questions 53 & 54 hint

Conceptual

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

2013_P1

Date: 26-08-17

Time: 07:30 AM to 10:30 AM

WAT-16

Max. Marks: 180

Name of the Student: _____

H.T. NO:

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PAPER-I

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as :
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

26-08-17_Jr. Super60-I_WAT-16_SYLLABUS

MATHS: SYLLABUS COVERED FROM 21-08-17 TO 24-08-17 (WAT-16)

PHYSICS: SYLLABUS COVERD FROM 21-08-17 TO 24-08-2017(WAT-16)

CHEMISTRY: MOT, metallic and hydrogen bonding

JEE-ADVANCE-2013-P1-Model

Time: 07:30 AM – 10:30 AM

IMPORTANT INSTRUCTIONS

Max Marks: 180

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 10)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 11 – 15)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 16 – 20)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 21 – 30)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 31 – 35)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 36 – 40)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

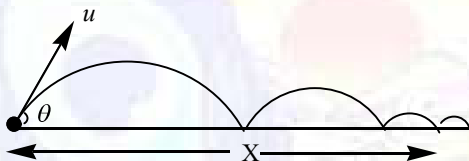
MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 41 – 50)	Questions with Single Correct Choice	2	0	10	20
Sec – II(Q.N : 51 – 55)	Questions with Multiple Correct Choice	4	-1	5	20
Sec – III(Q.N : 56 – 60)	Questions with Integer Answer Type	4	-1	5	20
Total				20	60

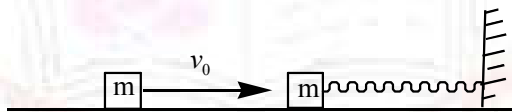
PHYSICS:**Max. Marks: 60****SECTION - I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A ball is projected with initial velocity u at an angle θ to the horizontal. Then horizontal displacement covered by ball as it collides third time to the ground would be, if coefficient of restitution is e :



- A) $(1+e)\frac{u^2 \sin 2\theta}{g}$ B) $e\frac{u^2 \sin 2\theta}{g}$ C) $(1-e)\frac{u^2 \sin 2\theta}{g}$ D) $(1+e+e^2)\frac{u^2 \sin 2\theta}{g}$
2. In the figure shown surface is frictionless and spring is in natural condition. If x_1, x_2 and x_3 are the maximum compression in spring for elastic, completely inelastic and inelastic ($e=0.5$) respectively then:

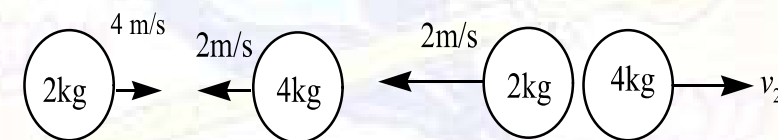


- A) $x_1 > x_2 > x_3$ B) $x_2 > x_3 > x_1$ C) $x_1 > x_3 > x_2$ D) $x_2 > x_1 > x_3$

3. A smooth sphere is moving on a horizontal surface with a velocity vector $(2\hat{i} + 2\hat{j}) \text{ m/s}$ immediately before it hit a vertical wall. The wall is parallel to y-z plane and coefficient of restitution between the sphere and the wall is $e = \frac{1}{2}$. The velocity of the sphere after it hits the wall is:

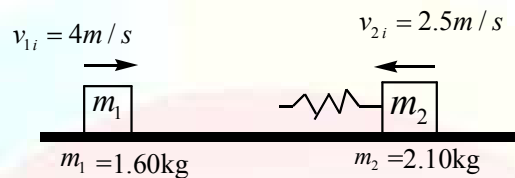
A) $\hat{i} - \hat{j}$ B) $-\hat{i} + 2\hat{j}$ C) $-\hat{i} - \hat{j}$ D) $2\hat{i} - \hat{j}$

4. Two balls of masses 2 kg and 4 kg are moved towards each other with velocities 4 m/s and 2 m/s respectively on a frictionless surface. After colliding the 2 kg ball returns back with velocity 2 m/s. Then find, coefficient of restitution of collision



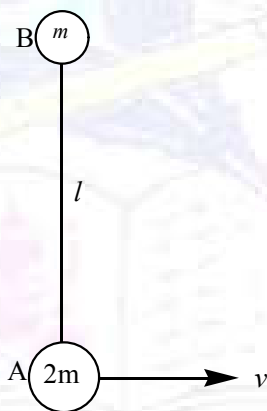
A) 0.5 B) 0.8 C) 1 D) 0.4

5. An elastic collision takes place between two masses m_1 and m_2 , moving on a frictionless surface as shown in fig: The spring constant is $k=600 \text{ N/m}$. What is the magnitude of velocity of block 2 at the instant block 1 is moving to right with a velocity 3 m/s? (i:initial)



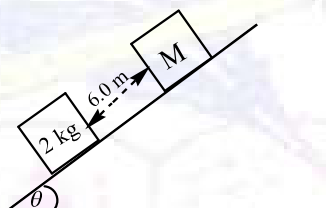
- A) 1.64 m/s B) 1.74 m/s C) 1.44 m/s D) 1.34 m/s

6. Two masses A and B connected with an inextensible string of length l lie on a smooth horizontal plane. A is given a velocity of $v \text{ m/s}$ along the ground perpendicular to line AB as shown in Fig. Find the tension in string during their subsequent motion.



- A) $\frac{2mv^2}{3l}$ B) $\frac{3mv^2}{2l}$ C) $\frac{2mv^2}{l}$ D) $\frac{mv^2}{3l}$

7. In an elastic head-on collision of bodies of mass m and $(x^2 - 9x + 21)m$, the second body is at rest and the maximum transfer of momentum takes place. Find the value of x
 A) 7 B) 6 C) 5 D) None of these
8. Two blocks of mass 2 kg and $M\text{ kg}$ are at rest on an inclined plane and are separated by a distance of 6.0 m as shown. The coefficient of friction between each block and the inclined plane is 0.25 . The 2 kg block is given a velocity of 10.0 ms^{-1} up the inclined plane. It collides with M , comes back and has velocity of 1.0 ms^{-1} when it reaches its initial position. The other block M after the collision moves 0.5 m up and comes to rest. Take $\sin\theta \cong \tan\theta = 0.05$ and $g = 10\text{ ms}^{-2}$. Then the approximate value of M is



- A) 2.5 kg B) 5 kg C) 10 kg D) 15 kg
9. A rocket, with an initial mass of 1000 kg , is launched vertically upwards from rest under gravity. The rocket burns fuel at the rate of 10 kg s^{-1} . The burnt matter is ejected vertically downwards with a speed of 2000 ms^{-1} relative to the rocket. If burning ceases

after one minute, find the maximum velocity of the rocket. (Assume g to be constant and have a value 10ms^{-2}) ($\ln(2.5) = 0.9162$)

- A) 1232.6 m/s B) 1332.6 m/s C) 1032.6 m/s D) 932.6 m/s

10. A rocket set for vertical firing weights 50 kg and contains 450 kg of fuel. It can have a maximum exhaust velocity of 2kms^{-1} with respect to rocket. What should be its minimum rate of fuel consumption to give an acceleration of 20 m/s^2

- A) 7.5kg/s B) 8.5kg/s C) 6.5kg/s D) 9.5kg/s

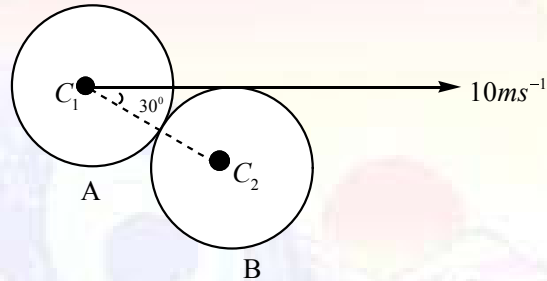
SECTION - II

Multiple Correct Answer(s) Type

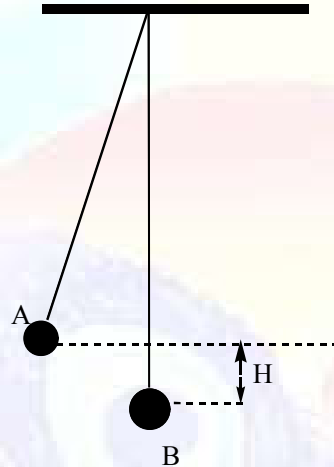
This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

11. A particle of mass m makes a head-on elastic collision with another particle of mass $2m$ initially rest. The velocity of the first particle before and after collision is given to be u_1 and v_1 and the velocity of second particle after the collision is v_2 . Which of the following statements is true in respect of this collision ?
- A) For all values of u_1, v_1 will always be less than u_1 in magnitude.
- B) The fractional loss in kinetic energy of the first particle is $\frac{8}{9}$
- C) The gain in kinetic energy of the second particle is $\left(\frac{8}{9}\right)^{\text{th}}$ of the initial kinetic energy the first particle.
- D) There is a net loss in the kinetic energy of the two particle system in the collision.

12. A ball A collides elastically with another identical ball B with a velocity of 10 ms^{-1} making an angle with the line joining their centres C_1 and C_2 . select correct statement(s) (there is no friction anywhere)

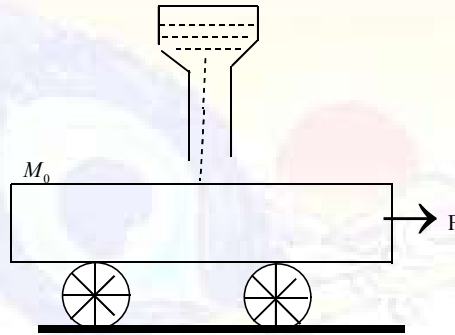


- A) Velocity of ball A after collision is 5 ms^{-1}
 B) Velocity of ball B after collision is $5\sqrt{3} \text{ ms}^{-1}$
 C) Kinetic energy will not be conserved here, because collision is not head on.
 D) Both the balls move at right angles after collision
13. Two small balls A and B of mass M and $3M$ hang from the ceiling by strings of equal length. The ball A is drawn aside so that it is raised to a height H . Now the ball A is released such that it collides with ball B. select the correct statement (s)



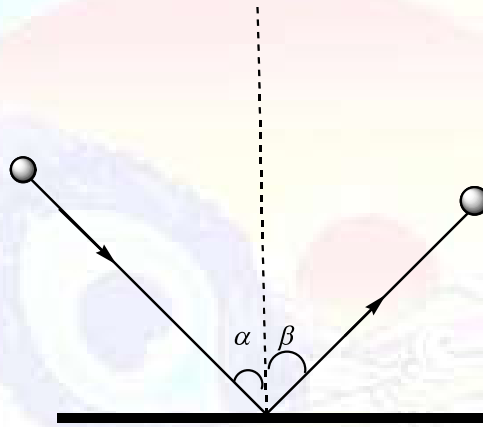
- A) If collision is perfectly elastic, ball B will rise upto a height $\frac{H}{4}$
- B) If the collision is perfectly elastic, ball A will rise upto a height $\frac{H}{4}$
- C) If the collision is perfectly inelastic, the combined mass will rise to a height $\frac{15H}{16}$
- D) If the collision is perfectly inelastic, the combined mass will rise to a height $\frac{H}{16}$
14. The figure shows a long cart moving on a smooth horizontal surface due to an external constant force of magnitude F . Initial mass of the cart is M_0 and its initial velocity is zero when the force F started to act. At $t = 0$, sand starts falling from a stationary hopper onto the cart at constant rate $\mu \text{ kg s}^{-1}$ and sticks to the cart. After time t_0 the sand

starts leaking from the bottom at the same constant rate $\mu \text{ kg s}^{-1}$. Eventually at time $t = 2t_0$, the sand stops falling from the hopper on to the cart and force F also stops acting. Based on above information, which of the following statement(s) is/are correct



- A) The velocity of the cart at time $t (< t_0)$ is $V = \frac{Ft}{M_0 + \mu t}$
- B) In the same model of above option if the cart is to be moved with constant velocity v , then the power supplied by external agent applying that force is μv^2
- C) In the same model of above option B rate of increase of the kinetic energy of the cart (with sand) is $\frac{1}{2} \mu v^2$
- D) In the same model of above option B the rate of increase of the kinetic energy of the cart (with sand) is $\frac{1}{3} \mu v^2$

15. A ball strikes the ground at an angle α and rebound at an angle β with the vertical as shown in the figure. Then (all surfaces are frictionless)

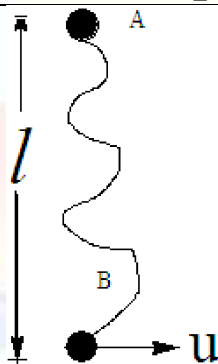


- A) coefficient of restitution is $\frac{\tan \alpha}{\tan \beta}$
- B) if $\alpha < \beta$ the collision is inelastic
- C) if $\alpha = \beta$ the collision is elastic
- D) if $\alpha > \beta$ the collision is inelastic

SECTION - III**Integer Answer Type**

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

16. A 6000 kg rockets is set for vertical firing. If the exhaust speed is 1000ms^{-1} , the amount of gas that must be ejected per second to supply the thrust needed to overcome the weight of the rocket is $(g = 10\text{ms}^{-2}) 10 \times x \text{ kg}$ the value of x is.... $(g = 10\text{m} / \text{s}^2)$
17. A sphere of mass m is moving with a velocity $3\hat{i} + 4\hat{j} + 5\hat{k}$ hits a smooth wall and rebounds with velocity $2\hat{i} + 3\hat{j} + 2\hat{k}$. Then the coefficient of restitution between the sphere and the wall is $n/6$. Find $n=?$
18. Two particles A and B each of mass m are attached by a light inextensible string of length $2l$. The whole system lies on a smooth horizontal table with B initially at a distance l from A. The particle at end B is projected across the table with speed u perpendicular to AB . Velocity of ball A just after the string is taut is $\frac{u\sqrt{x}}{4}$, the value of x is...



19. A particle of mass m having collided a stationary particle of mass $M = 5m$ deviate by an angle $\frac{\pi}{2}$ where as the particle M recoiled at angle $\theta = 30^\circ$ to the direction of the initial motion of the particle m . The percentage change in kinetic energy of this system after collision is $8 \times x\%$, the value of x is.....
20. A 1 kg ball, moving at 12ms^{-1} , collides head-on with a 2kg ball moving in the opposite direction at 24ms^{-1} . If the coefficient of restitution is $\frac{2}{3}$, then the energy lost in the collision is $60 \times x$ joule, the value x is...

CHEMISTRY:

Max. Marks: 60

SECTION - I

Single Correct Answer Type

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

21. Which of the following is false ?
- VanderWaals forces are responsible for the formation of molecular crystals.
 - Branching lowers the boiling point of isomeric organic compounds due to reduction of Van der Waals forces of attraction
 - In graphite, Van der Waals forces act between the hexagonal.
 - Diamond has layered structure of carbon atoms with VanderWaals forces between the carbon layers.
22. Which of the following statement(s) is/are true ?
- In N_2 , the doubly degenerate π_{2p} orbitals are completely filled.
 - In O_2 , the energy of σ_{2p_z} orbital is lower than the doubly degenerate π_{2p} orbitals.
 - Different molecular species with the same configuration have the same energy.
 - A π_{2p}^* orbital has two nodal planes.
- i, ii and iv
 - i and ii only
 - i, ii, iii and iv
 - ii, iii, iv

23. Correct bond order for $\overset{NO^-}{(1)}, \overset{NO^+}{(2)}, \overset{NO}{(3)}$
- A) $1 > 2 > 3$ B) $3 > 2 > 1$ C) $2 > 3 > 1$ D) $2 < 1 < 3$
24. s-p orbitals mixing is seen in
- A) F_2 B) C_2 C) O_2^+ D) O_2^-
25. Which of the following species is paramagnetic ?
- A) N_2 B) B_2 C) O_2^{2-} D) C_2
26. Which of the following statement is false ?
- A) The crystal lattice of ice does not form hydrogen bonds.
- B) The density of water increases when heated from 0°C to 4°C due to the change in the structure of the cluster of water molecules.
- C) Above 4°C , the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding.
- D) The density of water increases from 0°C to a maximum at 4°C because the entropy of the system increases.

27. Bond order of which of the following is equal to that of O_2 ?
 A) CO B) NO C) CN^+ D) CN^-
28. Metallic lustre is explained by
 A) diffusion of metal ions B) oscillation of loose electrons
 C) excitation of free protons D) existence of bcc lattice
29. In the compounds of the type ECl_3 where $E = B, P, As$ (or) Bi the angles $Cl-E-Cl$ for different elements E are in the order
 A) $B > P = As = Bi$ B) $B > P > As > Bi$ C) $B < P = As = Bi$ D) $B < P < As < Bi$
30. Amongst the following statements
 I : The structure of $[BrF_4]^+$ is regular tetrahedral.
 II : Bond order of O_2 decreases by the removal of an electron.
 III : Br_3^- and Br_3^+ are having same shape and same structure.
 IV : The hybrid orbital of phosphorous in PCl_5 is $sp^3 d_{xy}$.
 The incorrect statements is/are
 A) only II B) I, III, IV
 C) I, II, III D) I, II, III, IV

SECTION - II**Multiple Correct Answer(s) Type**

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

31. Select the correct statement (s) about the compound $\text{NO}[\text{BF}_4]$:
- A) It has 5σ and 2π bond
 - B) Nitrogen-oxygen bond length in this compound is longer than nitric oxide (NO)
 - C) It is a diamagnetic species
 - D) B – F bond length in this compound is lower than in BF_3
32. Which of the following statement is/are correct ?
- A) O_2 is paramagnetic, N_2 is also paramagnetic
 - B) O_2 is paramagnetic, N_2 is diamagnetic
 - C) O_2 is paramagnetic, O_2^{2-} is also paramagnetic
 - D) In $\text{O}_2 \rightarrow \text{O}_2^+$ and $\text{NO} \rightarrow \text{NO}^+$, bond length decreases.
33. Which of the following species have identical bond order ?
- A) CN^-
 - B) O_2^-
 - C) NO^+
 - D) Cl_2
34. Which of the following molecular orbital(s) has two nodal planes?
- A) $\Pi_{2\text{Py}}$
 - B) $\Pi_{2\text{Px}}^*$
 - C) $\sigma_{2\text{Pz}}^*$
 - D) $\Pi_{2\text{Py}}^*$

35. Which of the following is/are correct with respect to hydrogen bonding ?

- A) It creates infinite chains for HF_2^- .
- B) It creates hydrogen bonded chains for molecules like HF, HSO_4^- etc.
- C) It is responsible for creating 3_D network in compounds like $H_2O, H_2PO_4^-$.
- D) It helps in the dimerization of carboxylic acids.

SECTION - III

Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

36. Maximum number of hydrogen bonds a water molecule can form in ice ?

37. Number of hydrogen bonds in $H_9O_4^+$ species is

38. The sum of number of sigma and pi bonds formed between two carbon atoms in CaC_2 are:

39. Number of pi-bonds in B_2 is:

40. How many of the following statement(s) is/are true ?

- 1) During N_2^+ formation, from N_2 one electron is removed from the bonding molecular orbitals
- 2) During O_2^+ formation, from O_2 one electron is removed from the antibonding molecular orbital.
- 3) During O_2^- formation, from O_2 one electron is added to the bonding molecular orbital.
- 4) During CN^- formation, one electron is added to the bonding molecular orbital.
- 5) σ_{1s} molecular orbital has only one nodal plane.
- 6) σ^*_{1s} molecular orbital has only one nodal plane.

MATHS:**Max. Marks: 60****SECTION - I****Single Correct Answer Type**

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

41. Number of values of x for which $\sqrt{1+\cos 2x} = \sqrt{2} \cos^{-1}(\cos x)$ for $x \in \left(\frac{\pi}{2}, \pi\right)$, is
- A) 2 B) 0 C) 3 D) ∞
42. If α, β are the solutions of $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{2}$ then $\cos^{-1} \alpha + \cos^{-1} \beta =$
- A) $\frac{\pi}{2}$ B) $-\frac{\pi}{2}$ C) $\frac{3\pi}{2}$ D) $-\frac{3\pi}{2}$
43. $\tan^{-1} \frac{2}{4} + \tan^{-1} \frac{2}{9} + \tan^{-1} \frac{2}{16} + \dots + \tan^{-1} \frac{2}{(n+1)^2} + \dots$ to $\alpha =$
- A) $\cot^{-1} 2$ B) $\cot^{-1} 2 + \frac{\pi}{4}$ C) $\frac{\pi}{4} - \cot^{-1} 2$ D) $\frac{\pi}{2} + \cot^{-1} 2$
44. Number of solutions of $2\tan^{-1}(\tan x) = 6 - x$ is
- A) 4 B) 2 C) 3 D) 6
45. The least integral value of k for which $(K-2)x^2 + 8x + k + 4 > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) \forall x \in R$, is
- A) -7 B) -5 C) -3 D) 5

46. Let $f(x) = \frac{2}{\pi} (\sin^{-1}[x] + \tan^{-1}[x] + \cot^{-1}[x])$ (where $[.]$ denotes G.I.F), if A and B denote the domain and range of $f(x)$ respectively then the number of integers in $A \cup B$ is
 A) 1 B) 2 C) 3 D) 4
47. If $\cot^{-1}\left(\frac{n^2 - 10n + 26}{2\sqrt{3}}\right) > \frac{\pi}{6}, n \in \mathbb{N}$, then the minimum value of n is
 A) 3 B) 5 C) 6 D) 7
48. The number of real solutions of the equation $3\cos^{-1}x - \pi x - \frac{\pi}{2} = 0$ is
 A) 0 B) 1 C) 2 D) infinitely
49. The value of $\sin^{-1}\frac{1}{\sqrt{2}} + \sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right) + \dots + \sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right) + \dots$ to ∞ is equal to ($n \in \mathbb{N}$)
 A) π B) $\frac{\pi}{2}$ C) $\frac{\pi}{4}$ D) $\frac{3\pi}{2}$
50. If $(\sin^{-1}a)^2 + (\cos^{-1}b)^2 + (\sec^{-1}c)^2 + (\operatorname{cosec}^{-1}d)^2 = \frac{5\pi^2}{2}$ ($a, b, c, d \in \mathbb{R}$) then the value of $(\sin^{-1}a)^2 - (\cos^{-1}b)^2 + (\sec^{-1}c)^2 - (\operatorname{cosec}^{-1}d)^2$ is
 A) $-\pi^2$ B) $\frac{-\pi^2}{2}$ C) 0 D) $\frac{\pi^2}{2}$

SECTION - II**Multiple Correct Answer(s) Type**

This section contains **5 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE are correct**.

51. If

$$\sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7) = a\pi + b$$

Then which of the following is/are correct?

- A) $a - b = 53$ B) $3a + b = -1$ C) $4a + b = 12$ D) $a - b = 27$

52. If $(\sin^{-1} x)^2 + (\sin^{-1} y)^2 + (\sin^{-1} z)^2 = \frac{3\pi^2}{4}$ then the possible value of $x - y + z$ is equal to

- A) 1 B) -1 C) 3 D) -3

53. If $\sin^{-1}\left(a - \frac{a^2}{3} + \frac{a^3}{9} - \dots\right) + \cos^{-1}(1 + b + b^2 + \dots) = \frac{\pi}{2}$ ($|a| < 3$ and $|b| < 1$), then which of the following is/are correct?

- A) $b = \frac{3a-2}{2a}$ B) $b = \frac{2a-3}{3a}$ C) $a = \frac{2}{3-2b}$ D) $a = \frac{3}{2-3b}$

54. If $s_n = \cot^{-1} 3 + \cot^{-1} 7 + \cot^{-1} 13 + \cot^{-1} 21 + \dots$ to n terms then which of the following is/are correct?

- A) $s_{10} = \tan^{-1} \frac{5}{6}$ B) $s_{\infty} = \frac{\pi}{4}$ C) $s_6 = \sin^{-1} \frac{3}{5}$ D) $s_{20} = \cot^{-1}(101)$

55. If $\sin^{-1}(x^2 + 2x + 2) + \tan^{-1}(x^2 - 3x - k^2) > \frac{\pi}{2}$ ($x \in \mathbb{R}$), then k contains the set

A) (-1,0)

B) (0,1)

C) (1,2)

D) (0,2)

SECTION - III

Integer Answer Type

This section contains **5 questions**. The answer to each question is single digit integer, ranging from 0 to 9 (both inclusive).

56. If the equation $\sin^{-1}(x^2 + x + 1) + \cos^{-1}(\lambda x + 1) = \frac{\pi}{2}$ has exactly two solutions, then the range of λ is $[a, b)$ where $a + b =$

57. If $r = x + y + z$ ($x, y, z > 0$) and $\tan^{-1}\sqrt{\frac{xr}{yz}} + \tan^{-1}\sqrt{\frac{yr}{xz}} + \tan^{-1}\sqrt{\frac{zr}{xy}} = K\pi$ then $K =$

58. Let $f: [0, 4\pi] \rightarrow [0, \pi]$ be defined as $f(x) = \cos^{-1}(\cos x)$. The number of values of x satisfying the equation $f(x) = \frac{10-x}{10}$ is

59. If $\cot^{-1}\left(2^2 + \frac{1}{2}\right) + \cot^{-1}\left(2^3 + \frac{1}{2^2}\right) + \cot^{-1}\left(2^4 + \frac{1}{2^3}\right) + \dots \text{to } \infty = \cot^{-1}k$, then $K =$

60. If $\sin\left\{2\left[\sin^{-1}\frac{\sqrt{5}}{3} - \cos^{-1}\frac{\sqrt{5}}{3}\right]\right\} = \frac{k\sqrt{5}}{81}$, then the value of k is equal to



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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

WAT-16

Date: 26-08-17

Time: 07:30 AM – 10:30 AM

2013-P1

Max.Marks:180

KEY SHEET

PHYSICS

1	D	2	C	3	B	4	A	5	B	6	A
7	C	8	D	9	A	10	A	11	ABC	12	ABD
13	ABD	14	ABC	15	ABC	16	6	17	3	18	3
19	5	20	4								

CHEMISTRY

21	D	22	A	23	C	24	B	25	B	26	A
27	C	28	B	29	B	30	D	31	AC	32	BD
33	AC	34	BD	35	BCD	36	4	37	3	38	3
39	1	40	4								

MATHS

41	B	42	C	43	B	44	C	45	D	46	D
47	A	48	B	49	B	50	C	51	ABC	52	ABCD
53	BD	54	ABC	55	ABCD	56	1	57	1	58	3
59	2	60	8								

SOLUTIONS
PHYSICS

1. After each collision vertical velocity become e times whereas horizontal velocity remain same. So after each collision time of flight become e times of previous one.

$$\text{So that horizontal displacement} = R + eR + e^2R = (1 + e + e^2) \frac{u^2 \sin 2\theta}{g}$$

2. For elastic collision Velocity is interchanged

$$\therefore \frac{1}{2}mv_0^2 = \frac{1}{2}kx_1^2 \Rightarrow x_1 = \frac{\sqrt{mv_0}}{k}$$

For completely inelastic collision

$$mv_0 = 2mv \Rightarrow v = \frac{v_0}{2}$$

$$\text{Now } \frac{1}{2}kx_2^2 = \frac{1}{2}(2m)v^2 \Rightarrow x_2 = \sqrt{\frac{mv_0}{2k}}$$

For $e = 0.5$

$$-0.5 = \frac{v_2 - v_1}{0 - v_0}$$

$$v_2 - v_1 = \frac{v_0}{2}$$

$$\text{Also } mv_2 + mv_1 = mv_0$$

$$v_2 = 0.75v_0 \Rightarrow x_3 = \sqrt{\frac{3mv_0}{4k}}$$

3. Along wall momentum is conserved

$$4. \quad e = \frac{\text{velocity of separation}}{\text{velocity of approach}} = \frac{1 - (-2)}{4 - (-2)} = \frac{3}{6} = 0.5$$

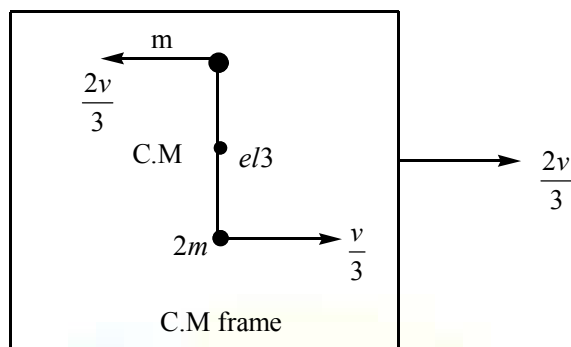
5. From momentum conservation, we obtain $m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$

$$(1.60)(4.00) + (2.10)(-2.50) = (1.60)(3.00) + (2.10)v_{2f}$$

$$v_{2f} = -1.74 \text{ m/s}$$

Negative sign implies that block 2 is still continuing in the same direction

6. Solve the problem in CM frame. In CM frame particles A and B moving along circular path with same ω



Velocity of $2m$ mass $= v - \frac{2v}{3} = \frac{v}{3}$

Angular velocity of particle A

$$\omega = \frac{v/3}{l/3} = \frac{v}{l}$$

$$T = M\omega^2 r = 2m \cdot \frac{v^2}{l^2} \cdot \frac{l}{3}$$

$$\text{or } T = \frac{2mv^2}{3l}$$

7 For maximum momentum transfer $m_1 = m_2$

8. Conceptual

9. Since, we know that

$$v = (u - gt) + v_r \ln\left(\frac{m_0}{m}\right)$$

According to the problem we have $u = 0, t = 60s, g = 10ms^{-2}, v_r = 2000ms^{-1}, m_0 = 1000kg$

and $m = 1000 - 10 \times 60 = 400kg$

$$\Rightarrow v = 0 - 600 + 2000 \ln\left(\frac{1000}{400}\right)$$

$$\Rightarrow v = 2000 \ln(2.5) - 600$$

The maximum velocity of the rocket is

$$v = 200(10 \ln 2.5 - 3) = 1232.6ms^{-1}$$

10. Net acceleration $a = 20ms^{-2}$

Since we have $ma = F_1 - mg$

$$\Rightarrow a = \frac{F_1}{m} - g$$

$$\Rightarrow a = \frac{v_r}{m} \left(\frac{-dm}{dt} \right) - g$$

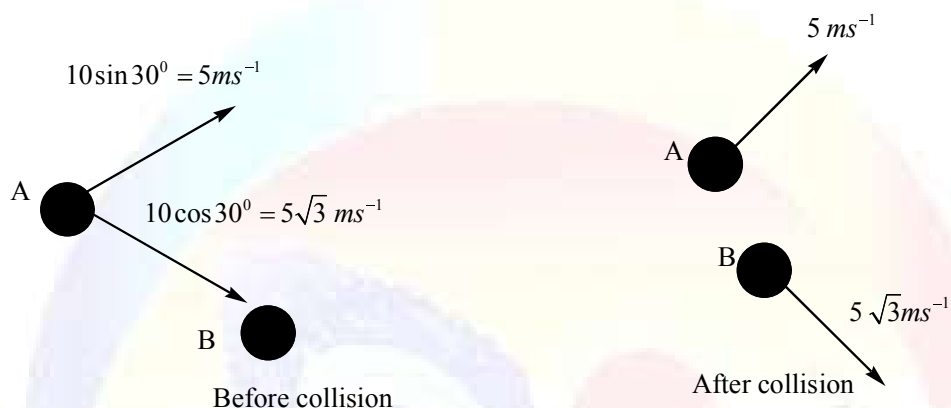
$$\Rightarrow \left(\frac{-dm}{dt} \right) = \frac{m(g + a)}{v_r}$$

$$\Rightarrow \left(\frac{-dm}{dt} \right) = \frac{(450 + 50)(10 + 20)}{2 \times 10^3} = 7.5kgs^{-1}$$

11. Impulse $j = \frac{m(v+4)}{t}$

Coefficient of restitution $e = \frac{v_2 - v_1}{u_1 - u_2} = \frac{v}{u}$

12.



Velocity components in common tangent direction will remain unchanged while velocity components in common normal direction are interchanged in case of an elastic

collision. Hence both A and B move at right angles after collision with $v_A = 5 \text{ ms}^{-1}$ and $v_B = 5\sqrt{3} \text{ ms}^{-1}$. Kinetic energy is conserved in an elastic collision. Whether it is a head on or an oblique collision

13. Before collision $v_A = \sqrt{2gH}$ and $v_B = 0$

$$v_A = \left(\frac{m_A - m_B}{m_A + m_B} \right) v_A$$

$$\Rightarrow v_A = \left(\frac{M - 3M}{M + 3M} \right) \sqrt{2gH} = -\frac{\sqrt{2gh}}{2}$$

$$\text{and } v_B = \left(\frac{2m_A}{m_A + m_B} \right) v_A$$

$$\Rightarrow v_B = \left(\frac{2M}{M + 3M} \right) \sqrt{2gH} = \frac{\sqrt{2gH}}{2}$$

$$\text{So, } h_A = \frac{v_A^2}{2g} = \frac{H}{4} \text{ and } h_B = \frac{v_B^2}{2g} = \frac{H}{4}$$

$$v_A = \frac{m_A v_A}{m_A + m_B} = \frac{M \sqrt{2gH}}{M + 3M} = \frac{\sqrt{2gH}}{4}$$

The combined mass will rise to height

$$h = \frac{v^2}{2g} = \frac{H}{16}$$

14. By Newton's Law

$$F - \mu v = (M + \mu t) \frac{dv}{dt}$$

$$\Rightarrow \int_0^v \frac{dv}{F - \mu v} = \int_0^t \frac{dt}{M + \mu t}$$

$$\Rightarrow v = \frac{Ft}{M_0 + \mu t}$$

For constant velocity $\frac{dv}{dt} = 0$

So, from (1) we get $F = \mu v^2$

$$K = \frac{1}{2} M v^2$$

For constant velocity $v = \text{constant}$

$$\Rightarrow \frac{dk}{dt} = \frac{v^2}{2} \frac{dn}{dt}$$

Since $M = M_0 + \mu t$

$$\Rightarrow \frac{dM}{dt} = \mu$$

$$\Rightarrow \frac{dk}{dt} = \frac{1}{2} \mu v^2$$

15. Let velocity of ball before collision and after collision be u and v respectively, then

$$e = \frac{v \cos \beta}{u \cos \alpha}$$

And by law of conservation of momentum along horizontal direction we get

$$u \sin \alpha = v \sin \beta$$

From (1) and (2) we get

$$e = \frac{\tan \alpha}{\tan \beta}$$

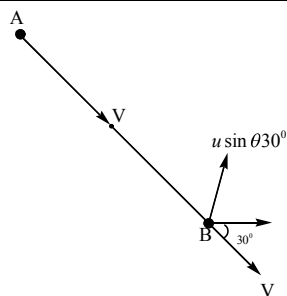
16. $M_0 g = \text{Thrust} = v \frac{dM}{dt}$

$$\Rightarrow \frac{dM}{dt} = \frac{M_0 g}{v}$$

$$\Rightarrow \frac{dM}{dt} = \frac{60000}{1000} = 60 \text{ kg s}^{-1}$$

17. $\frac{\vec{V} \cdot (\vec{V} - \vec{U})}{\vec{U} \cdot (\vec{V} - \vec{U})} = e$

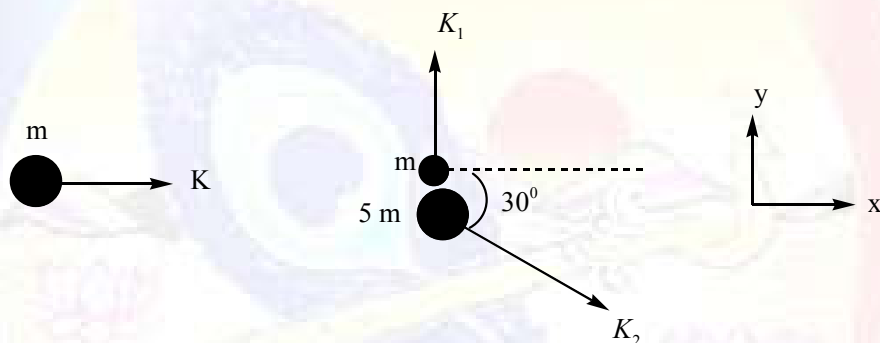
18. when the string becomes taut both particle begin to move with velocity components v in the direction AB. Applying Law of conservation of liner momentum in the direction AB we get $mu \cos 30^\circ = mv + mv$



$$\Rightarrow v = \frac{u\sqrt{3}}{4}$$

Hence the velocity of ball A just after the string becomes taut is $\frac{u\sqrt{3}}{4}$

19. Let K be the initial kinetic energy of mass m and K_1 and K_2 the kinetic energy after collision in the direction shown in figure initial momentum of the particle is $\sqrt{2mK}$
So applying conservation of linear momentum in vector form, we get



$$\sqrt{2mK} \hat{i} = \sqrt{2mK_1} \hat{j} + \sqrt{2mK_2} (\cos 30^\circ \hat{i} - \sin 30^\circ \hat{j})$$

$$\Rightarrow \sqrt{2mK} = \sqrt{10mK_2} \frac{\sqrt{3}}{2}$$

$$\sqrt{2mK_1} = \sqrt{10mK_2} \frac{1}{2}$$

Solving we get $K_2 = \frac{4}{15}K$ and $K_1 = \frac{K}{3}$

$$\Rightarrow K_1 + K_2 = \frac{9}{15}K = 0.6K$$

So, Loss in $KE = K - 0.6K = 0.4K$

Hence, KE will be decreased by 40%

20. $(1)(12) + (2)(-24) = 1v_1 + 2v_2$

$$\Rightarrow v_1 + 2v_2 = -36$$

Further

$$e = -\left(\frac{v_2 - v_1}{u_2 - u_1}\right)$$

$$\Rightarrow \frac{2}{3} = -\left(\frac{v_2 - v_1}{-24 - 12}\right)$$

$$\Rightarrow \frac{2}{3} = \frac{v_2 - v_1}{36}$$

$$\Rightarrow v_2 - v_1 = 24$$

Add(1) and (2)

$$3v_2 = -12$$

$$\Rightarrow v_2 = -4ms^{-1}$$

$$\Rightarrow v_1 = -28ms^{-1}$$

$$\Rightarrow \text{Loss} = \text{Total initial} - \text{Total final}$$

Since

$$\text{Total initial} = \frac{1}{2}(1)(144) + \frac{1}{2}(2)(16)$$

$$\Rightarrow E_i = 72 + 576 = 648J$$

Similarly

$$E_f = \frac{1}{2}(1)(784) + \frac{1}{2}(2)(16)$$

$$\Rightarrow E_f = 392 + 16$$

$$\Rightarrow E_f = 408J$$

$$\Rightarrow \text{Loss} = 648 - 408$$

$$\Rightarrow \text{Loss} = 240J$$

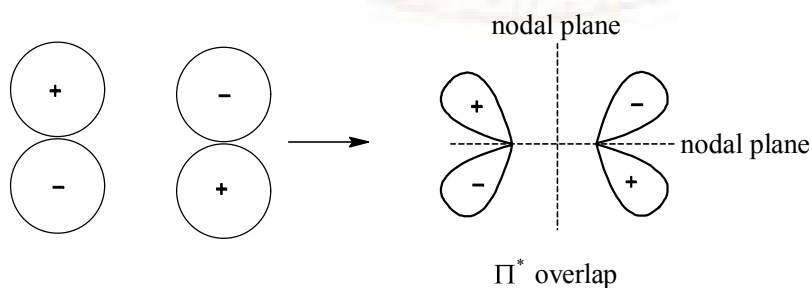
CHEMISTRY

21. Diamond is a crystalline covalent solid.
22. The σ_{2p_z} orbital in O_2 is higher than the doubly degenerate π_{2p} orbitals and more than π_{2p} orbitals in N_2 from MOT.
23. $B.O \Rightarrow NO = 2.5$
 $NO^+ = 3.0$; $NO^- = 2.0$
 $NO^+ > NO > NO^-$
24. $s-p$ mixing is observed in B_2, C_2, N_2 or in species with total electrons ≤ 14
25. B_2 has two unpaired electrons
26. Ice also has hydrogen bonds, a water molecule in ice is surrounded by 4 more water molecules through H-bonds
27. Bond order of $CN^+ = \frac{Nb - Na}{2}$

$$= \frac{8-4}{2} = 2$$

Bond order of $O_2 = 2$

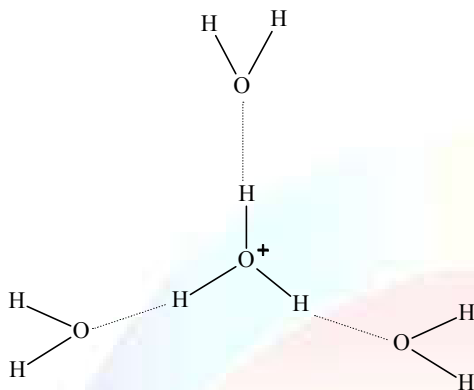
28. Metallic lustre is due to free valency electrons.
29. With decrease in electronegativity of central atom, bond angle decreases. BCl_3 has trigonal planar structure with 120° angles
30. I : BrF_4^+ : Trigonal bipyramidal structure and see-saw shape
 II : Bond order of O_2 increases by the removal of e^- , thus bond order increases.
 III : Br_3^- and Br_3^+ are of different shape and structure.
 IV : In PCl_5 , hybrid orbital of P is $sp^3d_{z^2}$
31. Bond order of NO^+ is = 3, one sigma bond and two pie bonds.
 No of pie bonds = 2
 No of sigma bonds = 5
 B.O. of $NO^+ = 3$
 B.O. of $NO = 2.5$
 NO^+ is the diamagnetic cation and BF_4^- is also a diamagnetic anion.
 B – F bonds are longer in BF_4^- than in BF_3 due to absence of $P\pi - d\pi$ back bonding in $[BF_4]^-$.
32. O_2 is paramagnetic due to 2 unpaired electrons while N_2 & O_2^{2-} are diamagnetic. In conversion of $O_2 \rightarrow O_2^+$ and $NO \rightarrow NO^+$, e^- from antibonding orbital ($CN^-, N_2, NO^+, C_2^{2-}$) is removed, thus bond order increases & bond length decreases.
33. Isoelectronic species has same bond order
- 34.



35. $\text{HF}_2^{(-)}$ exists in discrete form as salt, KHF_2 .

36. A water molecule can form 4 hydrogen bonds in ice.

37.



38. C_2^{2-} has one sigma and two pi bonds

39. B_2 has two pi electrons in bonding MO's, so it translates into one π -bond

40. During O_2^- formation, one electron is added to the antibonding molecular orbital. $\sigma^* 1s$ molecular orbital has only one nodal plane.

MATHEMATICS

41. $\sqrt{1+\cos 2x} = \sqrt{2} \cos^{-1} |\cos x| \Rightarrow \cos x = x \text{ for } x \in \left(\frac{\pi}{2}, \pi\right)$

42. $x(x+1) \geq 0 \ \& \ x^2+x+1 \leq 1 \Rightarrow x(x+1) = 0 \Rightarrow x = -1, 0$

43. $s_n = \sum \tan^{-1} \left(\frac{2}{n^2 + 2n + 1} \right) = \tan^{-1}(n+1) + \tan^{-1}(n+2) - \tan^{-1}1 - \tan^{-1}2$

$$s_\infty = \frac{\pi}{4} + \cot^{-1} 2$$

44. $\tan^{-1} \tan(x) = 3 - \frac{x}{2}$

No. of solutions is equal to no. of points of integer of $y = \tan^{-1}(\tan x)$ and $y = 3 - \frac{x}{2}$

45. $\cos^{-1} \cos 12 + \sin^{-1} \sin 12 = 0$

$$\therefore (k-2)x^2 + 8x + k + 4 > 0 \forall x \in R$$

$$\Rightarrow K-2 > 0 \ \& \ 64 - 4(K-2)(K+4) < 0$$

$$\Rightarrow K > 4$$

46. $A = [-1, 2), B = \{0, 1, 2\}$

47. $\frac{n^2 - 10n + 26}{2\sqrt{3}} < \sqrt{3}$

48. $\cos^{-1} x = \frac{\pi}{3}x + \frac{\pi}{6}$ Draw the graph

49. $s_n = \sin^{-1} \left[\frac{\sqrt{n} - \sqrt{n-1}}{\sqrt{n(n+1)}} \right] = \sum \tan^{-1} \frac{\sqrt{n} - \sqrt{n-1}}{1 + \sqrt{n}\sqrt{n-1}} = \sum (\tan^{-1} \sqrt{n} - \tan^{-1} \sqrt{n-1})$

$$s_\infty = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

50. $(\sin^{-1} a)^2 = \frac{\pi}{4}, (\cos^{-1} b)^2 = \pi^2, (\sec^{-1} c)^2 = \pi^2, (\cos e^{-1} d)^2 = \frac{\pi^2}{4}$

51. $\sin^{-1} \sin 8 = 3\pi - 8, \tan^{-1}(\tan 10) = 10 - 3\pi, \cos^{-1} \cos 12 = 4\pi - 12$

$$\cot^{-1} \cot^6 = 6 - \pi, \sec^{-1} \sec 9 = 9 - 2\pi, \cos e c^{-1} \cos e c 7 = 7 - 2\pi$$

52. $\sin^{-1} x = \pm \frac{\pi}{2}, \sin^{-1} y = \pm \frac{\pi}{2}, \sin^{-1} z = \pm \frac{\pi}{2}$

53. $a - \frac{a^2}{3} + \frac{a^3}{9} \dots = 1 + b + b^2 + \dots$

54. $s_n = \sum_{r=1}^n (\tan^{-1}(r+1) - \tan^{-1} r)$

55. Domain = $\{-1\}$

$$\therefore \frac{\pi}{2} + \tan^{-1}(4 - K^2) > \frac{\pi}{2}$$

$$4 - K^2 > 0$$

56. $x^2 + x + 1 = \lambda x + 1 \Rightarrow x = 0, \lambda - 1$

$$-1 \leq \lambda x + 1 \leq 1 \Rightarrow \lambda \in [0, 1]$$

57. Take $x = y = z$

58. Draw the graph of $y = \cos^{-1} \cos x$ and $y = \frac{10-x}{10}$

59. $\sum_{r=1}^{\infty} \tan^{-1} \left(2^{2^{r+1}} + \frac{1}{2^2} \right) = \sum_{r=1}^{\infty} (\tan^{-1}(2^{r+1}) - \tan^{-1}(2^r)) = \cot^{-1} 2$

60. $2 \left\{ \left[\sin^{-1} \frac{\sqrt{5}}{3} - \cos^{-1} \frac{\sqrt{5}}{3} \right] \right\}$

$$= \sin \left\{ 2 \left(\frac{\pi}{2} - 2 \cos^{-1} \frac{\sqrt{5}}{3} \right) \right\}$$

$$= \sin \left(4 \cos^{-1} \left(\frac{\sqrt{5}}{3} \right) \right)$$

$$= \frac{8\sqrt{5}}{81}$$

Success



JEE ADVANCED WEEKEND TEST



Sri Chaitanya IIT Academy., India.

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A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

WAT-18

Date: 16-09-17

Max.Marks:240

Name of the Student: _____

H.T. NO:

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2011-PAPER -II

IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

16-09-17_Jr.Super60-I_Jee-Adv_WAT-18_WEEKEND SYLLABUS

MATHS: SYLLABUS COVERED FROM 11-09-17 TO 15-09-17

PHYSICS: SYLLABUS COVERD FROM 11-09-17 TO 15-09-2017

CHEMISTRY: Alkaline earth metals, **Group 13:** Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups; unique behaviour Boron, Preparation, properties and uses of boron ,properties of boric acid, diborane, boron trifluoride, borax

JEE-ADVANCE-2011-P2-Model

Time: 03:00 Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 240

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 9 – 12)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 13 – 18)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 19 – 20)	Matrix Matching Type	8	0	2	16
Total				20	80

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 21 – 28)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 29 – 32)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 33 – 38)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 39 – 40)	Matrix Matching Type	8	0	2	16
Total				20	80

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 41 – 48)	Questions with Single Correct Choice	3	-1	8	24
Sec – II (Q.N : 49 – 52)	Questions with Multiple Correct Choice	4	0	4	16
Sec – III (Q.N : 53 – 58)	Questions with Integer Answer Type	4	0	6	24
Sec – IV (Q.N : 59 – 60)	Matrix Matching Type	8	0	2	16
Total				20	80

Sec: Jr. Super60-I

space for rough work

Page 2

SECTION-I

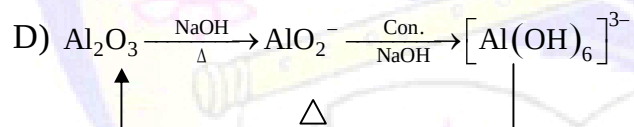
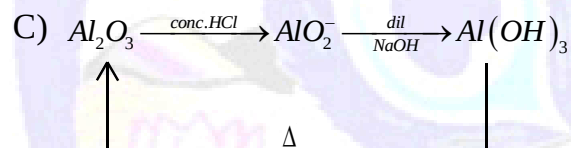
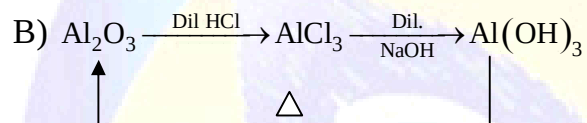
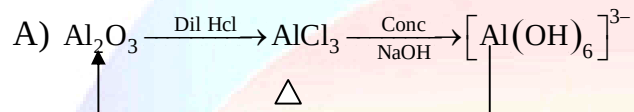
(SINGLE CORRECT CHOICE TYPE)

Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

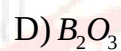
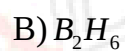
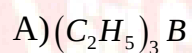
1. The aqueous solutions of lithium salts are poor conductor of electricity rather than other alkali metals because of
 - A) high ionization energy
 - B) high electronegativity
 - C) lower ability of Li^+ ions to polarize water molecules
 - D) Increase in size due to higher degree of hydration of Li^+ ion
2. Which of the following orders presents the correct sequence of the increasing basic nature of the given oxides?
 - A) $Al_2O_3 < MgO < Na_2O < K_2O$
 - B) $MgO < K_2O < Al_2O_3 < Na_2O$
 - C) $Na_2O < K_2O < MgO < Al_2O_3$
 - D) $K_2O < Na_2O < Al_2O_3 < MgO$

3. What happens when sodium metal is heated to a temperature 350°C in excess of dry air containing carbon dioxide gas?
- A) Na_2O_2 is formed
B) Na_2O is formed
C) First Na_2O_2 is formed which then converts into Na_2CO_3
D) First Na_2O_2 is formed which then converts into Na_2CO_3
4. $X + \text{C} + \text{Cl}_2 \xrightarrow[\text{of about } 1000\text{K}]{\text{High temperature}} Y + \text{CO}$; $Y + 2\text{H}_2\text{O} \rightarrow Z + 2\text{HCl}$ Compound Y is found in polymeric chain structure and is an electron deficient Molecule. Y must be
- A) BeO B) BeCl_2 C) BeH_2 D) MgCl_2
5. $\text{BaC}_2 + \text{N}_2 \xrightarrow{\Delta} (1)$
 $\text{CaC}_2 + \text{N}_2 \xrightarrow{\Delta} (2)$
(1) and (2) are
- A) $\text{BaCN}_2, \text{CaCN}_2$ B) $\text{Ba}(\text{CN})_2, \text{Ca}(\text{CN})_2$
C) $\text{Ba}(\text{CN})_2, \text{CaCN}_2$ D) None of these.

6. Which of the following is correct related to change aluminium oxide?



7. Borate salts when heated with conc. H_2SO_4 and $\text{C}_2\text{H}_5\text{OH}$ produce characteristic green colouration on flame due to the formation of volatile compound:



8. When borax is dissolved in water:
- A) $B(OH)_3$ is formed only
 - B) $[B(OH)_4]^-$ is formed only
 - C) both $B(OH)_3$ and $[B(OH)_4]^-$ are formed
 - D) $[B_3O_3(OH)_4]^-$ formed only

SECTION-II
(MORE THAN ONE TYPE)

Section - II (Multiple Correct Answers Type, Total Marks: 16) contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE may be correct. For each question you will be awarded 4 marks if you darken ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. There are no negative marks in this section.

9. Select correct statement(s)
- A) Li_2CO_3 is only sparingly soluble in water and no $LiHCO_3$ has been isolated.
 - B) K_2CO_3 cannot be made by a method similar to the ammonia- soda(solvay) process.
 - C) Li_2CO_3 and $MgCO_3$ both are thermally stable.
 - D) Due to alkaline nature of oxides and hydroxides of alkaline earth metals, these are found in the earth crust.

10. Which of the following is/are correct?

A) Ionization energy = $Ca > K$

B) Melting point = $Ca > K$

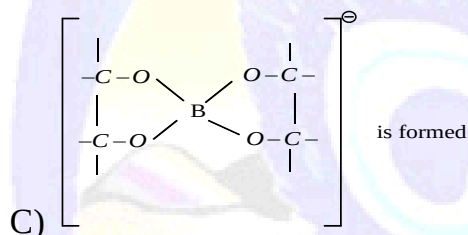
C) Solubility = $CaSO_4 > K_2SO_4$

D) Hydration energy = $Ca^{2+} > K^+$

11. On addition of cis 1-2 diol in the aqueous solution of boric acid:

A) H^+ ion concentration increases

B) H^+ ion concentration decreases



is formed

D) BO_3^{3-} is formed

12. The correct statements among the following

A) $NaBH_4$ is very much less rapidly hydrolysed by water than is $NaAlH_4$

B) The rate of hydrolysis of diborane by water vapour is proportional to $P_{B_2H_6}^2 \cdot P_{H_2O}^{\frac{1}{2}}$

C) 2, 6 – Dimethyl pyridine although stronger base than pyridine do not form adduct with trimethyl boron, however combine with triethyl aluminium

D) A saturated aqueous solution of boric acid do not effect the colour of bromocresol green (pH range 3.8-5.4) but changes its colour on adding excess of boric acid to a solution of KHF_2 due to alkaline.

SECTION-III
[INTEGER TYPE]

Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

13. Out of Be, Mg, Ca, Sr and Ba , number of elements which do not impart any colour to the flame, is
14. On heating a mixture of 8 moles of Li_2CO_3 and $KHCO_3$ in 3: 1 ratio, the number of moles of CO_2 liberated is
15. Na_2O_2 is a strong oxidizing agent. It oxidized chromic compounds into chromates. How many moles of Na_2O_2 are required to convert two moles of $Cr(OH)_3$ into sodium chromate?
16. In the compound $Na_2[B_4O_5(OH)_4] \cdot 8H_2O$, if the
- i) number of B-O-B bonds is 'x'
 - ii) number of B-B bonds is 'y'
 - iii) number of sp^2 hybridized B atoms is 'z'
- Calculate the value of $x+y+z$.

17. 1 mole of $B_2H_6(g)$ on hydrolysis yields.....moles of $H_2(g)$.
18. Find the number of B-O bonds in peroxoborate.

SECTION-IV
[Matrix Matching Type]

Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

19. Match the particulars/name of process listed in **column-I** with the formula of the compounds listed in **column II**.

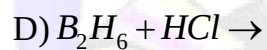
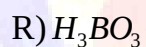
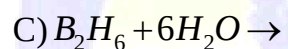
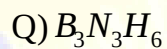
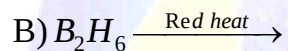
Column-I

- A) Solvay process
- B) Evolve CO_2 on heating
- C) Aqueous solution is not neutral towards litmus
- D) Used as air purifier in submarine

Column-II

- P) Na_2O
- Q) Na_2O_2
- R) $NaHCO_3$
- S) Na_2CO_3

20. Match the Reactions given in **column I** with the Products given in **column II**.

Column-I**Column-II**

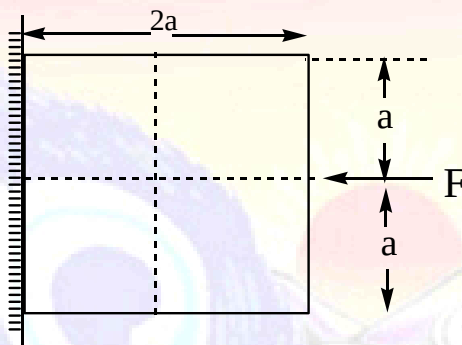
SECTION-1

(SINGLE CORRECT CHOICE TYPE)

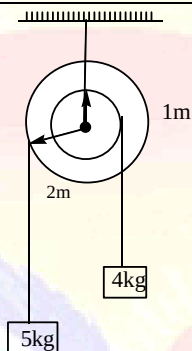
Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

21. Consider four bodies of equal mass - a ring, a cube, a disc and a sphere. Ring, disc and sphere have the same diameter, equal to the length of the cube on each edge. Ring and disc rotate about the axes passing through their centers and perpendicular to their planes. Sphere rotates about its diameter. Cube rotates about an axis passing through centre and parallel to edges. Which one has the largest moment of inertia?
- A) Ring B) Cube C) Disc D) Sphere.
22. A circular disc A of radius r is made from an iron plate of thickness t and another circular disc B of radius $4r$ is made from another iron plate of thickness $t/4$. The relation between the moments of inertia I_A and I_B is (I_A, I_B are the moments of inertia of the objects about normal axes passing through their centers)
- A) $I_B = 128I_A$ B) $I_A = I_B$
C) $I_B = 64I_A$ D) dependent on the actual values of t and r

23. A block of mass m is held stationary against a rough wall by applying a force F as shown in figure-. Which one of the following statements is incorrect?

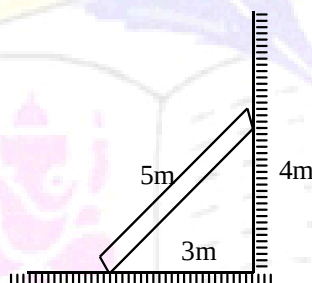


- A) frictional force $f = mg$
B) normal reaction $N = F$
C) F will not produce a torque about centre of the block
D) N will not produce any torque about centre of the block
24. The moment of inertia of the pulley system as shown in the figure is 4 kgm^2 . The radii of bigger and smaller pulleys are 2m and 1m respectively. Angular acceleration of the pulley system is: (take $g = 10\text{m/s}^2$)



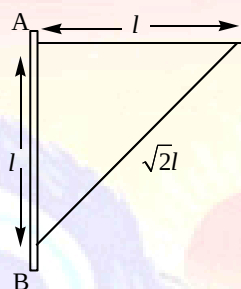
- A) 2.1 rad/s^2 B) 4.2 rad/s^2 C) 1.2 rad/s^2 D) 0.6 rad/s^2

25. A uniform ladder of length 5m is placed against the wall as shown in the figure. If coefficient of friction μ is the same for both the walls, what is the minimum value of μ for it not to slip?

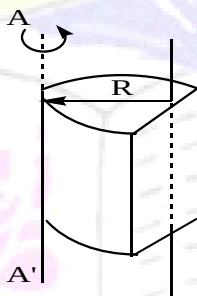


- A) $\mu = \frac{1}{2}$ B) $\mu = \frac{1}{4}$ C) $\mu = \frac{1}{3}$ D) $\mu = \frac{1}{5}$

26. Find moment of inertia of the triangular lamina of mass M about the axis of rotation AB shown in figure.



- A) $\frac{Ml^2}{6}$ B) $\frac{Ml^2}{3}$ C) $\frac{Ml^2}{12}$ D) $\frac{3}{4}Ml^2$
27. Find moment of inertia of the half cylinder of mass M shown in figure about the axis AA'.



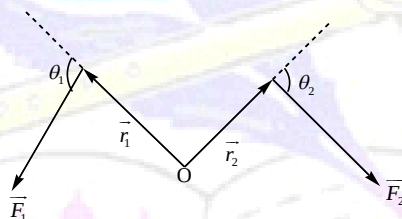
- A) $\frac{(9-16\pi)}{6\pi}MR^2$ B) $\left(\frac{9\pi-16}{6\pi}\right)MR^2$ C) $\frac{3}{2}MR^2$ D) $2MR^2$

28. Calculate the moment of inertia of a wheel about its axis which is having rim of mass $24M$ and twenty four spokes each of mass M and length l .
- A) $24Ml^2$ B) $32Ml^2$ C) $64Ml^2$ D) $16Ml^2$

SECTION-II
(MORE THAN ONE TYPE)

Section - II (Multiple Correct Answers Type, Total Marks: 16) contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE may be correct. For each question you will be awarded 4 marks if you darken ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. There are no negative marks in this section.

29. In the figure are shown the lines of action and moment arms of two forces about the origin O. Imagining these forces to be acting on a rigid body pivoted at O, all vectors shown being in the plane of the figure, the magnitude and direction of the resultant torque will be:

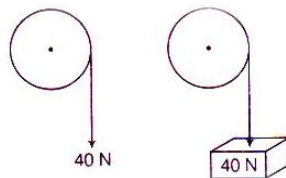


- A) $(F_2 r_2 \sin \theta_2 - F_1 r_1 \sin \theta_1)$ out of the plane of the page.
 B) $(F_1 r_1 \sin \theta_1 - F_2 r_2 \sin \theta_2)$ out of the plane of the page.
 C) $(F_2 r_2 \sin \theta_2 - F_1 r_1 \sin \theta_1)$ into the plane of the page.
 D) zero.

30. Two identical semicircular discs of mass 'm' each and radius 'R' are placed in the XY(horizontal) plane and the YZ (vertical) plane, respectively. They are so placed that they have their common diameter along the Y-axis and centre of curvatures are at the origin. Then, the moment of inertia(I_n) of the system about the appropriate axis is given by (I_n refers to moment of inertia about axis n-where n is X,Y,Z)

A) $I_x = \frac{1}{2}mR^2$ B) $I_y = \frac{1}{2}mR^2$ C) $I_z = \frac{3}{4}mR^2$ D) $I_x = I_y = I_z$

31. A cord is wrapped around the rim of a solid cylinder of radius 0.25 m and a constant force of 40 N is exerted on the cord. The cylinder is mounted on frictionless bearings and its moment of inertia is $4\text{ kg} \cdot \text{m}^2$



The angular velocity of the cylinder after 5 m of cord has been unwound, is ω_1 . If the 40N force is replaced by a 40 N weight and released from rest at the top, the angular

velocity of the cylinder after 5 m of cord has been unwound is ω_2 . (Given, $g = 10 \text{ ms}^{-2}$). Then,

A) $\omega_1 = 10 \text{ rad.s}^{-1}$

B) $\omega_2 = \frac{40}{\sqrt{17}} \text{ rad / s}$

C) $\omega_1 = 8.16 \text{ rad.s}^{-1}$

D) $\omega_2 = 8 \text{ rad.s}^{-1}$

32. A sphere is rotating about a fixed axis along a diameter with constant angular velocity. The which of the following options is/are correct:

A) The particles on the surface of the sphere do not have any linear acceleration.

B) The particles on the diameter mentioned above do not have any linear acceleration

C) Different particles on the surface have different angular speeds

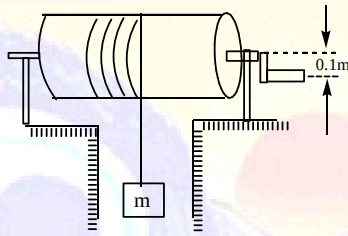
D) All the particles on the surface have the same linear speed

SECTION-III [INTEGER TYPE]

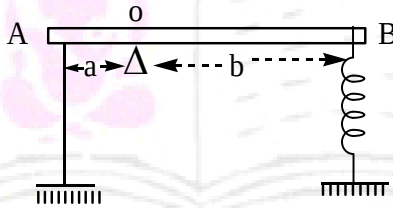
Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

33. The mechanism shown in figure is used to raise a wooden box of mass $m = 50 \text{ kg}$. A string is wrapped around a cylinder that turns on an axle. The cylinder has radius 0.25 m and moment of inertia 1 kg-m^2 about the axle. Magnitude of the force F applied

tangentially to the rotating crank handle is required to raise the box with an acceleration of 2.0 m/s^2 is $1500 + 16a \text{ N}$. Here we can neglect the moment of inertia of the axle and the crank. Then $a =$

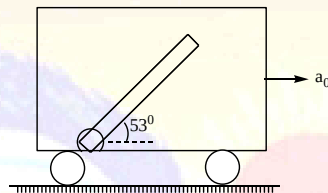


34. A uniform rod AB of mass 2 kg and length $l = 100 \text{ cm}$ is placed on a sharp support O such that $AO = a = 40 \text{ cm}$ and $OB = b = 60 \text{ cm}$. A spring of force constant $k = 600 \text{ N/m}$ is attached to end B as shown in figure. To keep the rod horizontal, its end A is tied with a thread such that the spring is elongated by $y = 1 \text{ cm}$. The reaction of support on the rod immediately after the thread is burnt is $5a \text{ N}$. Then $a =$

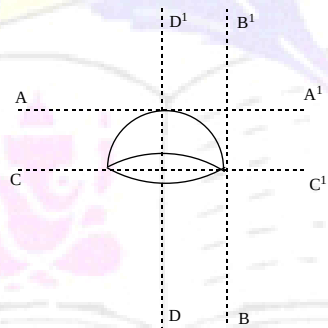


35. Two wooden disc, one with radius 2 cm and mass 1 kg and the other with radius 4 cm and mass 2 kg, are welded together coaxially and mounted on a frictionless axis through their common centre. A light string is wrapped around the edge of the smaller disc, and a 3 kg block is suspended from the free end of the string. The acceleration of the block after it is released is $\frac{2}{3}a \text{ m/s}^2$ then what is the value of a ? Assume that string does not slip on the disc.
36. A rod of length l is pivoted about an end. Find Moment of inertia of the rod about this axis if the linear mass density of rod varies as $\rho = ax^2 + b$ (x is distance from axis of rotation) is $\frac{al^5}{5(p)} + \frac{bl^3}{q}$. Then $p \times q =$
37. A uniform rod is pivoted to a trolley as shown in figure. Let α_1 is the angular acceleration of rod with respect to trolley when the trolley is moving with uniform

velocity V_o and α_2 is the angular acceleration of the rod with respect to trolley when the trolley is moving with uniform acceleration a_o so that $\vec{\alpha}_2 = -\vec{\alpha}_1$. Then $\frac{2a_o}{g} = \underline{\hspace{1cm}}?$



38. Consider a thin uniform shell as shown. Let I_1, I_2, I_3 and I_4 are the moments of inertia of the object about axes AA^1, BB^1, CC^1 & DD^1 respectively. $I_1 + I_2 + I_3 + I_4 = \frac{nI_1}{2}$. Then $n - 5 = \underline{\hspace{1cm}}$

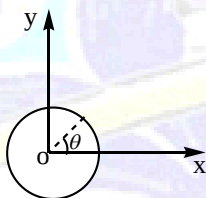


SECTION-IV

[Matrix Matching Type]

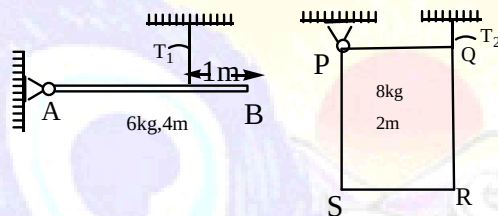
Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

39. A thin circular ring of mass M and radius R is placed at origin in xy plane. Let I_x, I_y , and I_z are the moments of inertia of it about x -axis, y -axis and z -axis respectively. If λ is the mass per unit length of ring then



- | | |
|---|-------------------------------|
| A) If $\lambda = \lambda_0 (1 + \cos \theta)$, $\lambda_0 = \text{constant}$ | P) $I_x = \frac{MR^2}{2}$ |
| B) If $\lambda = \lambda_0 (1 + x)$, $\lambda_0 = \text{constant}$ | Q) $I_y = \pi \lambda_0 R^3$ |
| C) If $\lambda = \lambda_0 (x^2 + y^2)$, $\lambda_0 = \text{constant}$ | R) $I_x = 2\pi \lambda_0 R^3$ |
| D) If $\lambda = \lambda_0 = \text{constant}$ | S) $I_z = MR^2$ |

40. A uniform thin rod AB of mass 6 kg, length 4m and a uniform thin square plate PQRS of mass 8 kg, side length 2 m are held horizontally as shown. Let F_1 & F_2 are the forces exerted by the pivots on rod & square plate and F_1^1 & F_2^1 are the forces exerted by the pivots on the objects just after the strings are cut. (take $g=10 \text{ m/s}^2$). Then



- | | |
|----------------------|----------|
| A) $2F_1$ | P) 340 N |
| B) F_2 | Q) 40 N |
| C) $8F_1^1$ | R) 120 N |
| D) $\sqrt{136}F_2^1$ | S) 680 N |

SECTION-1
(SINGLE CORRECT CHOICE TYPE)

Section-I (Single Correct Answer Type, Total Marks: 24) contains 8 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**. For each question you will be awarded 3 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks if no bubble is darkened. In all other cases, minus one (-1) mark will be awarded.

41. Let $L = \prod_{n=3}^{\infty} \left(1 - \frac{4}{n^2}\right)$; $M = \prod_{n=2}^{\infty} \left(\frac{n^3-1}{n^3+1}\right)$ and $N = \prod_{n=1}^{\infty} \frac{(1+n^{-1})^2}{1+2n^{-1}}$; then the value of

$L^{-1} + M^{-1} + N^{-1}$ is.

- A) 3 B) 6 C) 8 D) 2

42. The value of $\lim_{n \rightarrow \infty} \left[\left\{ 4 \sum_{r=0}^n (r+1)(r+2)(r+3) \right\}^{\frac{1}{4}} - n \right]$ is

- A) 0 B) Does not exist
C) 2/5 D) 5/2

43. If $f(x)$ is a polynomial satisfying $2 + f(x)f(y) = f(x) + f(y) + f(xy)$ for all $x, y \in \mathbb{R}$ and $f(2) = 5$ ($f(1) \neq 1$), then $\lim_{x \rightarrow 5} f(x) = \underline{\hspace{2cm}}$

- A) 35 B) 26 C) 5 D) 126

44. If $\lim_{x \rightarrow \infty} (a^x + e^x)^{1/x} = a; (a > 0)$ then
A) $a \in (1, \infty)$ B) $a \in (e, \infty)$ C) $a \in (1, e)$ D) None of these.
45. Let r^{th} term of a series be given by $t_r = \frac{r}{1 - 3r^2 + r^4}$. Then $\lim_{n \rightarrow \infty} \sum_{r=1}^n t_r$ is
A) 3/2 B) 1/2 C) -1/2 D) -3/2
46. Let $f(x) = \lim_{n \rightarrow \infty} \frac{1}{((3/\pi) \tan^{-1} 2x)^{2n} + 5}$. Then the set of values of x for which $f(x) = 0$ is
A) $|2x| > \sqrt{3}$ B) $|2x| < \sqrt{3}$ C) $|2x| \geq \sqrt{3}$ D) $|2x| \leq \sqrt{3}$
47. Let $f(x)$ be defined for all $x \in \mathbb{R}$ such that
 $\lim_{x \rightarrow 0} \left[f(x) + \ln \left(1 - \frac{1}{e^{f(x)}} \right) - \ln(f(x)) \right] = 0$ then $\lim_{x \rightarrow 0} f(x)$ is equal to
A) 0 B) 1 C) e D) 1/e
48. The value of $\lim_{x \rightarrow 0} \left\{ \sin^2 \left(\frac{\pi}{2 - ax} \right) \right\}^{\sec^2 \frac{\pi}{2 - bx}}$ is equal to
A) e^{a^2/b^2} B) e^{-a^2/b^2} C) 1 D) 0

SECTION-II
(MORE THAN ONE TYPE)

Section - II (Multiple Correct Answers Type, Total Marks: 16) contains 4 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which ONE or MORE may be correct. For each question you will be awarded 4 marks if you darken ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. There are no negative marks in this section.

49. If $f(x) = |x-1| - [x]$, where $[x]$ is the greatest integer less than or equal to x , then

A) $f(1^+) = -1, f(1^-) = 0$

B) $f(1^+) = f(1^-) = 0$

C) $\lim_{x \rightarrow 1} f(x)$ exists

D) $\lim_{x \rightarrow 1} f(x)$ does not exist.

50. If $\lim_{x \rightarrow 1} (2 - x + a[x-1] + b[1+x])$ exists, then a and b can take the values (where

$[.]$ denotes the greatest integer function)

A) $a = 1/3, b = 1$

B) $a = 1, b = -1$

C) $a = 9, b = -9$

D) $a = 2, b = 2/3$

51. Given a real valued function f such that $f(x) = \begin{cases} \frac{\tan^2 \{x\}}{(x^2 - [x]^2)}, & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot \{x\}}, & \text{for } x < 0 \end{cases}$

Where $[x]$ is the integer part and $\{x\}$ is the fractional part of x , then

A) $\lim_{x \rightarrow 0^+} f(x) = 1$

B) $\lim_{x \rightarrow 0^-} f(x) = \sqrt{\cot 1}$

C) $\cot^{-1}\left(\lim_{x \rightarrow 0^-} f(x)\right)^2 = 1$

D) $\tan^{-1}\left(\lim_{x \rightarrow 0^+} f(x)\right) = \frac{\pi}{4}$

52. Which of the following is/are correct.

A) $\lim_{x \rightarrow 0} \left(\frac{2^x + 2^{2x} + 2^{3x}}{3} \right)^{\frac{1}{x}} = 4$

B) $\lim_{x \rightarrow \infty} \left(\frac{x+6}{x+1} \right)^{x+4} = e^5$

C) $\lim_{x \rightarrow 0} \left((\sin x)^{\frac{1}{x}} + \left(\frac{1}{x} \right)^{\sin x} \right) = 1, (x > 0)$

D) $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 3} \right)^{\frac{1}{x}} = e^6$

SECTION-III [INTEGER TYPE]

Section-III (Integer Answer Type, Total Marks: 24) contains 6 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The bubble corresponding to the correct answer is to be darkened in the ORS. For each question you will be awarded 4 marks if you darken ONLY the bubble corresponding to the correct answer and zero marks otherwise. There are no negative marks in this section.

53. The integer value of n for which $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n}$ is a finite nonzero number is _

54. If $f(x) = \lim_{x \rightarrow \infty} \sum_{r=0}^n \frac{\tan\left(\frac{x}{2^{r+1}}\right) + \tan^3\left(\frac{x}{2^{r+1}}\right)}{1 - \tan^2\left(\frac{x}{2^{r+1}}\right)}$ then $\lim_{x \rightarrow 0} \frac{f(x) - \sin x}{x^3} = \frac{1}{k}$ Where k is equal to

55. If $L = \lim_{x \rightarrow 0+} \left\{ \lim_{n \rightarrow \infty} \left(\frac{[1^2(\sin x)^x] + [2^2(\sin x)^x] + \dots + [n^2(\sin x)^x]}{n^3} \right) \right\}$ where $[.]$ denotes the greatest integer function. Then $\frac{3}{L} = \underline{\hspace{2cm}}$

56. If $f(n, \theta) = \prod_{r=1}^n \left(1 - \tan^2 \frac{\theta}{2^r} \right)$, then $\lim_{\theta \rightarrow 0} \left(\lim_{n \rightarrow \infty} f(n, \theta) \right) = \underline{\hspace{2cm}}$

57. If $\lim_{x \rightarrow \infty} f(x)$ exists and is finite and nonzero and if $\lim_{x \rightarrow \infty} \left\{ f(x) + \frac{3f(x)-1}{f^2(x)} \right\} = 3$, then the value of $\lim_{x \rightarrow \infty} f(x)$ is $\underline{\hspace{2cm}}$

58. Let $f(x) = \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1}$, $g(x) = \frac{1-x}{1-\sqrt{x}}$ and $h(x) = 2$. The largest value of the non negative integer for which $\lim_{x \rightarrow 1} (f(x))^{g(x)} = \frac{1}{4}$ is m and $\lim_{x \rightarrow 1} (f(x))^{h(x)} = \frac{1}{4}$ is n then $m+n = \underline{\hspace{2cm}}$

SECTION-IV

[Matrix Matching Type]

Section-IV (Matrix-Match Type, Total Marks: 16) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s and t) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in q and r, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 2 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and zero marks otherwise. Thus, each question in this section carries a maximum of 8 marks. There are no negative marks in this section.

59. Match the following from Column –I to Column-II.

Column-I

Column-II

A) $\lim_{x \rightarrow 0} \left(\left[200 \frac{\tan x}{x} \right] + \left[200 \frac{\sin x}{x} \right] \right)$ equals

P)1

B) $\lim_{x \rightarrow 0} \left(\left\{ 100 \frac{\tan x}{x} \right\} + \left\{ 100 \frac{\sin x}{x} \right\} \right)$ equals

Q)399

C) $\lim_{x \rightarrow 0} \left(100 \left[\frac{\sin x}{x} \right] + 200 \left[\frac{x}{\tan^{-1} x} \right] \right)$ equals

R)199

D) $\lim_{x \rightarrow 0} \left(\left\{ \frac{\sin^{-1} x}{x} \right\} + \left[200 \frac{\tan^{-1} x}{x} \right] \right)$

S) 200

Here $[.]$ and $\{x\}$ are respectively greatest integer function and fractional part functions.

60. Let $\phi(x) = \frac{a_0x^m + a_1x^{m+1} + \dots + a_kx^{m+k}}{b_0x^n + b_1x^{n+1} + \dots + b_lx^{n+l}}$; where $a_0 \neq 0, b_0 \neq 0$, then $\lim_{x \rightarrow 0} \phi(x)$ is equal to for

($m, n > 0$ and $k, l \geq 0$)

Column-I

A) $m > n$

B) $m = n$

C) $m < n$ and $n - m$ is even, $a_0 / b_0 > 0$

D) $m < n$ and $n - m$ is even, $a_0 / b_0 < 0$

Column-II

P) ∞

Q) $-\infty$

R) a_0 / b_0

S) 0



Sri Chaitanya IIT Academy., India.

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Time: 07:30 AM – 10:30 AM

WAT-18

2011-P2

Date: 16-09-17

Max.Marks:240

KEY SHEET

CHEMISTRY

1	D	2	A	3	C	4	D	5	C
6	B	7	C	8	C	9	ABD	10	ABD
11	AC	12	ABCD	13	2	14	7	15	3
16	7	17	6	18	8	19	A-RS B-R C-PQRS D-Q	20	A-PQ B-PS C-PR D-P

PHYSICS

21	A	22	C	23	D	24	A	25	C
26	A	27	B	28	B	29	BC	30	BC
31	AB	32	B	33	5	34	4	35	6
36	3	37	3	38	6	39	A-PQS B-PQS C-PS D-PQS	40	A-Q B-Q C-R D-S

MATHS

41	C	42	D	43	B	44	B	45	C
46	A	47	A	48	B	49	AD	50	BC
51	ABCD	52	ABC	53	4	54	2	55	9
56	1	57	1	58	2	59	A-Q B-P C-S D-R	60	A-S B-R C-P D-Q